



MATHEMATICS FORM 1 SCHEMES OF WORK TERM 1

1-2

NATURAL NUMBERS

Place Value of Numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify, read and write natural numbers in symbols and words

Identify the place value of a number

Discussions

Solving problems in groups and individually

Illustrations

Charts showing place value of a number

The Abacus

Bank cheques and statements

Discovering secondary mathematics Book 1 Pages 1-3

Secondary mathematics KLB book 1 pages 1-2

Advantages in mathematics book 1 pages 1-2

3-4

NATURAL NUMBERS

Round off Numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Round off numbers to the nearest tens hundreds, thousands, millions and billions

Discussions

Solving problems involving rounding off numbers

Guiding

Illustrations

Charts to show the rounding off of numbers

Number line

Scales on a ruler, thermometer, veneer calipers

Discovering secondary mathematics book 1 page 3

KLB book 1 page 3

Secondary mathematics KIE book 1 page 2

Advancing in mathematics book 1 pages 3-4



5-6

NATURAL NUMBERS

Classification and operation on natural numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Classify numbers as odd, even and prime
- Solve word problems involving natural numbers
- Discussions
- Solving problems involving even, odd and prime numbers
- Carrying out operations on natural numbers
- Classification
- Guiding
- Charts to show the natural numbers
- Number line
- Place value charts
- Abacus
- Discovering secondary mathematics book 1 pages 4-8
- Secondary mathematics KLB book 1 pages 4-9
- Advancing mathematics book 1 pages 4-5
- Secondary mathematics KIE book 1 pages 9-11

6

1

FACTORS

Factors of composite numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find factors of various composite numbers
- Express composite numbers in factor form
- Solving problems involving factors of composite numbers
- Discussions
- Illustrations
- Charts to show the factorization of numbers
- Factor ? tree diagram
- Discovering Secondary mathematics book 1 pages 9
- Secondary mathematics KLB book 1 page 10
- Secondary mathematics KIE book 1 page 12-14
- Advancing in mathematics book 1 page 7

2

FACTORS

Prime Factors



OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Define the term prime factor
- Express numbers as products of prime factors
- Discussions
- Demonstrations
- Listing the prime factors of numbers
- Solving problems involving prime factors
- Guiding
- Charts to show the factorization of numbers
- Multiplication tables
- Discovering secondary mathematics book 1 page 10
- KLB book 1 pages 10-11
- Advancing mathematics book 1 page 9
- Secondary mathematics KIE book 1 page 12

3-4

FACTORS

Factors in power form

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Express factors in power form
- Solving problems involving factors expressed in power form
- Discussion
- Guiding the learner to express prime factors in power
- Charts to show the factorization of numbers
- Multiplication table
- Discovering secondary mathematics book 1 page 10
- KLB book 1 pages 10- 11
- Advancing mathematics book 1 page 9
- Secondary mathematics KIE book 1 page 12

5-6

DIVISIBILITY TEST

Divisibility of numbers by 2, 3, 4, 5

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Test the divisibility of numbers by 2, 3, 4, 5



Discussions
Solving problems involving divisibility of numbers by 2, 3, 4,5
Dividing numbers
Listing the prime factors of numbers
Illustrations
Divisibility test charts
Multiplication table
Prime numbers
Discovering secondary mathematics book 1 page 10
KLB book 1 pages 11-15
Advancing mathematics book 1 page 10-11
Secondary mathematics KIE book 1 page 14

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1-2
DIVISIBILITY TEST
Divisibility of numbers by 6, 8, 9

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Test the divisibility of numbers by 6, 8, 9 and use the knowledge of divisibility to solve problems
Discussions
Solving problems involving the divisibility of numbers of 6, 8, and 9
Illustrating to the learner how to test divisibility of numbers by 6, 8, 9, 10 and 11
Divisibility test charts
Multiplication table
Multiples of numbers
Factors of numbers
Prime numbers
Discovering secondary mathematics book 1 page 10 - 11
KLB book 1 pages 16 - 17
Advancing mathematics book 1 page 11
Secondary mathematics KIE book 1 page 14

3-4
DIVISIBILITY TEST
Divisibility of numbers 10 and 11

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Carry out the divisibility test of 10 and 11
Dividing numbers
Discussions



Solving problems

Divisibility test charts

Multiplication table

Multiples of numbers

Factors of numbers

Prime numbers

Discovering secondary mathematics book 1 page 10

KLB book 1 pages 19 - 21

Advancing mathematics book 1 page 12

Secondary mathematics KIE book 1 page 14

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GREATEST COMMON DIVISOR (GCD)

GCD of a set of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the GCD of a set of numbers

Apply GCD in real ? life situations

Discussions

Probing learners understanding of GCD

Reinforcing earlier knowledge

Solving problems involving GCD

Charts to show how to get GCD

Multiplication tables

Containers of different capacities

Discovering secondary mathematics book 1 page 10 - 11

KLB book 1 pages 22 - 23

Advancing mathematics book 1 page 13

Secondary mathematics KIE book 1 page 15

6

LEAST COMMON MULTIPLE (LCM)

Multiples of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

List the multiples of numbers

Discussions

Working out the multiples of numbers

Solving problems involving multiples of numbers

Guiding learner exercises to list down multiples of numbers.



Prime numbers
Multiplication tables
Natural numbers
Even numbers
Odd numbers
Containers of different capacities
Flickering light
Discovering secondary mathematics book 1 page 11
KLB book 1 pages 22 - 24
Advancing mathematics book 1 page 14
Secondary mathematics KIE book 1 page 15

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1-2
LEAST COMMON MULTIPLE (LCM)
LCM of a set of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the least common multiple of a set of numbers
Apply the knowledge of LCM in real ? life situations

Discussions
Working out the multiples of numbers
Illustrating and solving problems involving multiples of numbers
Solving word problems from everyday life situations
Multiplication tables
Natural numbers
Prime numbers
Multiples of numbers
Even numbers
Odd numbers
Bells Flickering lights
Alarms
Containers of different capacities
Discovering secondary mathematics book 1 page 11
KLB book 1 pages 25 - 26
Advancing mathematics book 1 page 15
Secondary mathematics KIE book 1 page 15

3-4
INTEGERS
Integers

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Define integers
Identify integers on a number line
Discussions
Reading numbers on a number line
Showing numbers on a number line
Solving problems involving integers and the number line
Definition
Number line
Stairways
Natural numbers
Whole Numbers
Discovering secondary mathematics book 1 page 14
KLB book 1 pages 27- 29
Advancing mathematics book 1 page 18
Secondary mathematics KIE book 1 page 1-3

5-6
INTEGERS
Operations of integers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Perform the four basic operations on integers using the number line
Demonstrations
Discussions
Illustrating and solving problems involving four basic operations on integers
Explanations
Discussing practical exercises
Charts showing integers
Number line
Stair case
Ladder
Thermometer
Real life situations
Discovering secondary mathematics book 1 page 16-20
KLB book 1 pages 30-36
Advancing mathematics book 1 page 22-29
Secondary mathematics KIE book 1 page 3-8

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1-2
INTEGERS
Combined operations

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Work out problems involving combined operations on integers in the correct order
Discussions
Demonstrations
Guiding the learner to perform operations on integers on correct order
Solving problems involving combined operations on integers
Number line
Stair case
Ladder
Thermometer
Discovering secondary mathematics book 1 page 20-22
Secondary mathematics KLB book 1 pages 37-39
Advancing mathematics book 1 page 30-33
Secondary mathematics KIE book 1 page 9-11
Discovering secondary mathematics teachers book 1 pages 6-7

3-4

FRACTIONS

The meaning of fractions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define fractions
Identify and write fractions in figures (proper and improper)
Definition
Discussions
Solving problems involving fractions
Charts
Illustrating operations on fractions
Discovering secondary mathematics book 1 page 11
Discovering secondary mathematics teachers guide pages 8-11

Secondary mathematics KLB book 1 pages 40-41
Advancing mathematics book 1 page 36
Secondary mathematics KIE book 1 page 18

5-6

FRACTIONS

Equivalent fractions

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Identify and write equivalent fractions

Discussions

Dividing equally

Measuring

Weighing

Solving problems

Sticks

Pieces of paper

Ruler

Real life - situation

Discovering secondary mathematics book 1 page 24

teachers book pages 8-11

KLB book 1 pages 42-44

Advancing mathematics book 1 pages 37-38

Secondary mathematics KIE book 1 page 19-20

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1-2

FRACTIONS

Naming fractions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Name fractions correctly and convert an improper fraction to a mixed number and vice versa

Discussions

Sharing equally

Solving problems

Converting fractions

Doing exercises

Counters such as seeds, bottle tops, stones

Pieces of paper

Sticks

Discovering secondary mathematics book 1 pages 25-26

Discovering secondary mathematics teachers guide pages 8-11

KLB book 1 page 44

Advancing mathematics book 1 page 38-39

Secondary mathematics KIE book 1 page 20

3-4

FRACTIONS

Adding and subtracting fractions

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Add and subtract fractions
Showing
Discussions
Adding
Subtracting
Converting Fractions
Oranges
Sticks
Pieces of paper
Counters
Discovering secondary mathematics book 1 pages 25-26
KLB book 1 page 45-48
Advancing mathematics book 1 page 41
Secondary mathematics KIE book 1 page 21-22

5-6

FRACTIONS

Multiplication and division of fractions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Perform multiplication and division of fractions
Discussions
Multiplication
Division
Converting fractions
Showing the learner how to manipulate fractions
Sticks
Stones
Seeds
Pieces of paper
Discovering secondary mathematics book 1 pages 27-29
Discovering secondary mathematics teachers guide pages 8-11
KLB book 1 page 49-54
Advancing mathematics book 1 page 42-45
Secondary mathematics KIE book 1 page 23-26

11

1-2

FRACTIONS

Order of operations

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Carry out combined operations on fractions on the correct order
Discussions
Addition?
Subtraction
Multiplication
Division
Conversion of fractions
Multiplication tables
Conversion tables
Real objects
Discovering secondary mathematics book 1 pages 29-30
Discovering secondary mathematics teachers guide pages 8-11
KLB book 1 page 54-57
Advancing mathematics book 1 page 47
Secondary mathematics KIE book 1 page 28-31

3-4

FRACTIONS

Application of fractions in a real life situation

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve world problems involving fractions in real life situations
Discussions
Solving problems involving fractions in real life situation
Multiplication tables
Conversion tables
Real objects
Counters
Discovering secondary mathematics book 1 pages 30-31
Discovering secondary mathematics teachers guide pages 8-11
Secondary mathematics KLB students book 1 page 57
Advancing mathematics book 1 page 47
Secondary mathematics KIE book 1 page 31

5-6

FRACTIONS

Revision

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Answer the questions in the student's book.



Further exercises

12 AND 13
REVISION AND EXAMINATION

MATHEMATICS FORM 1 SCHEMES OF WORK ? TERM 2
WEEK
LESSON
TOPIC
SUB - TOPIC
OBJECTIVES
LEARNING/TEACHING ACTIVITIES
LEARNING/TEACHING RESOURCES
REFERENCES
REMARKS
1
1-2
DECIMALS
Fractions and decimals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define decimals
Convert fractions into decimals
Definition
Discussions
Doing exercises
Illustration
Demonstrations
Dividing
Multiplying
Equivalent fractions
Multiplication tables
Real life situations
Discovering secondary mathematics Book 1 Pages 32-34
Discovering secondary mathematics teachers guide pages 12-15
Secondary mathematics KLB book 1 pages 58-60
Secondary mathematics KIE book 1 page 32-34
Advancing in mathematics book 1 pages 48

3-4
DECIMALS



Recurring decimals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify and write recurring decimals

Discussions on recurring decimals

Doing exercises

Dividing

Multiplying

Demonstrations

Explanation

Equivalent fractions

Multiplication tables

Real ? life situation

Discovering secondary mathematics book 1 page 34

KLB book 1 pages 61-62

Secondary mathematics KIE book 1 page 43-44

Advancing in mathematics book 1 pages 59

Discovering secondary mathematics teachers guide pages 12-15

5-6

DECIMALS

Recurring decimals and fractions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify recurring decimals

Convert recurring decimals into fractions

Guiding learner to identify recurring decimals

Discussion on recurring decimals

Doing exercises

Conversion

illustrations

Equivalent fractions

Percentages

Multiplication tables

Discovering secondary mathematics book 1 pages 35

Discovering secondary mathematics teachers guide pages 25-30

Secondary mathematics KLB book 1 pages 4-9

Advancing mathematics book 1 pages 59

Secondary mathematics KIE book 1 pages 44



1

DECIMALS

Rounding off decimals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Round off a decimal number to the required number of decimal places

Discussions

Estimation

Rounding off

Demonstrations

Doing exercises

Place value charts

Ruler

Tape measure

Objects

Discovering Secondary mathematics book 1 pages 36

Secondary mathematics KLB book 1 page 64

Secondary mathematics KIE book 1 page 42

Advancing in mathematics book 1 page 56-57

2

DECIMALS

Standard form

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Write numbers in standard form and apply in real life situations

Discussions

Writing whole numbers and decimal numbers in standard form

Doing exercises

Illustrations

explanations

place value charts

measuring instruments

objects

Discovering secondary mathematics book 1 page 36-37

KLB book 1 pages 62

Advancing mathematics book 1 page 51-55

Discovering secondary mathematics teachers guide pages 12-15

KLB teachers book 25-30

3

DECIMALS



Addition and subtraction of decimals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Add decimals
- Subtract decimals
- Discussions
- Demonstrations
- Explanations
- Adding
- Subtracting

- Place value charts
- Measuring instruments such as tape measure, ruler, meter rule
- Regular shaped objects
- Discovering secondary mathematics book 1 page 37-38
- KLB book 1 pages 63
- Advancing mathematics book 1 page 49
- Secondary mathematics KIE book 1 page 36-37
- KLB teachers book 25-30

4-5

DECIMALS

Multiplication and division

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Multiply decimals
- Divide decimals
- Discussions
- Demonstrations
- Multiplying
- Correcting errors caused by failure to manipulate decimal point correctly
- Doing exercises
- Mathematical table
- Multiplication table
- Place value chart
- Measuring instruments
- Regular shaped objects
- Discovering secondary mathematics book 1 page 38-42
- KLB book 1 pages 64
- Advancing mathematics book 1 page 42-45
- Secondary mathematics KIE book 1 page 38-40



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DECIMALS

Combined operation on decimals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Carry out operations in the correct order
Apply the knowledge of decimals to real life situations
Discussions
Application of decimals to real life situations
Adding
Multiplying
Dividing
Subtracting
Solving puzzles
Playing games
Place value charts
Multiplication tables
Mathematical tables
Tape measure
Metre rule
Ruler
Strings
Regular shaped objects
Discovering secondary mathematics book 1 page 42-43
KLB book 1 pages 71-72
Advancing mathematics book 1 page 60
Secondary mathematics KIE book 1 page 44
KLB teachers book 25-30
Golden tips (KCSE) mathematics page 14

3

1-2

SQUARES AND SQUARE ROOTS

Squares of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define the term square
Find squares of numbers by multiplication and factorization

Discussions
Multiplication
Memorizing
Doing short test
Solving puzzles



Multiplication tables
Mathematical tables
Calculators (scientific)
Discovering secondary mathematics book 1 page 44-46
KLB book 1 pages 73
Advancing mathematics book 1 page 61-62
Secondary mathematics KIE book 1 page 96-97
mathematics page 60

3

SQUARES AND SQUARE ROOTS

Squares of numbers greater than 1 and less than 10

OBJECTIVES:

By the end of the lesson, the learner should be able to:

read the mathematical table
find the squares of numbers from the mathematical table
Discussions
Reading the mathematical table
Emphasizing standard form
Doing short tests
Playing games
Mathematical tables
Multiplication tables
Calculators (scientific)
Discovering secondary mathematics book 1 page 45
KLB book 1 pages 75
Advancing mathematics book 1 page 62-63
Secondary mathematics KIE book 1 page 97 Golden tips (KCSE) mathematics
page 61

4

SQUARES AND SQUARE ROOTS

Squares of numbers greater than 10

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the square of numbers greater than 10 from the mathematical table.
Discussion
Exercises
Reading mathematical tables
Illustrations
Mathematical tables
Multiplication tables



Calculators (Scientific)

KLB book 1 pages 75

Discovering secondary mathematics book 1 page 45

Advancing mathematics book 1 page 63

Secondary mathematics KIE book 1 page 97

Golden tips (KCSE) mathematics page 61

5

SQUARES AND SQUARE ROOTS

Squares of numbers less than 1

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the squares of numbers less than 1 from mathematical tables

Discussions

Doing exercises

Reading mathematical tables

Solving puzzles

Mathematical tables

Multiplication tables

Calculators (Scientific)

KLB book 1 pages 76

Discovering secondary mathematics book 1 page 46

Advancing mathematics book 1 page 63

Secondary mathematics KIE book 1 page 97-98

Golden tips (KCSE) mathematics page 61

6

SQUARES AND SQUARE ROOTS

Finding square roots by factorization

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find square roots of numbers by factorization

Discussions

Factorizing numbers

Doing short tests

Demonstrations

Dividing numbers

Mathematical tables

Multiplication tables

Discovering secondary mathematics book 1 page 46-47

Secondary mathematics KLB book 1 pages 77

Advancing mathematics book 1 page 64-65



Secondary mathematics KIE book 1 page 99-101
Golden tips (KCSE) mathematics page 62

4

1-2

SQUARES AND SQUARE ROOTS

Square root tables

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Read the square root table
Read square roots of numbers $1 < A < 10$ from mathematical tables
Guiding the learner to read square roots from tables
Discussions
Doing exercises
Mathematical tables
Multiplication tables
Discovering secondary mathematics book 1 pages 47-48
KLB book 1 page 78
Advancing mathematics book 1 page 66
Secondary mathematics KIE book 1 page 102-103

3-4

SQUARES AND SQUARE ROOTS

Square roots of numbers less than one and greater than 100

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Get the Square roots of numbers less than one and greater than 100 from their mathematical tables
Discussions
Mathematical table
Doing exercises
Mathematical tables
Multiplication tables
Discovering secondary mathematics book 1 pages 48-49
KLB book 1 page 78
Advancing mathematics book 1 page 67-68
Secondary mathematics KIE book 1 page 102-103

5-6

SQUARES AND SQUARE ROOTS

Revision



OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Solve problems involving squares and square roots
- Discussions
- Illustrations
- Doing exercises
- Supervised practice
- Mathematical table
- Charts
- Discovering secondary mathematics book 1 pages 49
- KLB book 1 page 79
- Advancing mathematics book 1 page 69-72
- Secondary mathematics KIE book 1 page 104

5

1

ANGLES

Measuring and drawing angles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Measure and draw an angle using a protractor
- Discussions
- Demonstration
- Measuring angles
- Illustrations
- Protractor
- Ruler
- Models of figures
- Charts
- Discovering secondary mathematics book 1 pages 50-52
- KLB book 1 page 19757
- Advancing mathematics book 1 page 173-175
- Secondary mathematics KIE book 1 page 62

2

ANGLES

Types of angles

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Name and draw different types of angles

Discussions

Drawing angles

Measuring angles

Naming angles

Protractor

Ruler

Models of triangles, rectangles etc

Charts

Discovering secondary mathematics book 1 pages 51-52

Secondary mathematics KLB students book 1 page 197-198

Advancing mathematics book 1 page 175-177

Secondary mathematics KIE book 1 page 62-64

3-4

ANGLES

Angles on a straight line, vertically opposite angles and angles at a point

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify and draw angles on a line vertically opposite angles and angles at a point

Discussions

Drawing angles

Doing exercises

Naming angles

Protractor

Ruler

Models of triangles, rectangles etc

Discovering secondary mathematics book 1 pages 52-54

Secondary mathematics KLB students book 1 page 200-201

Advancing mathematics book 1 page 177-179

Secondary mathematics KIE book 1 page 66-67

5-6

ANGLES

Parallel lines corresponding alternate and interior angles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify and draw parallel lines, corresponding alternate and interior angles



Discussions
Drawing angles
Measuring angles
Identifying angles
Protractor
Ruler
Real objects

Discovering secondary mathematics book 1 pages 55-57
Secondary mathematics KLB students book 1 page 206-210
Advancing mathematics book 1 page 180-181
Secondary mathematics KIE book 1 page 70-72

6
1-2
POLYGONS
Triangles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define a polygon and identify and draw different triangles.
Discussions
Drawing triangles
Measuring Angles
Measuring lengths
Definitions
Protractor
Ruler
Models of different triangles

Discovering secondary mathematics book 1 pages 58-61
Secondary mathematics KLB students book 1 page 211
Advancing mathematics book 1 page 182-183
Secondary mathematics KIE book 1 page 75-76

3-4
POLYGONS
Interior and exterior angles of a polygons

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify and draw interior and exterior angles of a quadrilateral
Discussions
Drawing quadrilaterals
Measuring angles



Measuring lengths
Illustrations
Protractor
Ruler
Strings
Real Objects

Discovering secondary mathematics book 1 pages 63-66
Secondary mathematics KLB students book 1 page 212-213
Advancing mathematics book 1 page 182-183
Secondary mathematics KIE book 1 page 76-77

5-6
POLYGONS
Quadrilaterals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify and draw different quadrilaterals
Discussions
Drawing quadrilaterals
Measuring angles
Measuring lengths

Protractor
Ruler
Strings
Real Objects

Discovering secondary mathematics book 1 pages 60-63
Secondary mathematics KLB students book 1 page 219-220
Advancing mathematics book 1 page 185-186
Secondary mathematics KIE book 1 page 82-83

7
1-2
LENGTH
Units of length

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the units of measuring length and express length to a given significant figure
Discussions
Definitions



Rounding off
Measuring lengths
Solving problems involving units of length

Tape measure
Rulers
Strings
Measuring instruments

Discovering secondary mathematics book 1 pages 67-68
Secondary mathematics KLB students book 1 page 110
Advancing mathematics book 1 page 100-101
Secondary mathematics KIE book 1 page 106

3

LENGTH

Conversion of units of length

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Convert the units of length
Discussions
Conversions of units of length
Measuring length in different units
Solving problems

Tape measure
Rulers
Strings
Conversion charts for length
Real objects

Discovering secondary mathematics book 1 pages 68-71
Secondary mathematics KLB students book 1 page 110-112
Advancing mathematics book 1 page 100-101
Secondary mathematics KIE book 1 page 107-108

4

LENGTH

Perimeter of plain figures

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the perimeter of a plain figure



Discussions
Measuring length
Solving problems

Tape measure
Rulers
Strings
Conversion charts for length
Real objects

Discovering secondary mathematics book 1 pages 71-73
Secondary mathematics KLB students book 1 page 113-116
Advancing mathematics book 1 page 103-104
Secondary mathematics KIE book 1 page 109-112

5-6
LENGTH
Circumference

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the circumference of a circle
Discussions
Measuring length
Solving problems on circumference
Demonstrations

Circular objects
Tape measure
Rulers
Strings Conversion charts for length

Discovering secondary mathematics book 1 pages 73-76
Secondary mathematics KLB students book 1 page 116-119
Advancing mathematics book 1 page 105-106
Secondary mathematics KIE book 1 page 116-118

8
1-2
GEOMETRIC CONSTRUCTIONS
Construction of a perpendicular bisector of a line

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Use a pair of compasses and ruler only to construct a perpendicular bisector of a line

Discussions

Construction of a perpendicular bisector of a line

Drawing shapes

Plane figures

Geometrical sets

Polygonal shapes

Ruler

Pair of compasses

Discovering secondary mathematics book 1 pages 77-78

Secondary mathematics KLB students book 1 page 227

Advancing mathematics book 1 page 197-198

Secondary mathematics KIE book 1 page 153

3-4

GEOMETRIC CONSTRUCTIONS

Construction of parallel lines

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Construct parallel lines using a ruler and a set square or at a given distance

Discussions

Constructions

Demonstrations

Measuring angles

Set squares

Ruler

Pair of compass

Discovering secondary mathematics book 1 pages 78-79

Secondary mathematics KLB students book 1 page 235

Advancing mathematics book 1 page 202

Secondary mathematics KIE book 1 page 154-155

5-6

GEOMETRIC CONSTRUCTIONS

Dividing a line proportionally

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Divide a line proportionally using a ruler and a set square
Discussions
Dividing lines proportionally
Drawing straight lines
Measuring angles and lengths

Set squares
Ruler
Pair of compass

Discovering secondary mathematics book 1 pages 79-80
Secondary mathematics KLB students book 1 page 236
Advancing mathematics book 1 page 203
Secondary mathematics KIE book 1 page 156

9

1-2

GEOMETRIC CONSTRUCTIONS

Reproducing and bisecting angles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Reproduce and bisect an angle
Discussions
Demonstrations
Illustrations
Drawing angles

Ruler
Pair of compass

Discovering secondary mathematics book 1 page 80
Secondary mathematics KLB students book 1 page 233
Advancing mathematics book 1 page 202
Secondary mathematics KIE book 1 page 149-150

3-4

GEOMETRIC CONSTRUCTIONS

Constructing angles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Construct angles at 60 degrees
Discussions



Measuring angles
Bisecting angles
Constructing angles
Solving problems on construction of angles

Ruler
Pair of compass
Protractor
Real objects

Discovering secondary mathematics book 1 page 80-82
Secondary mathematics KLB students book 1 page 233-234
Advancing mathematics book 1 page 201-202
Secondary mathematics KIE book 1 page 149-150-151

5-6
GEOMETRIC CONSTRUCTIONS
Constructing polygons

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Construct regular and irregular polygons
Discussions
Measuring angles and lengths
Bisecting angles
Constructing angles

Ruler
Pair of compass
Protractor
Real objects

Discovering secondary mathematics book 1 page 83-84
Secondary mathematics KLB students book 1 page 237-240
Advancing mathematics book 1 page 207-208
Secondary mathematics KIE book 1 page 158-159

10
1-2
ALGEBRAIC EXPRESSIONS
Representing numbers by letters

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Use letters to represent numbers



Discussions
Doing exercises
Guessing
Demonstrations

Counters
Groups of items

Discovering secondary mathematics book 1 page 85-86
Secondary mathematics KLB students book 1 page 80-81
Advancing mathematics book 1 page 73-74
Secondary mathematics KIE book 1 page 47-48

3-4

ALGEBRAIC EXPRESSIONS

Simplifying algebraic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Simplify algebraic expressions
Discussions
Doing exercises
Grouping items together
Adding objects
Multiplying and dividing

Counters
Groups of items
charts

Discovering secondary mathematics book 1 page 86-88
Secondary mathematics KLB students book 1 page 82-84
Advancing mathematics book 1 page 74-75
Secondary mathematics KIE book 1 page 49-51

5

ALGEBRAIC EXPRESSIONS

Multiplying algebraic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Multiply algebraic expressions
Discussions
Simplification of algebraic expressions
Multiplying algebraic expressions
Grouping like items
Doing exercises

Groups of unlike items
Groups of like items

Discovering secondary mathematics book 1 page 87
Secondary mathematics KLB students book 1 page 89
Advancing mathematics book 1 page 75
Secondary mathematics KIE book 1 page 49-51

6
ALGEBRAIC EXPRESSIONS
Dividing algebraic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Divide algebraic expressions
Discussions
factorization
Grouping like items
Doing exercises
Puzzles
games

Groups of like items

Discovering secondary mathematics book 1 page 87-88
Secondary mathematics KLB students book 1 page 89
Advancing mathematics book 1 page 76
Secondary mathematics KIE book 1 page 49-51

11
1-2
ALGEBRAIC EXPRESSIONS
Use of brackets in algebraic expressions



OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Use of brackets in algebraic expressions
- Discussions
- Adding
- Subtracting
- Demonstrations
- Matching items
- Multiplying
- Dividing
- Doing exercises

Groups of like items/objects

- Discovering secondary mathematics book 1 page 88-89
- Secondary mathematics KLB students book 1 page 85
- Advancing mathematics book 1 page 77
- Secondary mathematics KIE book 1 page 52-53

3-4

ALGEBRAIC EXPRESSIONS

factorization in algebraic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Use factorization in algebraic expressions
- Discussions
- Factorizing
- puzzles
- Adding
- Substitutions
- Doing exercises
- Playing games

Groups of like items/objects

- Discovering secondary mathematics book 1 page 88-91
- Secondary mathematics KLB students book 1 page 90
- Advancing mathematics book 1 page 81-83
- Secondary mathematics KIE book 1 page 57-58



5-6

ALGEBRAIC EXPRESSIONS

Substitution and factorization

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Substitute and factorize algebraic expressions
- Discussions
- Dividing
- multiplying
- Adding
- Subtracting
- Solving puzzles
- Substitutions
- Doing exercises
- Playing games

Groups of like items/objects

- Discovering secondary mathematics book 1 page 91-94
- Secondary mathematics KLB students book 1 page 91-92
- Advancing mathematics book 1 page 80& 84
- Secondary mathematics KIE book 1 page 59-62

12

1-2

LINEAR EQUATIONS

Solving equations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Solve linear equations in one unknown
- Discussions
- Demonstrations
- Solving problems on linear equations
- Puzzles

Beam balance



See ? saw
Games

Discovering secondary mathematics book 1 page 95-96
Secondary mathematics KLB students book 1 page 160-162
Advancing mathematics book 1 page 134
Secondary mathematics KIE book 1 page 173-174

3-4
LINEAR EQUATIONS
Forming linear equations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Form linear equations
Discussions
Forming linear equations
Demonstrations
Doing exercises

Beam balance
Real life experience

Discovering secondary mathematics book 1 page 96-97
Secondary mathematics KLB students book 1 page 163-168
Advancing mathematics book 1 page 139
Secondary mathematics KIE book 1 page 176-178

5-6
LINEAR EQUATIONS
Simultaneous equations and forming simultaneous equations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve simultaneous equations by substitution and elimination and form simultaneous equations
Discussions
Solving simultaneous equations by substitution and elimination
Demonstrations



Forming simultaneous equations
Playing games

Beam balance
Real life experience

Discovering secondary mathematics book 1 page 99-102
Secondary mathematics KLB students book 1 page 168-169
Advancing mathematics book 1 page 137-138
Secondary mathematics KIE book 1 page 178-180

13
END ? TERM EXAMS

MATHEMATICS FORM 1 SCHEMES OF WORK ? TERM 3

WEEK

LESSON

TOPIC

SUB - TOPIC

OBJECTIVES

LEARNING/TEACHING ACTIVITIES

LEARNING/TEACHING RESOURCES

REFERENCES

REMARKS

1

1-2

COORDINATES AND GRAPHS

The Cartesian plane

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Draw and label the Cartesian plane

Discussions

Labeling the Cartesian plane

Writing scale

Drawing the axes of the Cartesian plane

Reading points from the Cartesian plane

Graph papers

Square board

Cartesian plane

Typographical maps

Ruler



Discovering secondary mathematics Book 1 Pages 103-104
Secondary mathematics KLB book 1 pages 182-183
Secondary mathematics KIE book 1 page 223-224
Advancing in mathematics book 1 pages 161-162

3-4

COORDINATES AND GRAPHS

Plotting graphs and linear graphs

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Plot points on a Cartesian plane
Read points on linear graphs on the Cartesian plane

Discussions
Labeling the Cartesian plane
Writing scale
Plotting points on the Cartesian plane
Reading points from the Cartesian plane
Drawing axes of a Cartesian plane
Graph papers
protractor
Square boards
Creo boards
Grid boards
Typographical maps
Ruler

Discovering secondary mathematics Book 1 Pages 1014-108
Secondary mathematics KLB book 1 pages 185-186
Secondary mathematics KIE book 1 page 226
Advancing in mathematics book 1 pages 163-164

5-6

COORDINATES AND GRAPHS

Plotting linear graphs and linear equations of lines

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Plot a linear graph
Work out the equation of a line

Discussions
Labeling the Cartesian plane



Writing scale
Plotting points on the Cartesian plane
Reading points from the Cartesian plane
Drawing axes of a Cartesian plane
Graph papers
protractor
Square boards
Creo boards
Grid boards
Typographical maps

Discovering secondary mathematics Book 1 Pages 108-111
Secondary mathematics KLB book 1 pages 185-186
Secondary mathematics KIE book 1 page 226
Advancing in mathematics book 1 pages 165-166

2

1-2

COORDINATES AND GRAPHS

Graphical solutions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve simultaneous linear equations graphically

Discussions
Writing scale
Labeling the Cartesian plane
Solving linear equations graphically
Illustrations
Drawing the axes of a Cartesian plane
Graph papers
protractor
Square boards
Typographical maps
Ruler
Grid boards

Discovering secondary mathematics Book 1 Pages 111-113
Secondary mathematics KLB book 1 pages 188-189
Secondary mathematics KIE book 1 page 229-232
Advancing in mathematics book 1 pages 167-170

3-4

AREA

Area of combined and conversion of units of area

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Define and work out area of combined rectangular shapes
Convert units of area from one form to another

Drawing rectangular shapes
Working out the area of rectangles
Discussions
Solving problems on conversion of units of area

Regular flat surfaces
Square paper
Square boards
Rectangular objects

Discovering secondary mathematics Book 1 Pages 114-117
Secondary mathematics KLB book 1 pages 123-124
Secondary mathematics KIE book 1 page 109
Advancing in mathematics book 1 pages 109

5-6
AREA
Area of a triangle and rectangle

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the area of a triangle and rectangle
Discussions
Measuring length
Calculating the area of a triangle
Triangular flat surfaces
Square paper
Square boards
Triangular objects
Discovering secondary mathematics Book 1 Pages 117-118
Secondary mathematics KLB book 1 pages 124-125
Secondary mathematics KIE book 1 page 110
Advancing in mathematics book 1 pages 109-110

3
1-2
AREA
Area of a parallelogram and trapezium

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Calculate the area of a
Parallelogram
trapezium

Discussions
Measuring length
Calculating the area of a parallelogram
Calculating the area of a trapezium
Square board
Trapezoidal objects
Shapes with a shape of a parallelogram
Discovering secondary mathematics Book 1 Pages 118-121
Secondary mathematics KLB book 1 pages 125-126
Secondary mathematics KIE book 1 page 110-111
Advancing in mathematics book 1 pages 111-113

3-4
AREA
Area of a circle

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the area of a circle

Discussions
Demonstrations
Measuring the radius/diameter
Calculating the area of a circle
Circular shapes
Or objects
Square paper
Square board
Discovering secondary mathematics Book 1 Pages 121-123
Secondary mathematics KLB book 1 pages 129-132
Secondary mathematics KIE book 1 page 123-125
Advancing in mathematics book 1 pages 115

5-6
AREA
Area of irregular plane

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Work out the area of an irregular plane figure
Discussions
Demonstrations
Measuring length
Estimating area
Converting units of areas
Irregular objects/shapes
Square paper
Square board
Discovering secondary mathematics Book 1 Pages 123-124
Secondary mathematics KLB book 1 pages 111
Secondary mathematics KIE book 1 page 115-116
Advancing in mathematics book 1 pages 141-142

4
1-2
AREA
Surface area of a cuboids

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Work out the surface area of a cuboids
Discussions
Demonstrations
Measuring length
Estimating area
Converting units of areas
Regular flat shapes
Square paper
Square board
Model cubes and cuboids
Discovering secondary mathematics Book 1 Pages 124-125
Secondary mathematics KLB book 1 pages 135
Secondary mathematics KIE book 1 page 130
Advancing in mathematics book 1 pages 118

3-4
AREA
Surface area of a prism

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Work out the surface area of a prism



Discussions
Demonstrations
Measuring length
Estimating area
Converting units of areas
Regular cylinders and prisms
Square paper
Square board
Model of a prism
Discovering secondary mathematics Book 1 Pages 125-126
Secondary mathematics KLB book 1 pages 135-136
Secondary mathematics KIE book 1 page 131-132
Advancing in mathematics book 1 pages 118

5-6
AREA
Surface area of a cylinder

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Work out the surface area of a cylinder
Discussions
Demonstrations
Measuring length
Estimating area
Converting units of areas
Doing exercises
Regular cylinders
Square paper
Square board
Model of a cylinder
Discovering secondary mathematics Book 1 Pages 126-128
Secondary mathematics KLB book 1 pages 137-138
Secondary mathematics KIE book 1 page 131-132
Advancing in mathematics book 1 pages 118

5
1-2
VOLUME & CAPACITY
Units of volume

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the units of volume in cubic units and convert one from another



Discussions
Measuring length, width and height
Measuring volume
Calculating the volume of a cuboids
Equipment for measuring volume
Discovering secondary mathematics Book 1 Pages 129
Secondary mathematics KLB book 1 pages 143
Secondary mathematics KIE book 1 page 131-136
Advancing in mathematics book 1 pages 121

3-4
VOLUME & CAPACITY
Volume of cuboids and cylinders

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the volume of cuboids and cylinders
Discussions
Demonstrations
Measuring lengths
Calculating volume
Cubes, cuboids and cylinders
Models of cubes cuboids and cylinders
Measuring instruments for volume
Discovering secondary mathematics Book 1 Pages 130-131
Secondary mathematics KLB book 1 pages 143-145
Secondary mathematics KIE book 1 page 136-138
Advancing in mathematics book 1 pages 121

5-6
VOLUME & CAPACITY
Capacity

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Show the relationship between volume and capacity and solve problems involving volume and capacity
Discussions
Demonstrations
Measuring capacity
Calculating capacity
Converting capacity to volume and vise versa
Measuring instruments for capacity
Discovering secondary mathematics Book 1 Pages 131-133



Secondary mathematics KLB book 1 pages 146
Secondary mathematics KIE book 1 page 136-138
Advancing in mathematics book 1 pages 123-124

6

1-2

MASS, WEIGHT AND DENSITY

Units of mass, weight and density

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define mass, weight and density
State their units and relate mass weight and density
Discussions
Measuring
Calculating
Converting
Masses
Measuring cylinders
Weights
Spring balance
Beam balance
lactometer
Discovering secondary mathematics Book 1 Pages 134-135
Secondary mathematics KLB book 1 pages 149-151
Secondary mathematics KIE book 1 page 140-141
Advancing in mathematics book 1 pages 126-127

3

TIME

Converting units of time

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Convert the units of time from one form to another
Discussions
Converting time, events
Calculating
Clock,
Watches
Conversion tables
Travel timetable for trains, busses, ships and aero planes
Discovering secondary mathematics Book 1 Pages 136-137
Secondary mathematics KLB book 1 pages 153-154
Secondary mathematics KIE book 1 page 142



Advancing in mathematics book 1 pages 129

4-5

TIME

The 12 hour and 24 hour clocks

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State time in the 12 hour and 24 hour clocks

Discussions

Timing, events

Reading time

Converting time

Clock,

Watches

Conversion tables

Travel timetable for trains, busses, ships and aero planes

Discovering secondary mathematics Book 1 Pages 138-141

Secondary mathematics KLB book 1 pages 154-155

Secondary mathematics KIE book 1 page 129-130

Advancing in mathematics book 1 pages 142

6

TIME

Travel timetables

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Read and interpret travel timetable

Solve problems involving travel timetable

Discussions

Solving problems

Reading the travel timetable

Travelling

Clock,

Watches

Conversion tables

Travel timetable for trains, busses, ships and aero planes

Discovering secondary mathematics Book 1 Pages 141-143

Secondary mathematics KLB book 1 pages 156-157

Advancing in mathematics book 1 pages 131-132

7

1-2



RATE, RATIO, PERCENTAGES AND PROPORTION
Rates and ratio

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Define rate and ratio
- Solve problems involving rates
- Use ratio to compare increase and decrease quantities
- Discussions
- Solving problems
- Sharing things equally
- Doing exercises
- Sharing quantities in given ratio
- Real life experience
- Currency
- Counters

- Discovering secondary mathematics Book 1 Pages 144-151
- Secondary mathematics KLB book 1 pages 96-98
- Secondary mathematics KIE book 1 page 162-164
- Advancing in mathematics book 1 pages 86-88

3-4

RATE, RATIO, PERCENTAGES AND PROPORTION
Proportion

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Change quantities in a given ratio and proportion
- Discussions
- Doing exercises
- Sharing out quantities in a given ratio
- Currency
- Counters
- Real life experience
- Discovering secondary mathematics Book 1 Pages 151-153
- Secondary mathematics KLB book 1 pages 97-102
- Secondary mathematics KIE book 1 page 165-166
- Advancing in mathematics book 1 pages 88-93

5-6

RATE, RATIO, PERCENTAGES AND PROPORTION
percentages



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Convert fractions and decimals to percentages
Calculate the percentage change in a quantity
Discussions
Doing exercises
100 square grid
100 items
Counters
Discovering secondary mathematics Book 1 Pages 153-157
Secondary mathematics KLB book 1 pages 105-106
Secondary mathematics KIE book 1 page 169-170
Advancing in mathematics book 1 pages 94-97

8

1-2

COMMERCIAL ARITHMETRIC

Currency conversion

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the currencies of different countries and convert currency from one form to another

Discussions
Solving problems involving currency exchange rates
Giving change and balance
Actual currency exchange rate table
Actual currency
Newspaper/magazine
Discovering secondary mathematics Book 1 Pages 158-162
Secondary mathematics KLB book 1 pages 171-173
Secondary mathematics KIE book 1 page 208-213
Advancing in mathematics book 1 pages 149-152

3-4

COMMERCIAL ARITHMETRIC

Profit and loss, Discount and commission

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Discussions
Doing exercises
Illustrations
Demonstrations
Resource person
Real life
Retail shops
Discovering secondary mathematics Book 1 Pages 162-165
Secondary mathematics KLB book 1 pages 175-178
Secondary mathematics KIE book 1 page 216-218
Advancing in mathematics book 1 pages 153-155

5-6

SCALE DRAWING AND ANGLES OF ELEVATION AND DEPRESSION
Indicating scale

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Read, interpret and indicate scale in linear statement ratio
Discussions
Solving problems
Drawing to scale
Writing scale
Interpreting scale
Determining scale

Ruler
Tape measure
Figure drawn to scale
photographs
Discovering secondary mathematics Book 1 Pages 116-168
Secondary mathematics KLB book 1 pages 248-250
Secondary mathematics KIE book 1 page 185
Advancing in mathematics book 1 pages 209-210

9

1-2

SCALE DRAWING AND ANGLES OF ELEVATION AND DEPRESSION
Angles of elevation & depression

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the angles of elevation and depression
Discussions



Drawing to scale
Doing exercises
Solving problems
Measuring angles/lengths
Ruler
Tape measure
Figure drawn to scale
photographs
Discovering secondary mathematics Book 1 Pages 168-172
Secondary mathematics KLB book 1 pages 256-260
Secondary mathematics KIE book 1 page 187-192
Advancing in mathematics book 1 pages 211

3-4

SCALE DRAWING AND ANGLES OF ELEVATION AND DEPRESSION
Bearing

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the bearing of a point from another point
Discussions
Drawing to scale
Measuring angles/lengths
Solving problems involving bearings

Ruler
Tape measure
Protractor
Set square
Plumb line
Discovering secondary mathematics Book 1 Pages 173-175
Secondary mathematics KLB book 1 pages 251-252
Secondary mathematics KIE book 1 page 193-195
Advancing in mathematics book 1 pages 211-213

5-6

BEARING AND SURVEYING
Methods of surveying

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply scale drawing in methods of surveying
Discussions
Drawing to scale



Measuring angles/lengths
Estimating area
Geometrical sets
Clinometers
Surveying equipment
Protractor
Ruler
Playfield
School compound
Discovering secondary mathematics Book 1 Pages 176-178
Secondary mathematics KLB book 1 pages 262-265
Secondary mathematics KIE book 1 page 199-202
Advancing in mathematics book 1 pages 213-216

10

1-2

BEARING AND SURVEYING

Area of irregular shapes

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the area of irregular shapes using surveying techniques

Discussions
Measuring lengths/objects
Drawing scale
Estimating area
Geometrical sets
Clinometers
Surveying equipment
Protractor
Ruler
Playfield
School compound
Discovering secondary mathematics Book 1 Pages 178-181
Secondary mathematics KLB book 1 pages 267-268
Secondary mathematics KIE book 1 page 205-206
Advancing in mathematics book 1 pages 216-220

3-4

COMMON SOLIDS

Regular solids

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Identify and sketch common solids
Discussions
Counting sides faces and vertices
Sketching solids
Models of common solids
Actual solids
Discovering secondary mathematics Book 1 Pages 182-185
Secondary mathematics KLB book 1 pages 271-276
Secondary mathematics KIE book 1 page 248-254
Advancing in mathematics book 1 pages 222-227

5-6
COMMON SOLIDS
Nets of solids

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Sketch, draw nets of solids and make models of solids from the nets
Discussions
Sketching
Drawing to scale
Drawing accurately
Making models

Models of common solids
Examples of the common solids
Discovering secondary mathematics Book 1 Pages 186-188
Secondary mathematics KLB book 1 pages 277-283
Secondary mathematics KIE book 1 page 255-260
Advancing in mathematics book 1 pages 228-229

11
1-2
COMMON SOLIDS
Surface area of solids

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the surface area of solid from nets
Discussions
Sketching
Making models
Drawing to scale
Models of common solids
Actual solids



Discovering secondary mathematics Book 1 Pages 188-189
Secondary mathematics KLB book 1 pages 284-285
Secondary mathematics KIE book 1 page 264-265
Advancing in mathematics book 1 pages 230-231

3-4

COMMON SOLIDS

Distance between two points on the surface area of solid

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the distance between two points on the surface of a solid

Discussions

Sketching

Making models

Drawing to scale

Measuring lengths/angles

Sketches of cubes and cuboids

charts

Discovering secondary mathematics Book 1 Pages 189-191

Secondary mathematics KLB book 1 pages 286-288

Advancing in mathematics book 1 pages 231-233

5-6

REVISION

12

1-6

END YEAR EXAMINATIONS

13

1-6

CLOSING OF SCHOOL

MATHEMATICS FORM 2 SCHEMES OF WORK TERM 1

WEEK

LESSON

TOPIC

SUB - TOPIC

OBJECTIVES

LEARNING/TEACHING ACTIVITIES

LEARNING/TEACHING RESOURCES

REFERENCES

REMARKS

1

1-2

CUBES AND CUBE ROOTS



Cubes of numbers by multiplication and from tables

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the cubes of numbers by multiplication
Find the cube roots of numbers from tables

Multiplying numbers
Reading mathematical tables
Discussions
Demonstrations
Exercises
Exercises in given class
Mathematical tables
Real life situation

Discovering secondary mathematics Book 2 Pages 1-3
Secondary mathematics KLB book 2 pages 1 and 2
KLB teachers? guide book 2 page 1
Golden tips mathematics pages 6 and 63

3

CUBES AND CUBE ROOTS

Cube roots of numbers by factor method

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the cube roots of numbers by factor method

Multiplying numbers
Reading mathematical tables
Discussions
Demonstrations
Exercises
Exercises in given class
Mathematical tables
Real life situation

Discovering secondary mathematics Book 2 Pages 5-6
Secondary mathematics KLB book 2 page 3
KLB teachers? guide book 2 page 1-2
Golden tips mathematics pages 62



4

CUBES AND CUBE ROOTS

Evaluation of cube and cube roots expressions and application of cubes and cube roots in real life situation

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Evaluate expressions involving cubes and cube roots
Apply the knowledge of cubes and cube roots in real life situations

Multiplying numbers
Reading mathematical tables
Discussions
Demonstrations
Exercises
Exercises in given class
Mathematical tables
Real life situation

Discovering secondary mathematics Book 2 Pages 5-6
Secondary mathematics KLB book 2 page 3 and 4
KLB teachers? guide book 2 page 2
Golden tips mathematics pages 63 and 64

5-6

RECIPROCAL

Reciprocals of numbers by division and from tables

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find reciprocals of numbers by division
Find reciprocals of numbers from tables
Multiplying numbers
Dividing numbers
Reading mathematical tables
Discussions
Demonstrations
Exercises
Exercises in given class
Mathematical tables

Discovering secondary mathematics Book 2 Pages 12-13
Secondary mathematics KLB book 2 page 5
KLB teachers? guide book 2 page 5
Golden tips mathematics pages 64



2

1-2

RECIPROCAL

Computation using reciprocals

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Use reciprocals of numbers in computation

Multiplying numbers

Dividing numbers

Reading mathematical tables

Discussions

Demonstrations

Exercises

Exercises in given class

Mathematical tables

Discovering secondary mathematics Book 2 Pages 12-13

Secondary mathematics KLB book 2 page 6

KLB teachers' guide book 2 page 5-6

Golden tips mathematics pages 64

3

INDICES AND LOGARITHMS

Indices (powers) and base

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define indices

Express numbers in index form

Express indices in number form

Multiplying numbers

Dividing numbers

Factorizing numbers

Reading mathematical tables

Discussions

Exercises in given class

Logarithm tables

Charts illustrations laws of indices

Discovering secondary mathematics Book 2 Page 7

Secondary mathematics KLB book 2 page 7

KLB teachers' guide book 2 page 7-8

Golden tips mathematics pages 44-46



4

INDICES AND LOGARITHMS
Laws of Indices

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State laws of indices regarding multiplication of indices
State laws of indices regarding zero index
State laws of indices regarding division of indices
Multiplying numbers
Dividing numbers
Factorizing numbers
Reading mathematical tables
Discussions
Exercises in given class
Logarithm tables
Charts illustrations laws of indices

Discovering secondary mathematics Book 2 Page 7-11
Secondary mathematics KLB book 2 page 7-8
KLB teachers' guide book 2 page 7-8
Golden tips mathematics pages 44-46

5-6

INDICES AND LOGARITHMS
Laws of Indices

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State laws of indices regarding negative indices
State laws of indices fractional indices
Apply the laws of indices in calculation
Multiplying numbers
Dividing numbers
Factorizing numbers
Reading mathematical tables
Discussions
Exercises in given class
Logarithm tables
Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 7-11
Secondary mathematics KLB book 2 page 8-13



KLB teachers? guide book 2 page 7-8
Golden tips mathematics pages 44-46

3

1-2

INDICES AND LOGARITHMS

Powers of 10 and common logarithms

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Relate the powers of 10 to common logarithms
Identify the parts of the logarithms i.e characteristic mantissa
Multiplying numbers
Dividing numbers
Factorizing numbers
Discussions
Exercises in given class
Mathematical tables
Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 15
Secondary mathematics KLB book 2 page 16-17
KLB teachers? guide book 2 page 7-8
Golden tips mathematics pages 52

3-4

INDICES AND LOGARITHMS

Logarithms of positive numbers less than one

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the logarithm of a number less than 1 from mathematical tables
Apply the logarithms of numbers less than one in computation
Multiplying numbers
Dividing numbers
Factorizing numbers
Discussions
Exercises in given class
Mathematical tables
Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 15
Secondary mathematics KLB book 2 page 18
KLB teachers? guide book 2 page 7-8



Golden tips mathematics pages 52

5

INDICES AND LOGARITHMS

Logarithms of numbers less than ten ($X < 10$)

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find the logarithm numbers less than 10 but greater than 1
- Apply the logarithms of numbers less than 10 but greater than 1 in computation
- Multiplying numbers
- Dividing numbers
- Factorizing numbers
- Discussions
- Exercises in given class
- Mathematical tables
- Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 16

Secondary mathematics KLB book 2 page 18

KLB teachers' guide book 2 page 7-8

Golden tips mathematics pages 54

6

INDICES AND LOGARITHMS

Logarithms of numbers greater than ten

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find the logarithm numbers greater than 10
- Apply the logarithms of numbers greater than 10 in computation
- Multiplying numbers
- Dividing numbers
- Factorizing numbers
- Discussions
- Exercises in given class
- Mathematical tables
- Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 16

Secondary mathematics KLB book 2 page 18



KLB teachers? guide book 2 page 7-8
Golden tips mathematics pages 54

4
1

INDICES AND LOGARITHMS

Antilogarithms

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find antilogarithms of numbers
- Apply the antilogarithms of numbers in numerical
- Multiplying numbers
- Dividing numbers
- Factorizing numbers
- Discussions
- Exercises in given class
- Mathematical tables
- Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 17
Secondary mathematics KLB book 2 page 19
KLB teachers? guide book 2 page 7-8
Golden tips mathematics pages 54

2

INDICES AND LOGARITHMS

Multiplication of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Use logarithms to work out the multiplication of numbers
- Multiplying numbers
- Dividing numbers
- Factorizing numbers
- Discussions
- Exercises in given class
- Mathematical tables
- Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 18
Secondary mathematics KLB book 2 page 20
KLB teachers? guide book 2 page 7-8
Golden tips mathematics pages 55



3

INDICES AND LOGARITHMS

division of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Use logarithms to work out the division of numbers

Multiplying numbers

Dividing numbers

Factorizing numbers

Discussions

Exercises in given class

Mathematical tables

Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 19

Secondary mathematics KLB book 2 page 20

KLB teachers? guide book 2 page 7-8

Golden tips mathematics pages 56

4

INDICES AND LOGARITHMS

Combines multiplication and division of numbers

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Combine multiplication and division of numbers to work out logarithm problems

Multiplying numbers

Dividing numbers

Factorizing numbers

Discussions

Exercises in given class

Mathematical tables

Charts illustrating laws of indices

Discovering secondary mathematics Book 2 Page 19

Secondary mathematics KLB book 2 page 20

KLB teachers? guide book 2 page 7-8

Golden tips mathematics pages 56

5

INDICES AND LOGARITHMS



Negative characteristics

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Use negative logarithms
- Multiplying numbers
- Dividing numbers
- Factorizing numbers
- Discussions
- Exercises in given class
- Mathematical tables
- Charts illustrating laws of indices

- Discovering secondary mathematics Book 2 Page 20
- Secondary mathematics KLB book 2 page 18
- KLB teachers? guide book 2 page 7-8
- Golden tips mathematics pages 55

6
INDICES AND LOGARITHMS
Application of logarithms

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Apply the knowledge of logarithms and indices in daily computation
- Find roots and squares of numbers using logarithms
- Multiplying numbers
- Dividing numbers
- Factorizing numbers
- Discussions
- Exercises in given class
- Mathematical tables
- Charts illustrating laws of indices

- Discovering secondary mathematics Book 2 Page 21
- Secondary mathematics KLB book 2 page 20
- KLB teachers? guide book 2 page 7-8
- Golden tips mathematics pages 53

5
1
GRADIENTS AND EQUATIONS OF STRAIGHT LINES
Gradient of a straight line



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define gradient of a straight line
Determine the gradient of a straight line through known points
Drawing linear graphs
Plotting co-ordinates on the Cartesian plane
Reading co-ordinates of points on the Cartesian plane

Square boards
Graph books
Straight edged ruler
Real life situation

Discovering secondary mathematics Book 2 Page 25-23
Secondary mathematics KLB book 2 page 27-34
KLB teachers? guide book 2 page 14-15
Golden tips mathematics pages 174

2

GRADIENTS AND EQUATIONS OF STRAIGHT LINES
equation of a straight line

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the equation of a straight line using gradient and a known point
Determine the equation of a straight line given two points
Drawing linear graphs
Plotting co-ordinates on the Cartesian plane
Reading co-ordinates of points on the Cartesian plane

Square boards
Graph books
Straight edge/ruler
Real life situation

Discovering secondary mathematics Book 2 Page 25-26
Secondary mathematics KLB book 2 page 34-35
KLB teachers? guide book 2 page 14-15
Golden tips mathematics pages 171

3-4

GRADIENTS AND EQUATIONS OF STRAIGHT LINES
General equation of a straight line



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Express the equation of a straight line in the form of $y=mx$
Interpret the equation $y=mx$

Drawing linear graphs
Plotting co-ordinates on the Cartesian plane
Reading co-ordinates of points on the Cartesian plane

Square boards
Graph books
Straight edge/rulers
Real life situation

Discovering secondary mathematics Book 2 Page 27
Secondary mathematics KLB book 2 page 34
KLB teachers? guide book 2 page 14-15
Golden tips mathematics pages 171

5-6

GRADIENTS AND EQUATIONS OF STRAIGHT LINES

The intercept of a straight line

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the x and the y intercept of a straight line
Express a double intercept equation of a straight line

Drawing linear graphs
Plotting co-ordinates on the Cartesian plane
Reading co-ordinates of points on the Cartesian plane

Square boards
Graph books
Straight edge/rulers
Real life situation

Discovering secondary mathematics Book 2 Page 28
Secondary mathematics KLB book 2 page 36
KLB teachers? guide book 2 page 14-15
Golden tips mathematics pages 171

6

1-2

GRADIENTS AND EQUATIONS OF STRAIGHT LINES

The gradient of parallel lines



OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find the gradient of parallel lines
- Relate parallel lines in terms of their gradients

- Drawing linear graphs
- Plotting co-ordinates on the Cartesian plane
- Reading co-ordinates of points on the Cartesian plane

- Square boards
- Graph books
- Straight edge/ rulers
- Real life situation

- Discovering secondary mathematics Book 2 Page 29
- Secondary mathematics KLB book 2 page 43-44
- KLB teachers' guide book 2 page 14-15
- Golden tips mathematics pages 175

3-4

GRADIENTS AND EQUATIONS OF STRAIGHT LINES

The gradient of perpendicular lines

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find the gradient of perpendicular lines
- Relate perpendicular lines in terms of their gradients

- Drawing linear graphs
- Plotting co-ordinates on the Cartesian plane
- Reading co-ordinates of points on the Cartesian plane

- Square boards
- Graph books
- Straight edge/ rulers
- Real life situation

- Discovering secondary mathematics Book 2 Page 30
- Secondary mathematics KLB book 2 page 41-43
- KLB teachers' guide book 2 page 14-15
- Golden tips mathematics pages 172

5-6



REFLECTION AND CONGRUENCE

Geometric transformation (reflection)

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- State the properties of reflection
- Construct and identify the images and the objects in a reflection using the properties
- Make geometrical deductions using reflection

- Observing objects in plane mirrors
- Identifying the objects and their images in a plan mirror
- Drawing
- Identifying lines of symmetry
- Identifying the mirror line in a plane mirror

Mirrors

- Cartesian plane
- Various symmetrical objects
- Tracing and graph papers
- Real life experiences

- Discovering secondary mathematics Book 2 Page 32
- Secondary mathematics KLB book 2 page
- KLB teachers' guide book 2 page 14-20
- Golden tips mathematics pages 230

7

1

REFLECTION AND CONGRUENCE

Lines and planes of symmetry

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Identify the line of symmetry in a reflection given the image and the object

- Observing objects in plane mirrors
- Identifying the objects and their images in a plan mirror
- Drawing
- Identifying lines of symmetry
- Identifying the mirror line in a plane mirror

Mirrors

- Cartesian plane
- Various symmetrical objects



Tracing and graph papers
Real life experiences

Discovering secondary mathematics Book 2 Page 32
Secondary mathematics KLB book 2 page 46-48
KLB teachers? guide book 2 page 19-20
Golden tips mathematics pages 230

2

REFLECTION AND CONGRUENCE

Lines and planes of symmetry

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify the line of symmetry in a reflection
Relate lines and planes of symmetry
Observing objects in plane mirrors
Identifying the objects and their images in a plan mirror
Drawing
Identifying lines of symmetry
Identifying the mirror line in a plane mirror

Mirrors

Cartesian plane
Various symmetrical objects
Tracing and graph papers
Real life experiences

Discovering secondary mathematics Book 2 Page 32
Secondary mathematics KLB book 2 page 46-48
KLB teachers? guide book 2 page 19-20
Golden tips mathematics pages 230

3-4

REFLECTION AND CONGRUENCE

Reflection in the Cartesian plane

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply the properties of a rotation in the Cartesian plane
Observing objects in plane mirrors
Identifying the objects and their images in a plan mirror
Drawing
Identifying lines of symmetry



Identifying the mirror line in a plane mirror

Mirrors

Cartesian plane

Various symmetrical objects

Tracing and graph papers

Real life experiences

Discovering secondary mathematics Book 2 Page 37

Secondary mathematics KLB book 2 page 48

KLB teachers' guide book 2 page 19-20

Golden tips mathematics pages 230

5-6

REFLECTION AND CONGRUENCE

Congruent triangles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify congruency

Solve problems involving congruency

Observing objects in plane mirrors

Identifying the objects and their images in a plane mirror

Drawing

Identifying lines of symmetry

Identifying the mirror line in a plane mirror

Mirrors

Cartesian plane

Various symmetrical objects

Tracing and graph papers

Real life experiences

Discovering secondary mathematics Book 2 Page 39

Secondary mathematics KLB book 2 page 64-65

KLB teachers' guide book 2 page 19-20

Golden tips mathematics pages 230

8

1-2

REFLECTION AND CONGRUENCE

Congruent triangles

OBJECTIVES:

By the end of the lesson, the learner should be able to:



Identify congruency
Solve problems involving congruency
Observing objects in plane mirrors
Identifying the objects and their images in a plan mirror
Drawing
Identifying lines of symmetry
Identifying the mirror line in a plane mirror

Mirrors
Cartesian plane
Various symmetrical objects
Tracing and graph papers
Real life experiences

Discovering secondary mathematics Book 2 Page 39
Secondary mathematics KLB book 2 page 64-65
KLB teachers? guide book 2 page 19-20
Golden tips mathematics pages 230

3
REFLECTION AND CONGRUENCE
Congruent figures

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify figures which are congruent through reflection

Observing objects in plane mirrors
Identifying the objects and their images in a plan mirror
Drawing
Identifying lines of symmetry
Identifying the mirror line in a plane mirror

Mirrors
Cartesian plane
Various symmetrical objects
Tracing and graph papers
Real life experiences

Discovering secondary mathematics Book 2 Page 40-41
Secondary mathematics KLB book 2 page 66
KLB teachers? guide book 2 page 19-20
Golden tips mathematics pages 230

4-5
ROTATION
The properties s of rotation



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define rotation as a transformation
State the properties of a rotation as a transformation

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry

Square boards
Graph papers
Geometrical instruments
Tracing paper and real life situations

Discovering secondary mathematics Book 2 Page 44-45
Secondary mathematics KLB book 2 page 73
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228

6

ROTATION

Center of angle of rotation

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the center of rotation
Determine the angle of rotation

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry

Square boards
Graph papers
Geometrical instruments
Tracing paper
real life situations

Discovering secondary mathematics Book 2 Page 46
Secondary mathematics KLB book 2 page 73
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228



9

1-2

ROTATION

Center of angle of rotation

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Rotate objects through a given angle of rotation and center of rotation
Establish the angle of rotation given an object and its image

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry

Square boards
Graph papers
Geometrical instruments
Tracing paper
real life situations

Discovering secondary mathematics Book 2 Page 46
Secondary mathematics KLB book 2 page 74
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228

3-4

ROTATION

Rotation in a Cartesian plane

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply the properties of rotation in the Cartesian plane

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry

Square boards
Graph papers
Geometrical instruments
Tracing paper
real life situations



Discovering secondary mathematics Book 2 Page 47
Secondary mathematics KLB book 2 page 75
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228

5-6

ROTATION

Rotational symmetry

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify point of rotational symmetry
State the order of rotational symmetry of plane figures
Identify the axis of rotational symmetry

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry

Square boards
Graph papers
Geometrical instruments
Tracing paper
real life situations

Discovering secondary mathematics Book 2 Page 49
Secondary mathematics KLB book 2 page 78
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228

10

1-2

ROTATION

Congruence and Rotation

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Deduce congruence from rotation

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry



Square boards
Graph papers
Geometrical instruments
Tracing paper
real life situations

Discovering secondary mathematics Book 2 Page 48
Secondary mathematics KLB book 2 page 84
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228

3-4
ROTATION
REVISION

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Answer all questions involving rotations
Apply rotation in real life situations

Rotating objects
Measuring angles/lengths
Drawing objects
Identifying the lines of symmetry

Square boards
Graph papers
Geometrical instruments
Tracing paper
real life situations

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Secondary mathematics KLB book 2 page 84-86
KLB teachers? guide book 2 page 24-25
Golden tips mathematics pages 228

5-6
SIMILARITY AND ENLARGEMENT
Similar figures

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify similar figures
Construct similar figures



Identifying similar figures
Tracing figures
Constructing similar figures
enlarging figures
Drawing figures on the Cartesian plane
measuring lengths/ angles

Geometrical instruments
Model maps
Photographs
Charts illustrating similarity and enlargement

Discovering secondary mathematics Book 2 Page 52
Secondary mathematics KLB book 2 page 87
KLB teachers? guide book 2 page 27-28
Golden tips mathematics pages 125

11

1-2

SIMILARITY AND ENLARGEMENT

Properties of enlargement

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the properties of enlargement as a transformation
Apply the properties of enlargement to construct objects and images

Identifying similar figures
Tracing figures
Constructing similar figures
enlarging figures
Drawing figures on the Cartesian plane
measuring lengths/ angles

Geometrical instruments
Model maps
Photographs
Charts illustrating similarity and enlargement

Discovering secondary mathematics Book 2 Page 52
Secondary mathematics KLB book 2 page 97
KLB teachers? guide book 2 page 27-28
Golden tips mathematics pages 125

3-4

SIMILARITY AND ENLARGEMENT

Enlargement



OBJECTIVES:

By the end of the lesson, the learner should be able to:

- State the scale factor
- State the center of enlargement

- Identifying similar figures
- Tracing figures
- Constructing similar figures
- enlarging figures
- Drawing figures on the Cartesian plane
- measuring lengths/ angles

- Geometrical instruments
- Model maps
- Photographs
- Charts illustrating similarity and enlargement

- Discovering secondary mathematics Book 2 Page 57-58
- Secondary mathematics KLB book 2 page 97
- KLB teachers? guide book 2 page 27-28
- Golden tips mathematics pages 125

5-6

SIMILARITY AND ENLARGEMENT

Enlargement on the Cartesian plane

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Apply enlargement on Cartesian planes

- Identifying similar figures
- Tracing figures
- Constructing similar figures
- enlarging figures
- Drawing figures on the Cartesian plane
- measuring lengths/ angles

- Geometrical instruments
- Model maps
- Photographs
- Charts illustrating similarity and enlargement

- Discovering secondary mathematics Book 2 Page 61-62



Secondary mathematics KLB book 2 page 97
KLB teachers? guide book 2 page 27-28
Golden tips mathematics pages 125

12

1-2

SIMILARITY AND ENLARGEMENT

Linear, area and volume scale factors

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine linear scale factor
Determine area scale factors
Determine volume scale factors
Relate area scale factor, volume scale factor, and linear scale factor

Identifying similar figures
Tracing figures
Constructing similar figures
enlarging figures
Drawing figures on the Cartesian plane
measuring lengths/ angles

Geometrical instruments
Model maps
Photographs
Charts illustrating similarity and enlargement

Discovering secondary mathematics Book 2 Page 62-65
Secondary mathematics KLB book 2 page 97-110
KLB teachers? guide book 2 page 27-28
Golden tips mathematics pages 125

3-4

SIMILARITY AND ENLARGEMENT

Areas of similar figures

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply volume area and linear scale factors in establishing areas of
similar figures

Identifying similar figures



Tracing figures
Constructing similar figures
enlarging figures
Drawing figures on the Cartesian plane
measuring lengths/ angles

Geometrical instruments
Model maps
Photographs
Charts illustrating similarity and enlargement

Discovering secondary mathematics Book 2 Page 62-64
Secondary mathematics KLB book 2 page 106-108
KLB teachers? guide book 2 page 27-28
Golden tips mathematics pages 125

5-6

SIMILARITY AND ENLARGEMENT

Volume of similar figures

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply knowledge of linear scale factor and volume scale factor to
determine values of similar figures

Identifying similar figures
Tracing figures
Constructing similar figures
enlarging figures
Drawing figures on the Cartesian plane
measuring lengths/ angles

Geometrical instruments
Model maps
Photographs
Charts illustrating similarity and enlargement

Discovering secondary mathematics Book 2 Page 64-65
Secondary mathematics KLB book 2 page 109-111
KLB teachers? guide book 2 page 27-28
Golden tips mathematics pages 125

13

1-2

SIMILARITY AND ENLARGEMENT

Application of scale factors in real life situations



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply knowledge of linear scale factor and volume scale factor to determine values of similar figures

Identifying similar figures
Tracing figures
Constructing similar figures
enlarging figures
Drawing figures on the Cartesian plane
measuring lengths/ angles

Geometrical instruments
Model maps
Photographs
Charts illustrating similarity and enlargement

Discovering secondary mathematics Book 2 Page 66
Secondary mathematics KLB book 2 page 109-111-112
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Golden tips mathematics pages 128

MATHEMATICS FORM 2 SCHEMES OF WORK ? TERM 2

WEEK

LESSON

TOPIC

SUB - TOPIC

OBJECTIVES

LEARNING/TEACHING ACTIVITIES

LEARNING/TEACHING RESOURCES

REFERENCES

REMARKS

1

1-2

THE PYTHOGOROUS THEOREM

Deriving the Pythagoras theorem

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Derive the Pythagoras theorem

Measuring lengths
Squaring numbers
Getting square roots of numbers



Drawing right angled triangles
Drawing squares
Working out the area of a square
Right angled triangles
Square paper
Ruler
protractor

Discovering secondary mathematics Book 2 Pages 67
Secondary mathematics KLB book 2 pages 119-120
KLB teachers? guide book 2 page 16-17
Golden tips mathematics pages

3-4
THE PYTHOGOROUS THEOREM
Applying the Pythagoras theorem

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve problems using the Pythagoras theorem

Measuring lengths
Squaring numbers
Getting square roots of numbers
Drawing right angled triangles
Drawing squares
Working out the area of a square
Right angled triangles
Square paper
Ruler
protractor

Discovering secondary mathematics Book 2 Pages 68-69
Secondary mathematics KLB book 2 pages 121
KLB teachers? guide book 2 page 16-17
Golden tips mathematics pages

5-6
THE PYTHOGOROUS THEOREM
Applying the Pythagoras theorem

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve problems using the Pythagoras theorem



Measuring lengths
Squaring numbers
Getting square roots of numbers
Drawing right angled triangles
Drawing squares
Working out the area of a square
Right angled triangles
Square paper
Ruler
protractor

Discovering secondary mathematics Book 2 Pages 68-69
Secondary mathematics KLB book 2 pages 121
KLB teachers? guide book 2 page 16-17
Golden tips mathematics pages

2

1

THE TRIGONOMETRIC RATIOS
The tangent of an angle

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the tangent of an angle

Measuring lengths/angles
Dividing numbers
Drawing right angles
Right corners
Ruler
protractor

Discovering secondary mathematics Book 2 Pages 70-71
Secondary mathematics KLB book 2 pages 123
KLB teachers? guide book 2 page 36
Golden tips mathematics pages 132

2

THE TRIGONOMETRIC RATIOS
The table of tangents

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Read the tangent of an angle from the tangent tables



Measuring lengths/angles
Dividing numbers
Drawing right angles
Reading mathematical tables
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Book 2 Pages 71-72
Secondary mathematics KLB book 2 pages 126
KLB teachers? guide book 2 page 36-37
Golden tips mathematics pages 132

3

THE TRIGONOMETRIC RATIOS
Using tangents in calculations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Use tangents of angles in calculation

Measuring lengths/angles
Dividing numbers
Drawing right angles
Reading mathematical tables
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Book 2 Pages 74-75
Secondary mathematics KLB book 2 pages 127
KLB teachers? guide book 2 page 38
Golden tips mathematics pages 133

4

THE TRIGONOMETRIC RATIOS
Application of tangents

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Apply tangents in real life situations



Measuring lengths/angles
Dividing numbers
Drawing right angles
Reading mathematical tables
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Book 2 Pages 74-75
Secondary mathematics KLB book 2 pages 127
KLB teachers? guide book 2 page 36-39
Golden tips mathematics pages 136

5-6
THE TRIGONOMETRIC RATIOS
Sines

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the sine of an angle
Read the sine of an angle from mathematical tables and apply sines of an angle in real life situations

Measuring lengths/angles
Dividing numbers
Drawing right angles
Reading mathematical tables
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Book 2 Pages 76-77
Secondary mathematics KLB book 2 pages 132
KLB teachers? guide book 2 page 17-19
Golden tips mathematics pages 132

3
1-2
THE TRIGONOMETRIC RATIOS
Cosines

OBJECTIVES:



By the end of the lesson, the learner should be able to:

- Determine the cosine of an angle
- Read the cosine of an angle from mathematical tables
- Apply cosines of angles in real life situations
- Measuring lengths/angles
- Dividing numbers
- Drawing right angles
- Reading mathematical tables
- Right corners
- Ruler
- Protractor
- Mathematical tables

- Discovering secondary mathematics Book 2 Pages 78-79
- Secondary mathematics KLB book 2 pages 132
- KLB teachers? guide book 2 page 17-19
- Golden tips mathematics pages 133

3-4

THE TRIGONOMETRIC RATIOS
Complimentary angles

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Establish the relationship of cosines and sines of complimentary angles
- Use the relationship of sines and cosines of complimentary angles
- Measuring lengths/angles
- Dividing numbers
- Drawing right angles
- Reading mathematical tables
- Right corners
- Ruler
- Protractor
- Mathematical tables

- Discovering secondary mathematics Book 2 Pages 81-83
- Secondary mathematics KLB book 2 pages 145
- KLB teachers? guide book 2 page 33-36
- Golden tips mathematics pages 133

5-6

THE TRIGONOMETRIC RATIOS
Trigonometric ratios of some angles



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the trigonometric ratios of some special angles
Measuring lengths/angles
Dividing numbers
Drawing right angles
Reading mathematical tables
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Book 2 Pages 83-84
Secondary mathematics KLB book 2 pages 146
KLB teachers? guide book 2 page 17-19
Golden tips mathematics pages 134

4

1-2

THE TRIGONOMETRIC RATIOS

The logarithms of sines, cosines and tangents

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Read tables of logarithms of sines cosines and tangents
Use the tables of logarithms of sines cosines and tangents to work out numerals.
Measuring lengths/angles
Dividing numbers
Drawing right angles
Reading mathematical tables
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Book 2 Pages 83-85
Secondary mathematics KLB book 2 pages 149
KLB teachers? guide book 2 page 17-19
Golden tips mathematics pages 136

3-4

AREA OF TRIANGLE



The formula
 $A = \frac{1}{2} ab \sin C$

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the formula
 $A = \frac{1}{2} ab \sin C$

Discussions
Drawing triangles
Measuring lengths/angles
Calculating areas
Right corners
Ruler
Protractor
Mathematical tables

Discovering secondary mathematics Students? Book 2 Pages 85-86
Teachers? Book 2 Pages 19-20
Secondary mathematics KLB book 2 pages 155-156
KLB teachers? guide book 2 page 43
Golden tips mathematics pages 138

5-6
AREA OF TRIANGLE

The formula
 $A = s(s-a)(s-b)(s-c)$

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Use the formula
 $A = s(s-a)(s-b)(s-c)$
To get the area of a triangle

Discussions
Drawing triangles
Measuring lengths/angles
Calculating areas
Right corners
Ruler
Protractor
Mathematical tables



Discovering secondary mathematics Students? Book 2 Pages 86-87
Teachers? Book 2 Pages 19-20
Secondary mathematics KLB book 2 pages 157
KLB teachers? guide book 2 page 43
Golden tips mathematics pages 70

5

1-2

AREA OF POLYGONS

Area of a parallelogram

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the area of a parallelogram using the fomular $A = bh$ and trigonometric ratios

Drawing parallelograms
Reading mathematical tables
Measuring lengths/angles
Discussions
Parallelograms
Squares/rectangles
Mathematical tables
Discovering secondary mathematics Students? Book 2 Pages 88-90
Teachers? Book 2 Pages 20-21
Secondary mathematics KLB book 2 pages 160
KLB teachers? guide book 2 page 45
Golden tips mathematics pages 69

3-4

AREA OF POLYGONS

Area of a trapezium and other polygons

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the area of a trapezium and other polygons

Drawing trapezium/polygons
Reading mathematical tables
Measuring lengths/angles
Discussions
Trapezium
polygons



Squares/rectangles
Mathematical tables
Discovering secondary mathematics Students? Book 2 Pages 90-92
Teachers? Book 2 Pages 20-21
Secondary mathematics KLB book 2 pages 162
KLB teachers? guide book 2 page 45
Golden tips mathematics pages 69

5-6
AREA OF A CIRCLE

Area of a sector

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the area of a sector of a circle

Drawing circles
Measuring radii/diameters
Calculating the area of a circle
Measuring angles
Discussions
circles
Discovering secondary mathematics Students? Book 2 Pages 93-94
Teachers? Book 2 Pages 21-22
Secondary mathematics KLB book 2 pages 167
KLB teachers? guide book 2 page 45
Golden tips mathematics pages 70

6
1-2
AREA OF PART OF A CIRCLE

Area of a segment

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the area of a segment of a circle

Drawing circles
Measuring radii/diameters
Calculating the area of a circle
Measuring angles
Discussions
circles



Discovering secondary mathematics Students? Book 2 Pages 95-96
Teachers? Book 2 Pages 21-22
Secondary mathematics KLB book 2 pages 169
KLB teachers? guide book 2 page 46
Golden tips mathematics pages 68

3-4

AREA OF PART OF A CIRCLE

Area of intersecting segments

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the area of intersecting segments of a circle

Drawing circles

Measuring radii/diameters

Calculating the area of a circle

Measuring angles

Discussions

circles

Discovering secondary mathematics Students? Book 2 Pages 97-98

Teachers? Book 2 Pages 21-22

Secondary mathematics KLB book 2 pages 173

KLB teachers? guide book 2 page 46

Golden tips mathematics pages 68

5-6

SURFACE AREA OF SOLIDS

Surface area of prisms

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the surface area of a prism

Drawing prisms

Measuring lengths

Opening prisms to form nets

Discussions

Calculating area

prisms

Discovering secondary mathematics Students? Book 2 Pages 99-100

Teachers? Book 2 Pages 23-34



Secondary mathematics KLB book 2 pages 177
KLB teachers? guide book 2 page 46
Golden tips mathematics pages 71

7

1-2

SURFACE AREA OF SOLIDS

Surface area of prisms

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the surface area of a prism

Drawing prisms

Measuring lengths

Opening prisms to form nets

Discussions

Calculating area

prisms

Discovering secondary mathematics Students? Book 2 Pages 99-100

Teachers? Book 2 Pages 23-34

Secondary mathematics KLB book 2 pages 177

KLB teachers? guide book 2 page 46

Golden tips mathematics pages 71

3-4

SURFACE AREA OF SOLIDS

Surface area of cones and frustum

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the surface area of a cone and a frustum

Drawing cones/frustums

Making spheres

Measuring diameters/radii of spheres

Discussions

spheres

Discovering secondary mathematics Students? Book 2 Pages 102-104

Teachers? Book 2 Pages 23-34

Secondary mathematics KLB book 2 pages 180

KLB teachers? guide book 2 page 51



Golden tips mathematics pages 71

5-6

SURFACE AREA OF SOLIDS

Surface area of spheres

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the surface area of a sphere

Sketching spheres

Making spheres

Measuring diameters/radii of spheres

Discussions

spheres

Discovering secondary mathematics Students? Book 2 Pages 104-106

Teachers? Book 2 Pages 23-24

Secondary mathematics KLB book 2 pages 183

KLB teachers? guide book 2 page 51

Golden tips mathematics pages 71

8

1-2

VOLUME OF SOLIDS

Volume of a prism

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the volume of a solid

Identifying prisms

Identifying cross sectional area

Drawing/sketching prisms

prisms

Discovering secondary mathematics Students? Book 2 Pages 107-110

Teachers? Book 2 Pages 24-26

Secondary mathematics KLB book 2 pages 186

KLB teachers? guide book 2 page 56

Golden tips mathematics pages 75



3-4

VOLUME OF SOLIDS

Volume of cylinders

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the volume of a cylinder

Drawing cylinders

Opening cylinders to form nets

discussions

cylinders

Discovering secondary mathematics Students? Book 2 Pages 110-111

Teachers? Book 2 Pages 24-26

Secondary mathematics KLB book 2 pages 191

KLB teachers? guide book 2 page 56

Golden tips mathematics pages 73

5-6

VOLUME OF SOLIDS

Volume of pyramids

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Find the volume of a pyramid

Drawing pyramids

Making pyramids

Opening pyramids to form nets

discussions

pyramids

Discovering secondary mathematics Students? Book 2 Pages 111-112

Teachers? Book 2 Pages 24-26

Secondary mathematics KLB book 2 pages 189

KLB teachers? guide book 2 page 56

Golden tips mathematics pages 75

9

1-2

VOLUME OF SOLIDS



Volume of cones and frustums

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find the volume of a cone
- Find the volume of a frustum

- Making cones/frustums
- Opening cones/frustums to form nets

Cones

frustums

Discovering secondary mathematics Students? Book 2 Pages 112-114

Teachers? Book 2 Pages 24-26

Secondary mathematics KLB book 2 pages 192

KLB teachers? guide book 2 page 56

Golden tips mathematics pages 75

3-4

VOLUME OF SOLIDS

Volume of spheres

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Find the volume of a sphere

- Identifying spheres

- Sketching spheres

- Measuring radii/diameters

- Discussions

spheres

Discovering secondary mathematics Students? Book 2 Pages 114-116

Teachers? Book 2 Pages 24-26

Secondary mathematics KLB book 2 pages 195

KLB teachers? guide book 2 page 57

Golden tips mathematics pages 75

5-6

QUADRATIC EXPRESSIONS AND EQUATIONS



Examples of algebraic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Expand algebraic expressions

Discussions

Multiplying numbers

Dividing numbers

Adding numbers

Subtracting numbers

Exercises given in class

Real life experiences

Worked out expressions

Discovering secondary mathematics Students? Book 2 Pages 117

Teachers? Book 2 Pages 27-29

Secondary mathematics KLB book 2 pages 203

KLB teachers? guide book 2 page 61

Golden tips mathematics pages 20

10

1-2

QUADRATIC EXPRESSIONS AND EQUATIONS

Quadratic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Form Quadratic expressions

Discussions

Multiplying numbers

Dividing numbers

Adding numbers

Subtracting numbers

Exercises given in class

Real life experiences

Worked out expressions

Discovering secondary mathematics Students? Book 2 Pages 117-118

Teachers? Book 2 Pages 27-29

Secondary mathematics KLB book 2 pages 203

KLB teachers? guide book 2 page 61



Golden tips mathematics pages 27

3-4

QUADRATIC EXPRESSIONS AND EQUATIONS

The Quadratic identities

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the three Quadratic identities

Discussions

Multiplying numbers

Dividing numbers

Subtracting numbers

Exercises given in class

Real life experiences

Worked out expressions

Discovering secondary mathematics Students? Book 2 Pages 118-119

Teachers? Book 2 Pages 27-29

Secondary mathematics KLB book 2 pages 204

KLB teachers? guide book 2 page 61

Golden tips mathematics pages 27

5-6

QUADRATIC EXPRESSIONS AND EQUATIONS

Factorizing Quadratic expressions

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Factorize Quadratic expressions

Discussions

Multiplying numbers

Dividing numbers

Subtracting numbers

Adding numbers

Exercises given in class

Real life experiences

Worked out expressions

Discovering secondary mathematics Students? Book 2 Pages 120-121

Teachers? Book 2 Pages 27-29



Secondary mathematics KLB book 2 pages 205
KLB teachers? guide book 2 page 63
Golden tips mathematics pages 28

11

1-2

QUADRATIC EXPRESSIONS AND EQUATIONS

The difference of two squares

OBJECTIVES:

By the end of the lesson, the learner should be able to:

solve Quadratic equations using the difference of two squares

Discussions

Multiplying numbers

Subtracting numbers

Adding numbers

Exercises given in class

Real life experiences

Worked out expressions

Discovering secondary mathematics Students? Book 2 Pages 121-122

Teachers? Book 2 Pages 27-29

Secondary mathematics KLB book 2 pages 204

KLB teachers? guide book 2 page 63

Golden tips mathematics pages 29

3-4

QUADRATIC EXPRESSIONS AND EQUATIONS

Solving quadratic equations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

solve quadratic equations

Discussions

Multiplying numbers

Subtracting numbers

Adding numbers

Exercises given in class

Real life experiences

Worked out expressions

Discovering secondary mathematics Students? Book 2 Pages 122-123

Teachers? Book 2 Pages 27-29



Secondary mathematics KLB book 2 pages 209
KLB teachers? guide book 2 page 64
Golden tips mathematics pages 29

5-6

QUADRATIC EXPRESSIONS AND EQUATIONS

Forming quadratic equations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Form quadratic equations
- Discussions
- Multiplying numbers
- Dividing numbers
- Subtracting numbers
- Adding numbers
- Exercises given in class

- Real life experiences
- Worked out expressions
- Discovering secondary mathematics Students? Book 2 Pages 123-124
- Teachers? Book 2 Pages 27-29
- Secondary mathematics KLB book 2 pages 208
- KLB teachers? guide book 2 page 64
- Golden tips mathematics pages 29

12

1-2

LINEAR INEQUALITIES

Inequalities on a number line

OBJECTIVES:

By the end of the lesson, the learner should be able to:

- Illustrate Inequalities on a number line
- Comparing numbers using the symbols for greater than and less than
- Drawing number lines
- counting whole numbers
- making scale on lines

- number lines
- graph papers
- square boards
- negative and positive numbers
- Discovering secondary mathematics Students? Book 2 Pages 125



Teachers? Book 2 Pages 29-30
Secondary mathematics KLB book 2 pages 213
KLB teachers? guide book 2 page 70
Golden tips mathematics pages 176

3-4

LINEAR INEQUALITIES
Solving linear Inequalities

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve linear Inequalities
Comparing numbers using the symbols for greater than and less than
Drawing number lines
counting whole numbers
making scale on lines

number lines
graph papers
square boards
negative and positive numbers
Discovering secondary mathematics Students? Book 2 Pages 126-127
Teachers? Book 2 Pages 29-30
Secondary mathematics KLB book 2 pages 215
KLB teachers? guide book 2 page 71
Golden tips mathematics pages 176

5-6

LINEAR INEQUALITIES
Compound Inequalities

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve Compound Inequalities
Comparing numbers using the symbols for greater than and less than
Drawing number lines
counting whole numbers
making scale on lines

number lines
graph papers
square boards
negative and positive numbers
Discovering secondary mathematics Students? Book 2 Pages 127-128



Teachers? Book 2 Pages 29-30
Secondary mathematics KLB book 2 pages 213
KLB teachers? guide book 2 page 71
Golden tips mathematics pages 177

13

1

LINEAR INEQUALITIES

Graphical representation of linear inequalities

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Represent linear inequalities graphically
Drawing graphs of inequalities
Determining the scale of a graph
Shading unwanted regions
Discussions

number lines
graph papers
square boards
negative and positive numbers
Discovering secondary mathematics Students? Book 2 Pages 128-129
Teachers? Book 2 Pages 29-30
Secondary mathematics KLB book 2 pages 219
KLB teachers? guide book 2 page 71
Golden tips mathematics pages 178

2-3

LINEAR INEQUALITIES

Inequalities with two variables

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve inequalities with two unknowns graphically
Drawing graphs of inequalities
Determining the scale of a graph
Shading unwanted regions
Discussions

number lines
graph papers
square boards
negative and positive numbers
Discovering secondary mathematics Students? Book 2 Pages 130-131



Teachers? Book 2 Pages 29-30
Secondary mathematics KLB book 2 pages 219
KLB teachers? guide book 2 page 71- 72
Golden tips mathematics pages 129

4-5

LINEAR INEQUALITIES

Graphical solutions of simultaneous inequalities

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Solve inequalities with two unknowns graphically
Drawing graphs of inequalities
Determining the scale of a graph
Shading unwanted regions
Discussions

number lines
graph papers
square boards

negative and positive numbers

Discovering secondary mathematics Students? Book 2 Pages 129-131

Teachers? Book 2 Pages 29-30

Secondary mathematics KLB book 2 pages 224

KLB teachers? guide book 2 page 71

Golden tips mathematics pages 178

6

LINEAR INEQUALITIES

Interpretation of the regions in an inequality graph

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Interpret regions in inequality graphs
Drawing graphs of inequalities
Determining the scale of a graph
Shading unwanted regions
Discussions

number lines
graph papers
square boards

negative and positive numbers

Discovering secondary mathematics Students? Book 2 Pages 131-135



Teachers? Book 2 Pages 29-30
Secondary mathematics KLB book 2 pages 224
KLB teachers? guide book 2 page 71
Golden tips mathematics pages 179

MATHEMATICS FORM 2 SCHEMES OF WORK ? TERM 3

WEEK

LESSON

TOPIC

SUB - TOPIC

OBJECTIVES

LEARNING/TEACHING ACTIVITIES

LEARNING/TEACHING RESOURCES

REFERENCES

REMARKS

1

1-2

ANGLE PROPERTIES OF A CIRCLE

Parts of a circle

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify the parts of a circle

Solve problems involving them

Discussions

Drawing circles

Measuring radii/diameters/angles

Identifying parts of a circle

Circles showing the different parts

Discovering secondary mathematics Students? Book 2 Pages 136-138

Teachers? Book 2 Pages 30-32

Secondary mathematics KLB book 2 pages 264

KLB teachers? guide book 2 page 91

Golden tips mathematics pages 163 and 102

3-4

ANGLE PROPERTIES OF A CIRCLE

Parts of a circle

OBJECTIVES:



By the end of the lesson, the learner should be able to:

Identify the parts of a circle
Solve problems involving them

Discussions
Drawing circles
Measuring radii/diameters/angles
Identifying parts of a circle
Circles showing the different parts

Discovering secondary mathematics Students? Book 2 Pages 138-140
Teachers? Book 2 Pages 30-32
Secondary mathematics KLB book 2 pages 264
KLB teachers? guide book 2 page 91
Golden tips mathematics pages 163 and 102

5-6

ANGLE PROPERTIES OF A CIRCLE
Cyclic quadrilateral

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the angle property of a cyclic quadrilateral

Discussions
Drawing circles
Measuring radii/diameters/angles
Identifying parts of a circle
Circles showing the different parts

Discovering secondary mathematics Students? Book 2 Pages
Teachers? Book 2 Pages
Secondary mathematics KLB book 2 pages 278
KLB teachers? guide book 2 page 92
Golden tips mathematics pages 163 and 104

2

1-2

VECTORS
Scalar quantities and translation

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define vectors



Define scalar quantities
Define transition

Show the direction of a vector
Writing the matrix of a vector
Drawing lines
Plotting the co-ordinates of points on the Cartesian plane
1X2 matrices
Graph papers
Square boards
Ruler

Discovering secondary mathematics Students? Book 2 Pages 145-146
Teachers? Book 2 Pages 33-34
Secondary mathematics KLB book 2 pages 284
KLB teachers? guide book 2 page 92
Golden tips mathematics pages 203

3-4
VECTORS
Equivalent and column vectors

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Identify equivalent vectors
Write column vectors

Show the direction of a vector
Writing the matrix of a vector
Drawing lines
Plotting the co-ordinates of points on the Cartesian plane
Adding numbers
1X2 matrices
Graph papers
Square boards
Ruler

Discovering secondary mathematics Students? Book 2 Pages 146-148
Teachers? Book 2 Pages 33-34
Secondary mathematics KLB book 2 pages 285
KLB teachers? guide book 2 page 94
Golden tips mathematics pages 204

5-6
VECTORS
Addition of vectors



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Add position vectors

Show the direction of a vector

Writing the matrix of a vector

Drawing lines

Plotting the co-ordinates of points on the Cartesian plane

Adding numbers

1X2 matrices

Graph papers

Square boards

Ruler

Discovering secondary mathematics Students? Book 2 Pages 148-149

Teachers? Book 2 Pages 33-34

Secondary mathematics KLB book 2 pages 286

KLB teachers? guide book 2 page 95

Golden tips mathematics pages 205

3

1-2

VECTORS

Subtracting vectors and zero vectors

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Subtract vectors

Identify the null (zero) vectors

Show the direction of a vector

Writing the matrix of a vector

Drawing lines

Plotting the co-ordinates of points on the Cartesian plane

subtracting numbers

1X2 matrices

Graph papers

Square boards

Ruler

Discovering secondary mathematics Students? Book 2 Pages 150-152

Teachers? Book 2 Pages 33-34

Secondary mathematics KLB book 2 pages 289

KLB teachers? guide book 2 page 95

Golden tips mathematics pages 205



3-4

VECTORS

Multiplication of vectors by scalar

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Multiply vectors by a scalar

Writing position vectors

Adding and subtracting vectors

Multiplying vectors by a scalar

1X2 matrices

Graph papers

Square boards

Ruler

Discovering secondary mathematics Students? Book 2 Pages 152-154

Teachers? Book 2 Pages 33-34

Secondary mathematics KLB book 2 pages 290

KLB teachers? guide book 2 page 95

Golden tips mathematics pages 203

5-6

VECTORS

The magnitude of a vector

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the magnitude of a vector

Writing position vectors

Adding and subtracting numbers

Squaring and getting the square root of numbers

1X2 matrices

Graph papers

Square boards

Ruler

Discovering secondary mathematics Students? Book 2 Pages 154-155

Teachers? Book 2 Pages 33-34

Secondary mathematics KLB book 2 pages 301

KLB teachers? guide book 2 page 95

Golden tips mathematics pages 205



4

1-2

VECTORS

The mid point of a vector

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Establish the equality of vectors and determine the midpoint of a vector

Writing position vectors

Adding and subtracting numbers

Squaring and getting the square root of numbers

1X2 matrices

Graph papers

Square boards

Ruler

Discovering secondary mathematics Students? Book 2 Pages 156-157

Teachers? Book 2 Pages 33-34

Secondary mathematics KLB book 2 pages 302

KLB teachers? guide book 2 page 96

Golden tips mathematics pages 208

3-4

REPRESENTATION OF DATA

Collection and representation of data

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Collect organize and represent data for easy interpretation

Collecting data

Measuring lengths/mass/age

Drawing graphs

Drawing tables

Using symbols to represent data

Discussion

Weighing balance

Ruler

Tape measure

Pieces of sticks

Arm length and foot length

Discovering secondary mathematics Students? Book 2 Pages 158-159

Teachers? Book 2 Pages 34-37



Secondary mathematics KLB book 2 pages 241-252
KLB teachers? guide book 2 page 77
Golden tips mathematics pages 184

5-6

REPRESENTATION OF DATA
pictograms

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Represent data in a pictogram

Collecting data
Measuring lengths/mass/age
Drawing graphs
Drawing tables
Using symbols to represent data
Discussion
Weighing balance
Ruler
Tape measure
Pieces of sticks
Arm length
foot length
graph papers

Discovering secondary mathematics Students? Book 2 Pages 160-161
Teachers? Book 2 Pages 34-37
Secondary mathematics KLB book 2 pages 253
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 192

5

1-2

REPRESENTATION OF DATA
Bar graphs

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Represent data in bar graphs

Collecting data
Measuring lengths/mass/age
Drawing graphs



Drawing tables
Using symbols to represent data
Discussion
Weighing balance
Ruler
Tape measure
Pieces of sticks
Arm length
foot length
graph papers

Discovering secondary mathematics Students? Book 2 Pages 161-164
Teachers? Book 2 Pages 34-37
Secondary mathematics KLB book 2 pages 252
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 188

3-4
REPRESENTATION OF DATA
line graphs

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Represent data in line graph

Collecting data
Measuring lengths/mass/age
Drawing graphs
Drawing tables
Using symbols to represent data
Discussion
Weighing balance
Ruler
Tape measure
Pieces of sticks
Arm length
foot length
graph papers

Discovering secondary mathematics Students? Book 2 Pages 164-165
Teachers? Book 2 Pages 34-37
Secondary mathematics KLB book 2 pages 255
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 191

5-6
REPRESENTATION OF DATA
Pie charts



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Represent data in a pie chart

Collecting data

Measuring lengths/mass/age

Drawing graphs

Drawing tables

Using symbols to represent data

Discussion

Weighing balance

Ruler

Tape measure

Pieces of sticks

Arm length

foot length

graph papers

Discovering secondary mathematics Students? Book 2 Pages 166-167

Teachers? Book 2 Pages 34-37

Secondary mathematics KLB book 2 pages 254

KLB teachers? guide book 2 page 78

Golden tips mathematics pages 187

6

1-2

REPRESENTATION OF DATA

Grouped and ungrouped data

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Group data into classes for easy representation

Interpret data

Collecting data

Grouping data into classes

Determining class intervals

Weighing balance

Ruler

Tape measure

Pieces of sticks

Arm length

foot length

graph papers

Discovering secondary mathematics Students? Book 2 Pages 167-170

Teachers? Book 2 Pages 34-37



Secondary mathematics KLB book 2 pages 241 and 247
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 193

3-4

REPRESENTATION OF DATA

Grouped and ungrouped data

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Group data into classes for easy representation

Collecting data

Grouping data into classes

Determining class intervals

Weighing balance

Ruler

Tape measure

Pieces of sticks

Arm length

foot length

graph papers

Discovering secondary mathematics Students? Book 2 Pages 167-170

Teachers? Book 2 Pages 34-37

Secondary mathematics KLB book 2 pages 241 and 247

KLB teachers? guide book 2 page 78

Golden tips mathematics pages 193

5-6

REPRESENTATION OF DATA

Frequency polygons

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Represent data in frequency polygons

Collecting data

Grouping data into classes

Determining class intervals

Weighing balance

Ruler

Tape measure

Pieces of sticks

Arm length



foot length
graph papers
Discovering secondary mathematics Students? Book 2 Pages 170-172
Teachers? Book 2 Pages 34-37
Secondary mathematics KLB book 2 pages 258
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 196

7

1-2

MEASURING OF CENTRAL TENDENCY

The mean

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the mean of a certain set of data

Measuring length/age mass

Adding numbers

Dividing numbers

Demonstrations

Exercises given in class

Weighing balance

Ruler

Tape measure

Counters

Mathematical tables

Multiplication tables

Discovering secondary mathematics Students? Book 2 Pages 173-174

Teachers? Book 2 Pages 38-40

Secondary mathematics KLB book 2 pages 243

KLB teachers? guide book 2 page 78

Golden tips mathematics pages 192

3-4

MEASURING OF CENTRAL TENDENCY

The mode

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the mode of a certain set of data

Measuring length/age mass

Adding numbers



Dividing numbers
Demonstrations
Exercises given in class
Weighing balance
Ruler
Tape measure
Counters
Mathematical tables
Multiplication tables

Discovering secondary mathematics Students? Book 2 Pages 174
Teachers? Book 2 Pages 38-40
Secondary mathematics KLB book 2 pages 244
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 195

5-6

MEASURING OF CENTRAL TENDENCY
The median

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Calculate the median of a given set of data

Measuring length/age mass
Adding numbers
Dividing numbers
Demonstrations
Exercises given in class
Weighing balance
Ruler
Tape measure
Counters
Mathematical tables
Multiplication tables

Discovering secondary mathematics Students? Book 2 Pages 174-176
Teachers? Book 2 Pages 38-40
Secondary mathematics KLB book 2 pages 244
KLB teachers? guide book 2 page
Golden tips mathematics pages 194

8

1-2

MEASURING OF CENTRAL TENDENCY
The use of
 Σf and Σfx



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Use $\bar{x}$ and $\bar{y}$ to calculate the mean and median of a given set of data

Measuring length/age mass
Adding numbers
Dividing numbers
Demonstrations
Exercises given in class
Weighing balance
Ruler
Tape measure
Counters
Mathematical tables
Multiplication tables

Discovering secondary mathematics Students? Book 2 Pages 176-177
Teachers? Book 2 Pages 38-40
Secondary mathematics KLB book 2 pages 249
KLB teachers? guide book 2 page 78
Golden tips mathematics pages 193

3-4

MEASURING OF CENTRAL TENDENCY
Grouped data

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Determine the mid-point, the mode, the mean and the median of grouped data

Measuring length/age mass
Adding numbers
Dividing numbers
Demonstrations
Exercises given in class
Weighing balance
Ruler
Tape measure
Counters
Mathematical tables
Multiplication tables

Discovering secondary mathematics Students? Book 2 Pages 176-177
Teachers? Book 2 Pages 38-40
Secondary mathematics KLB book 2 pages 247



KLB teachers? guide book 2 page 78
Golden tips mathematics pages 193

5-6

MEASURING OF CENTRAL TENDENCY
Revising

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Answer questions on the measure of central tendency

Asking and answering questions
Exercises given in class
Weighing balance
Ruler
Tape measure
Counters
Mathematical tables
Multiplication tables

Discovering secondary mathematics Students? Book 2 Pages 178
Teachers? Book 2 Pages 40
Secondary mathematics KLB book 2 pages 249-252
KLB teachers? guide book 2 page 79
Golden tips mathematics pages 199

9

1-2

LINEAR MOTION
Velocity and speed

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define displacement
Distinguish between displacement and distance
Differentiate between velocity and speed.

Tossing objects
Drawing graphs
Rolling objects
Observing vehicles
Graph papers
Stones
Pieces of paper
Moving vehicles/bicycles



Discovering secondary mathematics Students? Book 2 Pages 180-183
Teachers? Book 2 Pages 40-41
Secondary mathematics KLB book 2 pages 228
KLB teachers? guide book 2 page 74
Golden tips mathematics pages 80-81

3-4

LINEAR MOTION

Distance ? time graphs

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Plot and draw distance-time graphs

Plotting graphs
Drawing graphs
Graph papers
Stones
Pieces of paper
Moving vehicles/bicycles

Discovering secondary mathematics Students? Book 2 Pages 183-185
Teachers? Book 2 Pages 40-41
Secondary mathematics KLB book 2 pages 231
KLB teachers? guide book 2 page 74
Golden tips mathematics pages 82

5-6

LINEAR MOTION

Distance ? time graphs

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Plot and draw speed-time graphs

Plotting graphs
Drawing graphs
Graph papers
Stones
Pieces of paper
Moving vehicles/bicycles

Discovering secondary mathematics Students? Book 2 Pages 185-186



Teachers? Book 2 Pages 40-41
Secondary mathematics KLB book 2 pages 228-234
KLB teachers? guide book 2 page 74
Golden tips mathematics pages 83

10

1-2

LINEAR MOTION

Velocity and acceleration

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define velocity and acceleration
Distinguish between velocity and acceleration

Plotting graphs
Drawing graphs
Graph papers
Stones
Pieces of paper
Moving vehicles/bicycles

Discovering secondary mathematics Students? Book 2 Pages 186-187
Teachers? Book 2 Pages 40-41
Secondary mathematics KLB book 2 pages 230
KLB teachers? guide book 2 page 74
Golden tips mathematics pages 81

3-4

LINEAR MOTION

Velocity ? time graph

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Plot and draw velocity -time graphs

Plotting graphs
Drawing graphs
Graph papers
Stones
Pieces of paper
Moving vehicles/bicycles

Discovering secondary mathematics Students? Book 2 Pages 187-190
Teachers? Book 2 Pages 40-41



Secondary mathematics KLB book 2 pages 234
KLB teachers? guide book 2 page 74
Golden tips mathematics pages 83

5-6
LINEAR MOTION
Relative speed

OBJECTIVES:

By the end of the lesson, the learner should be able to:

State the problems involving relative speed

Plotting graphs
Drawing graphs
Graph papers
Stones
Pieces of paper
Moving vehicles/bicycles

Discovering secondary mathematics Students? Book 2 Pages 190-194
Teachers? Book 2 Pages 40-41
Secondary mathematics KLB book 2 pages 238
KLB teachers? guide book 2 page 75
Golden tips mathematics pages 82

REVISION AND EXAMINATIONS

MATHEMATICS FORM 4 SCHEMES OF WORK TERM 1

WEEK
LESSON
TOPIC
SUB - TOPIC
OBJECTIVES
LEARNING/TEACHING ACTIVITIES
LEARNING/TEACHING RESOURCES
REFERENCES
REMARKS
1
1
MATRIX AND TRANSFORMATION
Translation



OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define translation and describe an image and an object of a given translation

Reflecting objects in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects on the Cartesian plane
Multiplying, adding, subtracting, and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Rubber band
Models
Calculators
Peg

Discovering secondary mathematics Students? Book 4 Pages 1-3
Teachers? Book 4 Pages 1-3
Longman Explore mathematics students book 4 page 129
Secondary mathematics KLB book 4 pages 1
Golden tips mathematics pages 227

2

MATRIX AND TRANSFORMATION

Rotations

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define rotation and describe an image and an object under a given rotation

Reflecting objects in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects on the Cartesian plane



Multiplying, adding, subtracting, and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Rubber band
Models
Calculators
Peg

Discovering secondary mathematics Students? Book 4 Pages 1-2
Teachers? Book 4 Pages 1-3,23
Longman Explore mathematics students book 4 page 130
Secondary mathematics KLB book 4 pages 3
Golden tips mathematics pages 228

3

MATRIX AND TRANSFORMATION
Reflections

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define reflection
Describe the image and the object under a given reflection

Reflecting objects in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects on the Cartesian plane
Multiplying, adding, subtracting, and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Rubber band
Models
Calculators
Peg

Discovering secondary mathematics Students? Book 4 Pages 2
Teachers? Book 4 Pages 1-3,23
Longman Explore mathematics students book 4 page 135
Secondary mathematics KLB book 4 pages 2
Golden tips mathematics pages 230-234



4-5

MATRIX AND TRANSFORMATION

Enlargement

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Define reflection

Describe the image and the object under a given reflection

Reflecting objects in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects on the Cartesian plane

Multiplying, adding, subtracting, and dividing numbers

Discussions

Solving problems

Square boards

Graph papers

Rubber band

Models

Calculators

Peg

Discovering secondary mathematics Students? Book 4 Pages 3-4

Teachers? Book 4 Pages 1-3,23

Longman Explore mathematics students book 4 page 139

Secondary mathematics KLB book 4 pages 3

Golden tips mathematics pages 230-235

6-7

MATRIX AND TRANSFORMATION

Exercise

OBJECTIVES:

By the end of the lesson, the learner should be able to:

Answer questions on reflection, rotation and enlargement

Reflecting objects in a mirror

Rotating objects

Translating objects



Enlarging objects
Drawing images and objects on the Cartesian plane
Multiplying, adding, subtracting, and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Rubber band
Models
Calculators
Peg

Discovering secondary mathematics Students? Book 4 Pages 3-4
Teachers? Book 4 Pages 1-3,23
Longman Explore mathematics students book 4 page 139
Secondary mathematics KLB book 4 pages 8
Golden tips mathematics pages 241-243

MATHEMATICS FORM 3 SCHEMES OF WORK

WK
LESS
TOPIC
SUBTOPIC
OBJECTIVES
TEACHING/LEARNING ACTIVITIES
TEACHING RESOURCES
REFERENCES
REMARKS

1

1-2

Quadratic expression and equations

Factorizing quadratic expression

By the end of the lesson, the learner should be able to
identify and factorize quadratic expression

Substituting values
Drawing graphs
Writing algebraic expressions
Factorizing numbers
Dividing numbers
Discussions
Doing exercises

Square boards



Graph papers
Factors of production
Algebraic expression
calculators

KLB secondary mathematics book 3 page 1
Golden tips mathematics KCSE Revision Page 21

3-4

Quadratic expression and equation

Perfect squares

By the end of the lesson, the learner should be able to
Identify and simplify perfect square

Substituting values
Drawing graphs
Writing algebraic expressions
Factorizing number
Dividing number
Discussions
Solving problems

Square boards
Graph papers
Factor of numbers
Algebraic expressions
calculators

KLB secondary mathematics book 3page 1
Golden tips mathematics KCSE Revision Page 21

5-6

Quadratic expression and equation

Completion of squares

By the end of the lesson, the learner should be able
To complete the square of a quadratic equation

Discussions
Competing squares
Solving problems
Substituting values

Calculators
Graph paper



Square boards
Charts showing algebraic expression

KLB secondary mathematics book 3 page 1-4
Golden tips mathematics KCSE Revision Page 23

7 REVISION

2

1-2

Quadratic Expression and Equation

Solution on quadratic equations by completing the square

By the end of the lesson, the learner should be able to
Solve quadratic equations by completing the square

Discussions
Illustrations
Explanations
Factorizing numbers
Completing squares
Doing exercises

Square boards
Graph papers
Factors of number on a chart
Algebraic expression
calculator

KLB secondary mathematics book 3 page 1-4
Golden tips mathematics KCSE Revision Page 23

3

Quadratic Expressions and Equation

Solution of quadratic by completing the square

By the end of the lesson, the learner should be able to
Solve quadratic equations by completing the square



Drawing graphs
Writing algebraic expression
Factorizing number
Dividing numbers
Discussions
Solving problems

Square board
Graph papers
Factor of numbers
Chart showing algebraic expression
Calculators

KLB secondary mathematics book 3 page 5-7
Golden tips mathematics KCSE Revision Page 31-32

4-5

Quadratic expression and equation

Derivation of the quadratic formulae $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

By the end of the lesson, the learner should be able to
derive the quadratic formulae
 $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and use it to solve quadratic equations

Discussions
Derivation
Illustrations
Solving problems
Writing algebraic equation
Drawing graphs
Factorizing numbers

Square boards
Graph papers
Factors of numbers
Chart showing algebraic expression
calculators

KLB secondary mathematics book 3 page 7-8
Golden tips mathematics KCSE Revision Page 31

6-7

Quadratic Expression and equation

Forming quadratic equations



By the end of the lesson, the learner should be able to
Form quadratic equations and solve them

Substituting values
Writing algebraic expression
Factorizing numbers
Dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Factors of production
Charts showing algebraic expression
calculators

KLB secondary mathematics book 3 page 9-10
Golden tips mathematics KCSE Revision Page 26

3

1-2

Quadratic expression and equation

Solving simultaneous quadratic equations

By the end of the lesson, the learner should be able to
Solve the simultaneous quadratic equations

Illustrations
Doing exercise
Substituting values
Discussions
Factorizing numbers
Solving simultaneous quadratic equations

Factors of numbers
Square boards
Calculators
Chart showing algebraic expression

KLB secondary mathematics book 3 page 12
Golden tips mathematics KCSE Revision Page 26-27

3-4

Quadratic expressions and equation



Solution of quadratic equations by completing the square

By the end of the lesson, the learner should be able to
Solve quadratic equations by completing the square

Drawing graphs
Writing algebraic expression
Factors

Square board
Graph papers
Factorizing numbers

KLB secondary mathematics book 3 page 12-13
Golden tips mathematics KCSE Revision Page 31

5-6

Quadratic expression and equations

Graphs of quadratic equations

By the end of the lesson, the learner should be able to
Make tables of values for quadratic relations and draw the graph for the equation

Substituting values
Drawing graphs
Writing algebraic expressions
Factorizing numbers
Dividing numbers
Discussions solving problems

Square boards
Graph papers
Factors of numbers
Charts
calculators

KLB secondary mathematics book 3 page 12-13
Golden tips mathematics KCSE Revision Page 31

7

Quadratic Expression and equations

Graphical solutions of quadratic equations



By the end of the lesson, the learner should be able to
Solve quadratic equations graphically

Substituting values
Drawing graphs
Writing algebraic expressions
Factorizing numbers
Dividing numbers
discussions

calculators
square boards
graph papers
Algebraic expressions

KLB secondary mathematics book 3 page 15-17
Golden tips mathematics KCSE Revision Page 31-32

4

1-2

Quadratic expressions and equations

Graphical solutions of quadratic equations

By the end of the lesson, the learner should be able to
Solve simultaneous equations (one linear and quadratic)

Substituting values
Drawing Graphs
Writing algebraic expressions
Factorizing numbers
Dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Factors of numbers
Calculators
charts

KLB secondary mathematics book 3 page 19-21
Golden tips mathematics KCSE Revision Page 31

3-4

Approximations and errors



Significant figures and truncation of numbers

By the end of the lesson, the learner should be able to

Express a number to a given number of significant figures and truncate number

Measuring length and mass
Approximating numbers
Writing numbers
Solving problems

Scientific calculators
Mathematical tables
Place value charts
Measuring instruments
Real- life experiments

KLB secondary mathematics book 3 page 24
Golden tips mathematics KCSE Revision Page 244

5

Approximations and Errors

Rough estimates and the calculator

By the end of the lesson, the learner should be able to

Make rough estimations of numbers and use of calculators

Measuring length/mass
Approximating numbers
Writing numbers
Discussions
Doing exercises

Calculators
Mathematical tables
Place-value charts
Measuring instruments
Real-life experience

KLB secondary mathematics book 3 page 25-30
Golden tips mathematics KCSE Revision Page 244

6-7



Approximation and Errors

Errors

By the end of the lesson, the learner should be able to
Define errors, truncation errors, errors in measurements, absolute errors, relative errors and percentage errors

Measuring length/mass
Approximating errors
Writing numbers
Truncating
Solving problems

Calculators
Mathematical table
Place value charts
Measuring instruments
Real life experience

KLB secondary mathematics book 3 page 31-35
Golden tips mathematics KCSE Revision Page 244

5

1

Approximation and Errors

Propagation of errors

By the end of the lesson, the learner should be able to
Determine the possible errors made from addition and subtraction

Measuring length/mass
Approximating errors
Writing numbers
Solving problems

Calculators
Mathematical tables
Place value charts
Measuring instruments
Real-life experience

KLB secondary mathematics book 3 page 35-36
Golden tips mathematics KCSE Revision Page 244



2-3

Approximations and errors

Propagation of errors

By the end of the lesson, the learner should be able to

- Determine the possible error made from multiplication
- division

- Measuring length/mass

- Approximating numbers

- Writing numbers

- Solving problems

- Calculators

- Mathematical tables

- Place-value charts

- Measuring instruments

- Real-life experience

KLB secondary mathematics book 3 page 36-38

Golden tips mathematics KCSE Revision Page 244

4-5

Trigonometry

The unit -circle

By the end of the lesson, the learner should be able to

- Define and draw the unit circle and use it to find the trigonometric ratios in terms of coordinates of point for $0^\circ < \theta < 360^\circ$

- Drawing circle

- Plotting the coordinates

- Reading the coordinates of points

- Measuring lengths/ angles

- The unit

- Calculator

- Graph paper

- Square board

- Mathematical tables

KLB secondary mathematics book 3 page 41-44

Golden tips mathematics KCSE Revision Page 132



6-7

Trigonometry

Trigonometric ratios in the unit circle

By the end of the lesson, the learner should be able to
Use the unit circle to find trigonometric ratios of angles

Drawing circles
Plotting the coordinates of points
Reading the coordinates of points
Measuring lengths/angles

The unit circle
Calculator
Graph papers
Square boards
Mathematical tables

KLB secondary mathematics book 3 page 41-44
Golden tips mathematics KCSE Revision Page 132

6

1-2

Trigonometry

Ratios of angles greater than 90°

By the end of the lesson, the learner should be able to
Find the trigonometric ratios of angles greater than 90°

Drawing circle
Plotting the coordinates of points
Reading the coordinates of points
Measuring length/angles

Calculator
The unit circle
Graph papers
Square boards
Mathematical tables

KLB secondary mathematics book 3 page 44-47
Golden tips mathematics KCSE Revision Page 136-138



3

Trigonometry

Ratios of negative angles

By the end of the lesson, the learner should be able to
Find the trigonometric ratios of negative angles

Drawing circles
Plotting points
Reading the coordinates of points
Measuring length/angles

The unit circle
Calculator
Graph papers
Square board
Mathematical table

KLB secondary mathematics book 3 page 48-49
Golden tips mathematics KCSE Revision Page 134

3-4

Trigonometry

Using trigonometric tables and calculator

By the end of the lesson, the learner should be able to
Use mathematical tables and calculators to find the trigonometric ratios
of angles in the range of $0^\circ < \theta < 360^\circ$

Drawing circles
Plotting the coordinates of points
Recording the coordinates of points
Measuring length/angles

Unit circle
Calculator
Graph paper
Square boards
Mathematical tables

KLB secondary mathematics book 3 page 51-55
Golden tips mathematics KCSE Revision Page 136



5-6

Trigonometry

Finding the angle given the ratio

By the end of the lesson, the learner should be able to

Find the size of an angle given the trigonometric ratio for the angle

Drawing circles

Plotting the coordinates of points

Measuring lengths/angles

The unit circle

Calculator

Graph paper

Square board

Mathematical tables

KLB secondary mathematics book 3 page 51-55

7

Trigonometry

The ratio of angles greater than 360°

By the end of the lesson, the learner should be able to

Find the trigonometric ratio of angle $>360^\circ$ from mathematical tables and calculator

Plotting the co-ordinates of points

Drawing circles

Reading the coordinates of points

Measuring length angles

The unit circle

Calculator

Graph paper

Square board

Mathematical tables

pages 36-37

Explore Mathematics book 3 page 57

KLB secondary mathematics book 3 page 49-50

7

1-2



Trigonometry

The radian measure

By the end of the lesson, the learner should be able to
Define the radian measure and convert radians to degrees and vice versa

Drawing circles
Plotting the coordinates
Reading the coordinates of points
Measuring length/angles

The unit circle
Calculator
Charts
Graph papers
Square boards
Mathematical tables and boards

KLB secondary mathematics book 3 page 58-61
Golden tips mathematics KCSE Revision Page 150-151

3-4

Trigonometry

Application of radians in calculations

By the end of the lesson, the learner should be able to
Radian in calculation

Drawing circles
Plotting the coordinates of points
Reading the coordinates of points
Measuring length/angle

Mathematical tables
Graph papers
Square boards
Calculators and boards

KLB secondary mathematics book 3 page 58-61
Golden tips mathematics KCSE Revision Page 134

5-6

Trigonometry

Trigonometric graphs



By the end of the lesson, the learner should be able
Draw the graph of $y=\sin x$, $y=\cos x$ and $y=\tan x$

Drawing graphs
Discussions
Illustrations
Plotting coordinates of points
Reading coordinates of points

Arid boards
Calculators
Square boards
Mathematical tables
Graph paper
The unit circle

KLB secondary mathematics book 3 page 62-65
Golden tips mathematics KCSE Revision Page 151

7

Trigonometry

Trigonometric graphs

By the end of the lesson, the learner should be able to
Draw the graph of $y=\sin x$, $y=\cos x$ and $y=\tan x$

Drawing graphs
Discussions
Reading coordinates of points
Plotting the coordinates of points
Illustrations
Doing exercises

Arid boards
Graph papers
Calculators
Mathematical tables
Square boards

Discovering secondary mathematics book 3 pages
KLB secondary mathematics book 3 page 61
Golden tips mathematics KCSE Revision Page 151

8

1-2



Trigonometry

The sine rule

By the end of the lesson, the learner should be able to

- Derive the sin rule

- Use sine rule to solve problems involving the sides and angles of triangles

- Drawing triangle

- Measuring angles/length

- Derivations

- Solving problems

- Triangles

- Calculator

- Graph papers

- Mathematical tables

KLB secondary mathematics book 3 page 65-70

Golden tips mathematics KCSE Revision Page 138

3-4

Trigonometry

The sine rule

By the end of the lesson, the learner should be able to

- Solve problems involving sides and angles of triangles

- Drawing triangles

- Measuring angles/lengths

- Solving problems

- Discussions

- illustrations

- triangular figures

- square boards

- mathematical tables

- graph papers

- calculators

KLB secondary mathematics book 3 page 65-70

Golden tips mathematics KCSE Revision Page 138

5-6

Trigonometry



The cosine rule

By the end of the lesson, the learner should be able to

Derive cosine rule

Use cosine rule to solve problems involving the sides and angles of a triangle

Drawing triangles

Solving problems

Illustrations

Derivations

Discussions

Measuring angles/length

Triangular item/objects

Mathematical tables

Triangles

Graph papers

calculators

KLB secondary mathematics book 3 page 71-75

Golden tips mathematics KCSE Revision Page 139-140

7

Surds

Rational , irrational numbers and simplifying surds

By the end of the lesson, the learner should be able to

Define rational and irrational numbers and simplifying surds

Simplifying numbers

Definitions

Creating the square roots of numbers

Squaring numbers

discussions

square of numbers

mathematical tables

multiplication tables

calculator

square roots of numbers

KLB secondary mathematics book 3 page 78-79

Golden tips mathematics KCSE Revision Page 46

9



1-2

Surds

Like and unlike surds

By the end of the lesson, the learner should be able to
Define the order of surds and identify like and unlike surds

Definitions

Illustrations

Simplifying numbers

Grating square roots of numbers

Squaring numbers

Chart showing squares of numbers

Mathematical tables

Multiplication tables

Calculators

Factors of numbers

KLB secondary mathematics book 3 page 79-80

Golden tips mathematics KCSE Revision Page 46-47

3

Surds

Multiplication involving surds and division

By the end of the lesson, the learner should be able to
Carry out multiplication involving surds

Multiplying numbers

Squaring numbers

Discussions

Illustrations

Doing exercises

Mathematical tables

Multiplication tables

Calculators

Charts showing factors of numbers

KLB secondary mathematics book 3 page 80-82

Golden tips mathematics KCSE Revision Page 48

4-5



Surds

Rationalizing denominators

By the end of the lesson, the learner should be able to
Rationalize denominators in surds

Simplifying numbers
Rationalizing numbers
Squaring numbers
Illustrations
Doing exercises

Mathematical table
Multiplication table
Charts showing factors of numbers
calculators

KLB secondary mathematics book 3 page 85-87
Golden tips mathematics KCSE Revision Page 48

6-7

Logarithms

The logarithmic notation

By the end of the lesson, the learner should be able to
Derive the logarithmic relation from index form and vice versa

Computing using the calculator
Reading mathematical table
Writing numbers in standard form
Writing numbers in index form
discussions

chart showing laws of logarithms
mathematical table
calculators

KLB secondary mathematics book 3 page 89-90
Golden tips mathematics KCSE Revision Page 52

10

1

Logarithms



The law of logarithm

By the end of the lesson, the learner should be able to

State the laws of logarithms and use them to solve problems involving logarithms

Stating

Illustrations

Discussions

Doing exercises

Writing numbers in Standard form

Chart showing laws of logarithms

Mathematical tables

calculators

KLB secondary mathematics book 3 page 90-93

Golden tips mathematics KCSE Revision Page 52-54

2-3

Logarithms

Simplifying logarithmic expressions

By the end of the lesson, the learner should be able to

simplify logarithmic expressions using the laws of logarithms

Simplifying logarithmic expression

Discussions

Illustrations

Doing exercise

Reading mathematical tables

Computing using the calculator

Mathematical tables

Calculators

Chart showing laws of logarithms

KLB secondary mathematics book 3 page 90-93

Golden tips mathematics KCSE Revision Page 54-56

4-5

Logarithms

Solving logarithmic equations



By the end of the lesson, the learner should be able to
Solve logarithmic equations

Computing using calculator
Recording mathematical table
Solving logarithmic equations
Writing numbers in index form

Mathematical tables
Calculators
Chart showing logarithmic laws
indices

KLB secondary mathematics book 3 page 95-96
Golden tips mathematics KCSE Revision Page 54-56

11-13 REVISION, END OF TERM EXAMS MARKING AND CLOSING OF SCHOOL

MATHEMATICS FORM THREE
SCHEMES OF WORK
TERM TWO

1

1-2

Commercial Arithmetic

Simple interest

By the end of the lesson, the learner should be able to
Define principle rate and interest and calculate simple interest

Calculating interest
Multiplying dividing, subtracting and adding numbers
Discussions
Demonstrations
Providing theories
Reading newspapers

Income tax schedules/bands
Calculators
Mathematical tables
Multiplication tables
Advertisement in local dailies
Higher purchase terms

KLB secondary mathematics book 3 page 98-100
Golden tips mathematics KCSE Revision Page 89



3

Commercial Arithmetic

Compound interest

By the end of the lesson, the learner should be able to
Calculate compound interest using step by step method

Calculating interest
Multiplying dividing numbers
Deriving formulae
Providing theories
Discussions
Demonstrations
Doing exercises
Reading newspapers

Income tax schedule/bands
Calculators
Mathematical tables
Multiplication tables
Advertisement on local dailies

KLB secondary mathematics book 3 page 102-106

4-5

Commercial Arithmetic

Deriving the compound interest formulae

By the end of the lesson, the learner should be able to
Derive the compound interest formula and use it to solve problems
involving compound interest

Deriving formulae
Providing theories
Doing exercises
Reading newspaper
Multiplying
Calculating interests
Discussions
demonstrations

income tax schedule
calculators
mathematical tables



newspaper
chart showing hire purchase terms

KLB secondary mathematics book 3 page 104-106
Golden tips mathematics KCSE Revision Page 89-90

6-7

Commercial Arithmetic

Half yearly and quarterly interest

By the end of the lesson, the learner should be able to
Apply the compound interest formulae to calculate interest half yearly
and quarterly

Calculating interest
Multiplying, dividing
Discussions
Deriving formulae
Demonstrations
Doing exercises

Income tax schedule
Calculators
Charts showing hire purchase terms
Newspapers
Mathematical tables

KLB secondary mathematics book 3 page 104-106

1-2

Commercial Arithmetic

Appreciation and depreciation

By the end of the lesson, the learner should be able to
Define appreciation and depreciation and work out problems involving them

Discussions
Illustrations
Calculating appreciation and depreciation
Deriving formulae
Definitions
multiplying



mathematical table
calculators
multiplication tables
newspapers
chart showing hire purchase terms
income tax schedule

KLB secondary mathematics book 3 page 108-110
Golden tips mathematics KCSE Revision Page 90-91

3-4

Commercial arithmetic

Hire purchase

By the end of the lesson, the learner should be able to
Calculate hire purchase

Calculating hire purchase
Discussions
Demonstrations
Illustrations
Solving problems involving hire purchase
Doing exercises
Reading newspapers

Income tax schedule
Newspapers
Calculators
Chart showing hire purchase terms
Mathematical tables
Multiplication tables
KLB secondary mathematics book 3 page 110-112
Golden tips mathematics KCSE Revision Page 91

5-6

Commercial arithmetic

Income tax

By the end of the lesson, the learner should be able to
Calculate the income tax given the tax bonds

Calculating interest
Discussion
Calculating income tax
Doing exercise



Reading relevant newspaper cuttings
Illustrations
Lecture from resource persons

Income tax schedule
Mathematical table
Multiplication table
Newspapers
calculators
KLB secondary mathematics book 3 page 112-116
Golden tips mathematics KCSE Revision Page 91-94

7

Commercial Arithmetic

Income tax

By the end of the lesson, the learner should be able to
Calculate the income tax given the tax bonds

Calculating interest
Discussions
Calculating income tax
Doing exercises
Illustrations
Reading relevant newspaper cuttings
Solving problems involving income tax

Income tax schedule
Mathematical table
Multiplication table
Newspapers
calculator
KLB secondary mathematics book 3 page 112-116
Golden tips mathematics KCSE Revision Page 91-94

3

1-2

Circles chords and triangles

Properties of chord

By the end of the lesson, the learner should be able to
State the properties of chords

Drawing circles
Drawing chords
Drawing tangents/radii/diameter



Measuring length/ angles
discussions

chart illustrating the properties of chord
protractor
calculator
ruler
pair of compass

KLB secondary mathematics book 3 page 124
Golden tips mathematics KCSE Revision Page 162

3-4

Circles chords and tangents

The lengths of arcs

By the end of the lesson, the learner should be able to
Calculate the length of an arc and a chord

Drawing circles
Drawing chords/tangents/diameters
Measuring lengths/angles
Doing exercises
discussions

pulleys and wheels
charts illustrating the properties of the chords
calculators
protractor
ruler
pair of compass

KLB secondary mathematics book 3 page 124
Golden tips mathematics KCSE Revision Page 162

5

Circles and tangents

Equal chords

By the end of the lesson, the learner should be able to
Identify equal chords

Drawing chords/circles/tangents
Discussions
Doing exercises
Illustrations



Measuring length/angles

Charts illustrating properties of chords

Calculators

Protractors

Ruler

Pair of compass

Pulleys and wheels

KLB secondary mathematics book 3 page 131-132

Golden tips mathematics KCSE

Revision Page 165

6-7

Circles chords and tangents

Equal chords

By the end of the lesson, the learner should be able to

Identify equal chords

Drawing chords/circles/tangents

Discussions

Doing exercises

Illustrations

Measuring length/angles

Charts illustrating properties of chords

Calculators

Protractors

Ruler

Pair of compass

Pulleys and wheels

KLB secondary mathematics book 3 page 132-135

Golden tips mathematics KCSE Revision Page 165-166

4

1-2

Circles chords and tangents

Constructing tangents to a circle

By the end of the lesson, the learner should be able to

Construct a tangent to a circle

Drawing circles/chords/radii

Measuring angles



Measuring lengths
Discussions
Demonstrations
Doing exercises

Ruler
Protractor
A pair of compasses
Calculator
Charts illustrating properties of chords

KLB secondary mathematics book 3 page 139-142
Golden tips mathematics KCSE Revision Page 113

3-4

Circles chords and tangents

Angles in alternate segments

By the end of the lesson, the learner should be able to
Relate angles in alternate segments

Drawing circles/tangents/diameters
Discussions
Measuring lengths angles
Demonstrations
Doing exercises

Ruler
Protractor
A pair of compasses
Calculator
Charts illustrating alternate segments

KLB secondary mathematics book 3 page 142-144
Golden tips mathematics KCSE Revision Page 114

5-6

Circles Chords and tangents

Common tangents

By the end of the lesson, the learner should be able to
Construct direct and transverse common tangents to two circles

Drawing tangents/chords/radii
Measuring angles/lengths
Discussions



Demonstrations
Doing exercises

Protractor
Ruler
Pulleys and wheels
Calculator
Chart illustrating common tangents

KLB secondary mathematics book 3 page 142-144
Golden tips mathematics KCSE Revision Page 114

7

Circles chords and tangents

Common tangents

By the end of the lesson, the learner should be able to
Construct direct transverse common tangents to two circles

Drawing tangents chords/radii/diameters
Measuring lengths/angles
Demonstrations
Discussions
Doing exercises

Ruler
Protractor
Calculator
Charts illustrating common tangents
KLB secondary mathematics book 3 page 148-154
Golden tips mathematics KCSE Revision Page 114

5

1

Circles chords and tangents

Inscribed circles

By the end of the lesson, the learner should be able to
Construct inscribed circles

Drawing circles
Inscribing circles
Measuring lengths
Discussions
Illustration
Doing exercises



Ruler
A pair of compass
Calculator
Chart illustrating inscribed circle
protractor

KLB secondary mathematics book 3 page 142-144
Golden tips mathematics KCSE Revision Page 120

2-3

Circles chords and tangents

Circumcircle and Enscribed circles

By the end of the lesson, the learner should be able to construct
Circumcircles
Enscribed circles

Constructing circumcircles
Constructing enscribed circles
Discussions
Demonstrations
Measuring length/angles
Doing exercises

Ruler
Pair of compasses
Calculator
Chart illustrating circumcircles and enscribed circles
protractor

KLB secondary mathematics book 3 page 142-144
Golden tips mathematics KCSE Revision Page 121

4

Circles, chords and tangents

The centric of a triangle and orthocenter of it

By the end of the lesson, the learner should be able to
Locate the centroid of triangle and orthocenter of a circle

Drawing circles, triangles/radii
Measuring length, angles
Discussions
Demonstrations
Doing exercises



Locating centroid of a triangle

Ruler

Protractor

Calculator

Chart illustrating centroid of a triangle

Triangular shape

KLB secondary mathematics book 3 page 142-144

Golden tips mathematics KCSE Revision Page 121

5-6

Matrices

Definition order and notation of matrices

By the end of the lesson, the learner should be able to

Define a matrix given the order of matrix and use the matrix notation

Identifying matrices

Definitions

Forming rows and columns

Forming matrices

Solving problems

Discussions

Doing exercises

KLB secondary mathematics book 3 page 168-170

Golden tips mathematics KCSE Revision Page 219

7

Matrices

Adding and subtracting matrices

By the end of the lesson, the learner should be able to

Add and subtract matrices

Forming matrices

Adding and subtracting matrices

Solving problems

Discussions

illustrations

mathematical table

multiplication table



calculator
counters

KLB secondary mathematics book 3 page 168-170
Golden tips mathematics KCSE Revision Page 219

6

1-2

Matrices

Multiplying a matrix by a scalar

By the end of the lesson, the learner should be able to
Multiply a matrix by a scalar

Identifying matrices
Forming matrices
Multiplying matrices by scalar
Solving problems
Discussions
illustrations

mathematical table
multiplication table
calculator
counter

KLB secondary mathematics book 3 page 170-171
Golden tips mathematics KCSE Revision Page 219

3-4

Matrices

Multiplying a matrix by a matrix

By the end of the lesson, the learner should be able to
Determine compatibility in multiplication of matrices

Identifying matrices
Forming matrices
Multiplying
Discussions
Solving problems
Doing exercises

Mathematical table
Multiplication table



Calculator
counters

KLB secondary mathematics book 3 page 174-179
Golden tips mathematics KCSE Revision Page 220

5

Matrices

Types of matrices

By the end of the lesson, the learner should be able to
Identify null and equal matrices

Identifying matrices
Forming matrices
Doing exercises
Solving problems
Forming rows and columns

Counters
Calculator
Mathematical table
Multiplication table

KLB secondary mathematics book 3 page 174-179
Golden tips mathematics KCSE Revision Page 221

6-7

Matrices

The determinant and inverse of a 2×2 matrix

By the end of the lesson, the learner should be able to
Determine the determinant and inverse of a matrix

Calculating determinant
Forming matrix
Discussion
Solving problems
Multiplying
Doing exercises

Charts
Mathematical tables



Multiplication table
Counters
calculators

KLB secondary mathematics book 3 page 174-179
Golden tips mathematics KCSE Revision Page 222

7

1-2

Matrices

Solving simultaneous equations using matrices

By the end of the lesson, the learner should be able to
Solve simultaneous equations using matrices

Identifying matrices
Solving problems
Doing exercises
Discussions
Multiplying , dividing

Charts
Calculators
Counters
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 188-190
Golden tips mathematics KCSE Revision Page 223

3-4

Matrices

Solving simultaneous equations using matrices

By the end of the lesson, the learner should be able to
Solve simultaneous equations using matrices

Identifying matrices
Solving problems
Doing exercises
Discussions
Multiplying dividing, adding and subtracting



Charts
Calculator
Multiplications tables
Mathematical tables

KLB secondary mathematics book 3 page 188-190
Golden tips mathematics KCSE Revision Page 223

5-6

Formulae and variations

Formulae and change of subject

By the end of the lesson, the learner should be able to
Rewrite a given formulae by changing the subject

Reading formulae
Writing formulae
Adding, subtracting, multiplying, dividing.
Discussions
Solving problems

Square boards
Algebraic
Quadratic equations
Calculators
Real-life experience
Linear equations

KLB secondary mathematics book 3 page 191-193

7

Formulae and variations

Direct variations

By the end of the lesson, the learner should be able to
Define direct variation form and solve equations involving direct
variation
Definition
Discussions
Illustrations
Solving problems
Doing exercises



Real-life experience
Calculators
Square boards
Linear equations
Graph papers

KLB secondary mathematics book 3 page 188-190
Golden tips mathematics KCSE Revision Page 223

8

1-2

Formulae and variations

Inverse variations

By the end of the lesson, the learner should be able to
Define inverse variations form and solve equations involving indirect variations

Reading formulae
Writing formulae
Adding, dividing subtracting and multiplying
Discussions
Solving problems

Square boards
Real-life experience
Graph papers
Linear equations
Charts
calculators

KLB secondary mathematics book 3 page 197-200
Golden tips mathematics KCSE Revision Page 251

4-5

Formulae and variations

Joint variations

By the end of the lesson, the learner should be able to
Define joint variations form and solve equations involving joint variations

Reading formulae



Discussions
Doing exercises
Solving problems
Multiplying dividing
Subtracting and adding numbers

Calculator
Linear equations
Square board
Charts
Graph papers
Quadratic equations

KLB secondary mathematics book 3 page 204-205
Golden tips mathematics KCSE Revision Page 251

6-7

Formulae and variations

Partial variations

By the end of the lesson, the learner should be able to
Define partial variations form and solve equations involving partial variations

Definitions
Solving problems
Doing exercises
Discussions
Writing formulae
Reading formulae

Charts
Calculators
Real-life experience
Linear equations
Quadratic equation
Algebraic expressions
KLB secondary mathematics book 3 page 201-203
Golden tips mathematics KCSE Revision Page 252

9

1-2

Sequence and series0

Number patterns



By the end of the lesson, the learner should be able to
Identify number patterns and determine the missing numbers in a pattern

Adding, subtracting, multiplying and dividing
Arranging numbers to form pattern
Drawing patterns
Discussions
demonstration

even numbers
rectangle numbers
calculators
square numbers
triangular numbers

KLB secondary mathematics book 3 page 207-209
Golden tips mathematics KCSE Revision Page 255

3-4

Sequence and series

sequence

By the end of the lesson, the learner should be able to
Define a sequence and determine the missing term in a sequence

Arranging numbers to form pattern
Discussions
Demonstrations
Drawing patterns
Arranging items to form patterns
Adding and subtracting numbers

Rectangle numbers
Chart showing even, odd, prime and whole numbers
Calculators
Square numbers
Triangular numbers

KLB secondary mathematics book 3 page 207-209
Golden tips mathematics KCSE Revision Page 255

5

Sequence and series



Determining a term in a sequence

By the end of the lesson the learner should be able to
Define and determine a term in a sequence

KLB secondary mathematics book 3 page 207-209
Golden tips mathematics KCSE Revision Page 255

6-7

Sequence and series

Arithmetic sequence

By the end of the lesson, the learner should be able to
Identify an arithmetic sequence and solve problems involving arithmetic
sequence

Adding and subtracting numbers
Arranging items to form patterns
Drawing patterns
Discussions
Solving problems
demonstrations

square numbers
triangular numbers
rectangle numbers
chart showing even, odd, prime and whole numbers
KLB secondary mathematics book 3 page 209-210
Golden tips mathematics KCSE Revision Page 255-256

10

1-2

Sequence and series

Geometric sequence

By the end of the lesson, the learner should be able to
Identify a geometric sequence and solve problems involving geometric
sequences

Multiplying and dividing numbers
Arranging items to form patterns
Discussions



Solving problems
Demonstration
Doing exercises

Square numbers
Triangular numbers
Chart showing, odd, even prime and whole
Rectangle numbers

KLB secondary mathematics book 3 page 211-213
Golden tips mathematics KCSE Revision Page 257-258

3-4

Sequence and series

Arithmetic progress

By the end of the lesson, the learner should be able to
Recognize an arithmetic progression and solve problems involving AD's

Adding, subtracting, multiplying and dividing numbers
Arranging items to form patterns
discussions

rectangle numbers
triangular numbers
chart showing even, odd, prime and whole numbers
square numbers

KLB secondary mathematics book 3 page 214-215
Golden tips mathematics KCSE Revision Page 256

6-7

Sequence and series

Geometric Progression (G.P)

By the end of the lesson, the learner should be able to
Recognize a geometric progression and solve problems involving GP's

Adding and subtracting, multiplying and dividing numbers
Arranging items to form pattern
Drawing patterns
Discussions
demonstrations



rectangle numbers
chart showing even, odd, prime and whole numbers
calculators
square numbers
triangular numbers

KLB secondary mathematics book 3 page 216-218
Golden tips mathematics KCSE Revision Page 257-258

11-13

1-7

REVISION/END TERM EXAMS , MARKING AND CLOSING OF SCHOOL

FORM THREE MATHEMATICS
SCHEME OF WORK
TERM THREE

1

1-2

Vectors

Coordinates in two and three dimensions

By the end of the lesson, the learner should be able to
Locate a point in two dimension coordinate system
Locate a point in three coordinate system

Plotting points on the cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors

Charts illustrating coordinates in 2D and 3D
Graph paper
Square board
Model of cuboid
Calculator
Multiplying tables
Mathematical tables

KLB secondary mathematics book 3 page 216-218
Golden tips mathematics KCSE Revision Page 257-258
Mathematics for secondary schools form 3 (N.M patel) pages 198-199



3-4

Vectors

Column vectors in 2D and 3D

By the end of the lesson, the learner should be able to

Locate a vector as a column and position vectors in 3D

Represent a vector as a column and position vectors in 3D

Plotting points on the Cartesian plane

Reading points on the Cartesian plane

Adding, subtracting, multiplying and dividing vectors

Solving problems

Discussions

Charts illustrating

Coordinates in 2D and 3D

Graph papers

Square boards

Model of a cuboid

Calculator

Multiplying tables

Mathematical tables

KLB secondary mathematics book 3 page 221-223

Golden tips mathematics KCSE Revision Page 203-204

Mathematics for secondary schools form 3 (N.M patel) pages 198-199

5-6

vectors

Operations on vectors

By the end of the lesson, the learner should be able to

Carry out additions, subtraction, multiplication and division on vectors

Plotting points on the Cartesian plane

Reading points on the Cartesian plane

Adding, subtracting, multiplying and dividing vectors

Solving problems

discussions

charts illustrating coordinates in 2D and 3D

graph papers

square boards

model of cuboid

calculator



multiplication tables
mathematical tables

KLB secondary mathematics book 3 page 223-228
Golden tips mathematics KCSE Revision Page 204-212
Mathematics for secondary schools form 3 (N.M patel) pages 194-200

7

vectors

Unit vectors in 2D

By the end of the lesson, the learner should be able to
Represent unit vectors in 2D

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problems
discussions

charts illustrating coordinates in 2D and 3D
graph papers
square board
model of a cuboid
calculator
multiplication tables

KLB secondary mathematics book 3 page 221-224
Golden tips mathematics KCSE Revision Page 208-209
Mathematics for secondary schools form 3 (N.M patel) pages 198-199

2

1

Vectors

Unit vectors in 2D

By the end of the lesson, the learner should be able to
Represent unit vectors in 2D

Plotting on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problems



Chart illustrating coordinates in 2D and 3D
Graph papers
Square boards
Model of cuboid
Calculator
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 221-224
Golden tips mathematics KCSE Revision Page 208-209
Mathematics for secondary schools form 3 (N.M patel) pages 198-199

2-3

vectors

Units vectors in 3D

By the end of the lesson, the learner should be able to
Represent unit vectors in 3D

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problems
discussions

charts illustrating coordinates in 2D and 3D
graph papers
calculator

KLB secondary mathematics book 3 page 226-228
Golden tips mathematics KCSE Revision Page 210-211
Mathematics for secondary schools form 3 (N.M patel) pages 206

4-5

Vectors

The magnitude of a vectors

By the end of the lesson, the learner should be able to
determine the magnitude of a vectors

Plotting on the Cartesian plane
Reading points on the Cartesian plane



Adding, subtracting, multiplying and dividing vectors
Solving problems
discussions

charts illustrating coordinates in 2D and 3D
graph papers
square boards
model of cuboid
calculator
multiplication tables
mathematical tables

KLB secondary mathematics book 3 page 229-230
Golden tips mathematics KCSE Revision Page 210
Mathematics for secondary schools form 3 (N.M patel) pages 201-203

6-7

Vectors

Parallel vectors

By the end of the lesson, the learner should be able to
Identify parallel vectors
Solve problems involving parallel vectors

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problems
discussions

charts illustrating coordinates in 2D and 3D
Graph papers
Square board
Calculators
Model of cuboid
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 229-230
Golden tips mathematics KCSE Revision Page 210
Mathematics for secondary schools form 3 (N.M patel) pages 201-203

3

1-2

Vectors



Collinearly in vectors

By the end of the lesson, the learner should be able to
Use the vector method to show co linearity of points

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problems
discussions

charts illustrating coordinates in 2D and 3D
graph papers
square board
model of cuboid
calculator
multiplication table, mathematical tables

KLB secondary mathematics book 3 page 231-237
Golden tips mathematics KCSE Revision Page 267
Mathematics for secondary schools form 3 (N.M patel) pages 204-205

3-4

vector

Mid-point of vector

By the end of the lesson, the learner should be able to
Determine the mid-point of a vector

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problem
discussions

charts illustrating coordinates in 2D and 3D
Graph papers
Square boards
Model of a cuboid
Calculator
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 221-250
Golden tips mathematics KCSE Revision Page 208
Mathematics for secondary schools form 3 (N.M patel) pages 204-205



5

vectors

Proportional division of a line

By the end of the lesson, the learner should be able to
Use the vector method to divide a line proportionally

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problem
discussions

charts illustrating coordinates in 2D and 3D
Graph papers
Square boards
Model of a cuboid
Calculator
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 239-248
Golden tips mathematics KCSE Revision Page 212
Mathematics for secondary schools form 3 (N.M patel) pages 205-206

6

vectors

The ratio theorem

By the end of the lesson, the learner should be able to
State the ratio theorem
Use the ratio theorem in vectors

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problem
discussions

charts illustrating coordinates in 2D and 3D
Graph papers
Square boards
Model of a cuboid



Calculator
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 239-248
Golden tips mathematics KCSE Revision Page 212
Mathematics for secondary schools form 3 (N.M patel) pages 205-206

7

Vectors

Application of vectors in geometry

By the of the lesson, the learner should be able to
Apply vector methods in geometry

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors
Solving problem
discussions

charts illustrating coordinates in 2D and 3D
Graph papers
Square boards
Model of a cuboid
Calculator
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 248-250
Golden tips mathematics KCSE Revision Page 213-218

4

3

Binominal expansions

Expanding and simplifying binominal expansion

By the end of the lesson, the learner should be able to
Expand binominal expansions up to the power of form by multiplication

Plotting points on the Cartesian plane
Reading points on the Cartesian plane
Adding, subtracting, multiplying and dividing vectors



Solving problem
discussions

Pascal's triangle
mathematical tales
multiplication tables
calculators

KLB secondary mathematics book 3 page 256-258
Golden tips mathematics KCSE Revision Page 261-262

4-5

Binominal expansions

Binominal expansion and the Pascal's triangle

By the end of the lesson, the learner should be able to
Build up Pascal's triangle up to the 11th row
Use Pascal's triangle to expand binominal expressions

Writing factors of numbers
Multiplying and dividing factors of numbers
Expanding binominal expansions
Discussions
Solving problems

Pascal's triangle
Mathematical tables
Multiplication tablets
calculators

KLB secondary mathematics book 3 page 256-259
Golden tips mathematics KCSE Revision Page 262-264

6-7

Binominal expansions

Binominal expansion and the Pascal's triangle

By the end of the lesson, the learner should be able to
Build up Pascal's triangle up to the 11th row
Use Pascal's triangle to expand binominal expressions

Writing factors of numbers
Multiplying and dividing factors of numbers
Expanding binominal expansions



Discussions
Solving problems

Pascal's triangle
Mathematical tables
Calculators
Multiplication tables

KLB secondary mathematics book 3 page 256-259
Golden tips mathematics KCSE Revision Page 262-264

5

1-2

Probability

The meaning of probability

By the end of the lesson, the learner should be able to
Define probability

Playing probability
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g cards
Calculator
Coins
Cards
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 262
Mathematics for secondary schools (N.M. patel) form 3 pages 219
Golden tips mathematics KCSE Revision Page 262-265

3-4

Probability

Experimental probability

By the end of the lesson, the learner should be able to
Determine probability from experiment



Determine probability from real-life situations

Playing probability
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g card
Calculators
Coins
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 262-265
Mathematics for secondary schools (N.M. patel) form 3 pages 222-223

5-6

Probability

Theoretical probability

By the end of the lesson, the learner should be able to
Determine theoretical probability

Playing probability
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g card
Calculators
Coins
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 266-272
Mathematics for secondary schools (N.M. patel) form 3 pages 224-225
Golden tips mathematics pages 266-268



Probability

Probability space

By the end of the lesson, the learner should be able to
Construct a probability space

Playing probability
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g cards
Calculation
Coins
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 266
Mathematics for secondary schools (N.M. patel) form 3 pages 224
Golden tips mathematics pages 267

6

1-2

Probability

Probability space

By the end of the lesson, the learner should be able to
Construct a probability space

Playing probability games
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g cards
Calculators
Cons
Dies
Multiplication tables
Mathematical cables

KLB secondary mathematics book 3 page 266



Mathematics for secondary schools (N.M. patel) form 3 pages 224
Golden tips mathematics pages 267

3-4

Probability

Continuous probability

By the end of the lesson, the learner should be able to
Differentiate discrete and continuous probabilities

Playing probability games
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g cards
Calculators
Coins
Cards
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 262-289
Mathematics for secondary schools (N.M. patel) form 3 pages 225
Golden tips mathematics pages 268

5

Probability

Continuous probability

By the end of the lesson, the learner should be able to
Differentiate discrete and continuous probabilities

Playing probability games
Picking from a back
Tossing coins
Guessing
Discussions
Solving problems



Probability games e.g cards
Calculators
Coins cards
Dies
Multiplication tables
Mathematical tables
KLB secondary mathematics book 3 page 262-289
Mathematics for secondary schools (N.M. patel) form 3 pages 225
Golden tips mathematics pages 268

6-7

Probability

Mutually exclusive events

By the end of the lesson, the learner should be able to
Identify mutually exclusive events
Solve problems involving mutually exclusive events

Playing probability games
Picking from a back
Tossing coins
Guessing
Discussions
Solving problems

Probability games e.g cards
Calculator
Coins
Dies
Multiplication tables
Mathematical tables
KLB secondary mathematics book 3 page 272-274
Mathematics for secondary schools (N.M. patel) form 3 pages 227-228
Golden tips mathematics pages 268

7

1

Probability

Mutually exclusive events

By the end of the lesson, the learner should be able to
Identify exclusive events
Solve problems involving mutually exclusive



Playing probability games e.g cards
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Playing probability games e.g cards
Calculators
Coins
Cards
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 272-274
Mathematics for secondary schools (N.M. patel) form 3 pages 227-228
Golden tips mathematics pages 268

2-3

Probability

Independent events

By the end of the lesson, the learner should be able to
Identify independent events
Solve problems involving independent events

Playing probability game e.g cards
Picking from a bag
Tossing coins
Guessing
Discussions
Solving problems

Playing probability games e.g cards
Calculators
Coins
Cards
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 274-282
Mathematics for secondary schools (N.M. patel) form 3 pages 228-229
Golden tips mathematics pages 268-269



4

Probability

Independent events

By the end of the lesson, the learner should be able to

Identify independent events

Solving problems involving independent events

Playing probability games

Picking from a bag

Tossing coins

Discussion

Solving problems

Probability games e.g cards

Calculator

Coins

Cards

Multiplication tables

Mathematical tables

KLB secondary mathematics book 3 page 274-282

Mathematics for secondary schools (N.M. patel) form 3 pages 228-229

Golden tips mathematics pages 268-269

5-6

probability

Tree diagrams

By the end of the lesson, the learner should be able to

Use tree diagrams to determine probabilities of events

Playing probability games e.g cards

Picking from a bag

Tossing coins

Discussions

Solving problems

Probability games e.g cards

Calculators

Coins

Cards

Multiplication tables

Mathematical tables



KLB secondary mathematics book 3 page 282-287
Mathematics for secondary schools (N.M. patel) form 3 pages 232-238
Golden tips mathematics pages 269-272

7

Probability

Tree diagrams

By the end of the lesson, the learner should be able to
Use tree diagrams to determine probabilities of events

Playing probability games
Picking from a bag
Tossing coins
Guessing
Discussions
Problem solving

Playing games e.g cards
Calculators
Coins
Cards
Dies
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 282-287
Mathematics for secondary schools (N.M. patel) form 3 pages 232-238
Golden tips mathematics pages 269-272

8

1-2

Probability

Tree diagram

By the end of the lesson, the learner should be able to
Determine probability of events

Playing probability games
Picking from a bag
Tossing



Guessing
Discussions
Problem solving

Playing games e.g cards
Calculators
Coins
Cards
Dice
Multiplication tables
Mathematical tables

KLB secondary mathematics book 3 page 282-287
Mathematics for secondary schools (N.M. patel) form 3 pages 232-238
Golden tips mathematics pages 269-272

3-4

Compound Proportion and rate of work

Direct and inverse proportion

By the end of the lesson, the learner should be able to
Solve problems involving direct proportions
Solving problems involving inverse proportion

Sharing equally
Sharing according to a given ratio
Multiplying and dividing numbers
Discussions
Solving problems

Calculator
Mathematical tables
Multiplication tables

KLB secondary mathematics book 3 page 282-287
Mathematics for secondary schools (N.M. patel) form 3 pages 232-238
Golden tips mathematics pages 269-272

5

Compound proportion and rate of work

Direct and inverse proportions

By the end of the lesson, the learner should be able to
Solve problems involving direct proportions



Solve problems involving inverse proportions

Sharing equally

Sharing according to a given ratio

Multiplying and dividing numbers

Discussions

Solving problems

Calculators

Mathematical tables

Multiplication tables

KLB secondary mathematics book 3 page 282-287

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Golden tips mathematics pages 38-40

6-7

Compound proportion and rate of work

Compound proportions

By the end of the lesson, the learner should be able to

Solve problems involving compound proportions

Sharing equally

Sharing according to a given ratio

Multiplying and dividing number

Discussions

Solving problems

Calculator

Mathematical tables

Multiplication tables

KLB secondary mathematics book 3 page 288-294

Mathematics for secondary schools (N.M. patel) form 3 pages 246-248

Golden tips mathematics pages 39-40

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1

Compound proportion and rate of work

Compound proportions

By the end of the lesson, the learner should be able to

Solve problems involving compound proportions



Sharing equally
Sharing according to a given ratio
Multiplying and dividing numbers
Discussions
Problems solving

Calculators
Mathematical tables
Multiplication tables

KLB secondary mathematics book 3 page 288-294
Mathematics for secondary schools (N.M. patel) form 3 pages 246-248
Golden tips mathematics pages 39-40

2-3

Compound proportion and rate of work

Rate of work

By the end of the lesson, the learner should be able to
Solve problems involving rate of work

Sharing equally
Sharing according to a given ratio
Multiplying and dividing numbers
Discussions
Solving problems

Calculator
Mathematical tables
Multiplication tables

KLB secondary mathematics book 3 page 294-298
Mathematics for secondary schools (N.M. patel) form 3 pages 243-245
Golden tips mathematics pages 39-40

9

4

Compound proportion and rate of work

Rate of work

By the end of the lesson, the learner should be able to



Solve problems involving the rate of work

Sharing equally

Sharing according to a given ratio

Calculators

Mathematical tables

Multiplication tables

KLB secondary mathematics book 3 page 294-298

Mathematics for secondary schools (N.M. patel) form 3 pages 243-245

Golden tips mathematics pages 39-40

5-6

Compound proportion and rate of work

Proportion in mixtures

By the end of the lesson, the learner should be able to

Solve problems involving proportion in mixtures

Sharing equally

Sharing according to a given ratio

Multiplying and dividing numbers

Discussions solving problems

Calculators

Mathematical tables

Multiplication tables

KLB secondary mathematics book 3 page 294-298

Mathematics for secondary schools (N.M. patel) form 3 pages 249-250

Golden tips mathematics pages

40

7

Compound proportion and rate of work

Proportions in mixtures

By the end of the lesson, the learner should be able to

Solve problems involving proportions in mixtures

Sharing equally

Sharing according to a given ratio

Multiplying and dividing numbers

Discussions



Solving problems

Calculators

Mathematical tables

Multiplication tables

KLB secondary mathematics book 3 page 294-298

Mathematics for secondary schools (N.M. patel) form 3 pages 249-250

Golden tips mathematics pages

40

10

1-2

Graphical methods

Graphs of cubic functions

By the end of the lesson, the learner should be able to

Make tables of values from given cubic functions

Equations using graphs

Making tables values

Drawing graphs

Reading values from graphs

discussions

Calculator

Graph papers

Square boards

KLB secondary mathematics book 3 page 299-304

Mathematics for secondary schools (N.M. patel) form 3 pages 251-258

Golden tips mathematics pages 273-276

3-4

Graphical methods

Graph of cubic functions

By the end of the lesson, the learner should be able to

Make tables of values from given cubic functions

Find solutions of cubic equations using graphs

Making table values

Drawing graphs

Solving problems using graphs



Calculators
Graph papers
Square boards

KLB secondary mathematics book 3 page 299-304
Mathematics for secondary schools (N.M. patel) form 3 pages 251-258
Golden tips mathematics pages 273-276

5-6

Graphical methods

Rate of change

By the end of the lesson, the learner should be able to
Determine the average rate of change of a function over a given interval
Determine the average rate of change of a function over a point

Making table of values
Drawing graphs
Reading values from graphs
Solving problems using graphs
discussions

calculator
graph papers
square boards

KLB secondary mathematics book 3 page 304-315
Mathematics for secondary schools (N.M. patel) form 3 pages 258-263

7

Graphical methods

Rate of change

By the end of the lesson, the learner should be able to
Determine the average rate of a function over a given point
Determine the average rate of change of a function over an interval

Making tables of values
Drawing graphs
Reading values from graphs
Solving problems using graphs
discussions

calculators
mathematical table
multiplication tables



graph papers
square boards

Discovering secondary mathematics book 3 students book 3 pages 153-155
Teacher's book 3 pages 33-35,90
KLB secondary mathematics book 3 page 304-315
Mathematics for secondary schools (N.M. patel) form 3 pages 258-263

11

1

Graphical methods

Rate of change

By the end of the lesson, the learner should be able to
Determine the average rate of change rate of change of a function over a
given interval
Determine the average rate of change of a function over a given point

Making tables of values
Drawing graphs
Reading values from graphs
Solving problems
discussions

calculators
mathematical tables
multiplication on tables
graph papers
square boards

KLB secondary mathematics book 3 page 304-305
Mathematics for secondary schools (N.M. patel) form 3 pages 258-263

2-3

Graphical methods

The equation of a circle

By the end of the lesson, the learner should be able to
Identify the equation of a circle
Solve problems involving the equation of a circle

Making tables of values
Drawing graphs



Reading values from graphs
Solving problems using graphs
discussions

calculators
mathematical tables
multiplication tables
graph papers
square boards

KLB secondary mathematics book 3 page 325-329
Mathematics for secondary schools (N.M. patel) form 3 pages 272-279
Golden tips mathematical pages 277-279

4-5

Graphical methods

The equation of a circle

By the end of the lesson, the learner should be able to
Identify the equation of a circle
Solve problems involving the equation of a circle

Making tables of values
Drawing graphs
Reading values from graphs
Solving from graphs
Solving problems
Using graphs
discussions

calculators
mathematical tables
multiplication tables
graph papers
square boards

KLB secondary mathematics book 3 page 325-329
Mathematics for secondary schools (N.M. patel) form 3 pages 272-279
Golden tips mathematical pages 277-279

6-7

Graphical methods

Empirical graphs



By the end of the lesson, the learner should be able to
Draw graphs of empirical data
Interpret graphs of empirical data

Making tables of values
Drawing graphs
Reading values from graphs
Solving problems using graphs
discussions

calculators
mathematical tables
multiplication tables
graph papers
square boards

KLB secondary mathematics book 3 page 315-324
Mathematics for secondary schools (N.M. patel) form 3 pages 266-272
Golden tips mathematical pages 277

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1-2

Graphical methods

Empirical graphs

By the end of the lesson, the learner should be able to
Draw graphs of empirical data
Interpret graphs of empirical data

Making of values
Drawing graphs
Solving problems using graphs
discussions

calculators
mathematical tables
multiplication tables
square boards

KLB secondary mathematics book 3 page 315-324
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Golden tips mathematical pages 277

3-4

Graphical graphs



Changing how linear relationship to linear relationships

By the end of the lesson, the learner should be able to
Change non-linear relationships to linear relationships

Making tables of values
Drawing graphs
Solving problems using graphs
discussions

calculators
mathematical tables
multiplication
graph papers
square boards

KLB secondary mathematics book 3 page 318-324
Mathematics for secondary schools (N.M. patel) form 3 pages 267-272

5-6

Graphical methods

Changing non-linear relationship to linear relationships

By the end of the lesson, the learner should be able to
Change non-linear relationships to linear relationships

Making tables of values
Drawing graphs
Reading values from graphs
Solving problems using graph
discussions

calculators
graph papers
square boards

KLB secondary mathematics book 3 page 318-324
Mathematics for secondary schools (N.M. patel) form 3 pages 267-272

7

Graphical methods

Changing linear relationships to linear relationship



By the end of the lesson, the learner should be able to
Change non-linear relationships

Making tables of values
Reading values from tables
Solving problems using graphs
discussions

calculators
graphs papers
square boards

KLB secondary mathematics book 3 page 318-324
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END YEAR EXAMINATIONS

MATHEMATICS FORM IV
SCHEMES OF WORK
TERM 1

1

1-2

Matrix and transformation

translating

By the end of the lesson, the learner should be able to
Define translating and describe an image and an object under a given
translation

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Rubber band
Models



Calculators
Peg boards
strings

Longman explore mathematics students book 4 pages 129
KLB secondary mathematics book 4 page 1
Golden tips
mathematics pages 227

2

Matrices and transformation

rotations

By the end of the lesson, the learner should be able to
Define rotation and describe an image and an object under a given
rotation

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
Rubber band
Models
Calculators
Peg boards
strings

Longman explore mathematics students book 4 pages 130
KLB secondary mathematics book 4 page 3
Golden tips
mathematics pages 228

3

Matrices and transportation

Reflection

By the end of the lesson, the learner should be able to
Define reflections
Describe the image and the object under a given reflection



Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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Teacher's book 4 pages 1-3,23
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KLB secondary mathematics book 4 page 2
Golden tips
mathematics pages 230-234

4-5

Matrices and transformation

Enlargement

By the end of the lesson, the learner should be able to
Define reflection
Describe an image and its objects under a given reflection

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
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Teacher's book 4 pages 1-3,25
Longman explore mathematics students book 4 pages 139
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mathematics pages 235

6-7

Matrices and transformation

Exercise

By the end of the lesson, the learner should be able to
Able to answer questions on reflection, rotation translation and
enlargement

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
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Teacher's book 4 pages 1-3,24
Longman explore mathematics students book 4 pages 140
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Golden tips
mathematics pages 235

2

1-2

Matrix and transformation

sheers

By the end of the lesson, the learner should be able to
Define sheers



Describe an image and an object under a given sheer

Reflecting object in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects in the Cartesian plane

Multiplying, adding, subtracting and dividing numbers

Discussions

Solving problems

Square boards

Graph papers

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mathematics pages 236

3-4

Matrix and transformation

sheers

By the end of the lesson, the learner should be able to

Define sheer and describe an image and objects under a given sheer

Reflecting object in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects in the Cartesian plane

Multiplying, adding, subtracting and dividing numbers

Discussions

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mathematics pages 236

5-6

Matrices and transformation

stretches

By the end of the lesson, the learner should be able to
Define stretch
Describe an image and an object under a given stretch

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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Teacher's book 4 pages 1-3,24-25
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mathematics pages 237

7

Matrix and transformation

Stretches

By the end of the lesson, the learner should be able to
Define stretch and describe an image and an object under a given stretch



Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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mathematics pages 237

3

1

Matrix and transformation

Transformation matrix

By the end of the lesson, the learner should be able to
Define stretch and describe an image and an object under a given stretch

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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Golden tips

mathematics pages 237

2-3

Matrix and transformations

Transformation matrix

By the end of the lesson, the learner should be able to

Identify a transformation matrix given the image and the object and vice versa

Reflecting object in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects in the Cartesian plane

Multiplying, adding, subtracting and dividing numbers

Discussions

Solving problems

Square boards

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Calculators

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mathematics pages 239

4-5

Matrix and transformation

Transforming matrix

By the end of the lesson, the learner should be able to

Identify a transformation matrix given the image and the object and vice versa



Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

Square boards
Graph papers
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Teacher's book 4 pages 1-3,25-26
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mathematics pages 239

6-7

Matrix and transformation

Transformation matrix

By the end of the lesson, the learner should be able to
Identify a transformation matrix, given the image and the object and the
vice versa

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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mathematics pages 239

4

1-2

Matrix and transformation

Isometric and non-isometric transformation

By the end of the lesson, the learner should be able to

Define isometric and non-isometric transformation

Reflecting object in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects in the Cartesian plane

Multiplying, adding, subtracting and dividing numbers

Discussions

Solving problems

Square boards

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mathematics pages 234

3-4

Matrix and transformations

Successive transformations

By the end of the lesson, the learner should be able to

Perform successive transformations on an object and describe the image

Reflecting object in a mirror



Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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Golden tips
mathematics pages 238

5

Matrix and transformation

Successive transformations

By the end of the lesson, the learner should be able to
Perform successive transformation in an object and describe the image

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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6-7

Matrix and transformation

Matrix successive transformations

By the end of the lesson, the learner should be able to

Identify and determine a single matrix for successive transformations

Reflecting object in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects in the Cartesian plane

Multiplying, adding, subtracting and dividing numbers

Discussions

Solving problems

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5

1-2

Matrix and transformations

Matrix of successive transformations

By the end of the lesson, the learner should be able to

Identify and determine a single matrix for successive transformation



Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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Teacher's book 4 pages 1-3,27-28
Longman explore mathematics students book 4 pages 145
KLB secondary mathematics book 4 page 21
Golden tips
mathematics pages 239

3-4

Matrix and transformations

Inverse transformation

By the end of the lesson, the learner should be able to
Determine the inverse of a transformation matrix

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

Square boards
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mathematics pages 239

5-6

Matrices and transformations

Inverse of a transformation

By the end of the lesson, the learner should be able to

Determine the inverse of a transformation matrix

Reflecting object in a mirror

Rotating objects

Translating objects

Enlarging objects

Drawing images and objects in the Cartesian plane

Multiplying, adding, subtracting and dividing numbers

Discussions

Solving problems

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Teacher's book 4 pages 1-3,29

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7

Matrices and transformations

Area and the determinant of matrix

By the end of the lesson , the learner should be able to



Establish and use the relationship between area scale factors and determinant of a matrix

Reflecting object in a mirror
Rotating objects
Translating objects
Enlarging objects
Drawing images and objects in the Cartesian plane
Multiplying, adding, subtracting and dividing numbers
Discussions
Solving problems

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mathematics pages 240

6

1-2

statistics

The mean of ungrouped data

By the end of the new lesson, the learner should be able to
State the measures of central tendency
Calculate the mean of ungrouped data using the assumed mean method

Collecting data
Analyzing data
Representing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables



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Teacher's book 4 pages 3-6,32
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mathematics pages 192

3-4

statics

The mean of ungrouped data

By the end of the lesson, the learner should be able to
State the measure of central tendency
Calculate the mean of ungrouped data using assumed mean method

Collecting data
Analyzing data
Representing data
Assuming numbers
Discussions
Solving problems

Calculator
Square board
Graph paper
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 24-27
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages 15
KLB secondary mathematics book 4 page 38
Golden tips
mathematics pages 192

5-6

Statistics

The mean of ungrouped data

By the end of the lesson, the learner should be able to
Determine the mean of ungrouped data using an assumed mean or otherwise

Collecting data
Presenting data



Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 26-28
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages
KLB secondary mathematics book 4 page 38
Golden tips
mathematics pages 193

7

Statistics

The median of discrete data

By the end of the lesson, the learner should be able to
Determine the median of discrete data

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 29
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages 29
KLB secondary mathematics book 4 page 38
Golden tips
mathematics pages 194

7

1-2



Statics

The median of ungrouped data

By the end of the lesson, the learner should be able to
Determine the median of ungrouped frequently distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 29
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages 29
KLB secondary mathematics book 4 page 38
Golden tips
mathematics pages 194

3-4

Statistics

The median of ungrouped frequency distributions

By the end of the lesson, the learner should be able to
Determine the median of an ungrouped frequency distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 29
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages 29



KLB secondary mathematics book 4 page 38
Golden tips
mathematics pages 194

5-6

statistics

The median of grouped frequency distribution

By the end of the lesson, the learner should be able to
Determine the median of a grouped frequency distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical table

Discovering secondary mathematics book 4 students book 4 pages 30-31
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages 30
KLB secondary mathematics book 4 page 39
Golden tips
mathematics pages 194

7

Statistics

The median of grouped frequency distribution

By the end of the lesson, the learner should be able to
Determine the median of a grouped frequency distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems



Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 32-33
Teacher's book 4 pages 3-6,33-38
Longman explore mathematics students book 4 pages 30
KLB secondary mathematics book 4 page 38
Golden tips
mathematics pages 194-195

8

1

Statistics

The cumulative frequency curve

By the end of the lesson, the learner should be able to
Make a cumulative table
Draw cumulative curve from the data

Collecting data
Presenting data
Analyzing data
Assuming number
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 32-33
Teacher's book 4 pages 3-6,33-38
Longman explore mathematics students book 4 pages 27-29
KLB secondary mathematics book 4 page 48
Golden tips
mathematics pages 196

2-3

statistics

The quartiles



By the end of the lesson, the learner should be able to
Define quartile of a frequency distribution
Calculate the quartile of frequency distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 32-33
Teacher's book 4 pages 3-6,33-38
Longman explore mathematics students book 4 pages 31-32
KLB secondary mathematics book 4 page 46
Golden tips
mathematics pages 195

4-5

Statistics

The cumulative percentages

By the end of the lesson, the learner should be able to
Define cumulative percentage
Calculate the cumulative percentage

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 36-38
Teacher's book 4 pages 3-6,40
Longman explore mathematics students book 4 pages32
KLB secondary mathematics book 4 page 47
Golden tips
mathematics pages 195



6-7

statistics

Exercise

By the end of the lesson, the learner should be able to
Answer questions in previous exercises

Collecting data
Presenting data
Analyzing data
Assuming number
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 38
Teacher's book 4 pages 3-6,40
Longman explore mathematics students book 4 pages33
KLB secondary mathematics book 4 page 43
Golden tips
mathematics pages 199-202

9

1-2

Statistics

The range and the quartile range

By the end of the lesson, the learner should be able to
Define range, interquartile range, and the quartile deviation
Calculate range, interquartile range and the quartile deviation of
distribution

Collecting data
Presenting data
Analyzing data
Assuming data
Discussions
Solving problems



Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 38-42
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages31
KLB secondary mathematics book 4 page 47
Golden tips
mathematics pages 198

3-4

Statistics

The range and the quartile range

By the end of the lesson, the learner should be able to
Define and calculate the range, the interquartile range and the quartile
denotations of distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 38-42
Teacher's book 4 pages 3-6,32
Longman explore mathematics students book 4 pages31
KLB secondary mathematics book 4 page 47
Golden tips
mathematics pages 198

5-6

Statistics

The mean deviation

By the end of the lesson, the learner should be able to



Define mean deviation
Calculate the mean deviation, the absolute deviation and absolute value
of a distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 42-44
Teacher's book 4 pages 3-6,40
Longman explore mathematics students book 4 pages 161
KLB secondary mathematics book 4 page 56
Golden tips
mathematics pages 198

7

Statistics

The mean deviation

By the end of the lesson, the learner should be able to
Define mean deviation
Calculate the mean deviation, the absolute value, and the mean absolute
deviation of distribution

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 42-44
Teacher's book 4 pages 3-6,40
Longman explore mathematics students book 4 pages 161-162
KLB secondary mathematics book 4 page 56
Golden tips



mathematics pages 198

10

1-2

statics

The variance and the standard deviation

By the end of the lesson, the learner should be able to

Define variance and standard deviation

Calculate the variance and standard deviation

Collecting data

Presenting data

Analyzing data

Assuming numbers

Discussions

Solving problems

Calculators

Square boards

Graph papers

Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 44-47
Teacher's book 4 pages 3-6,40

Longman explore mathematics students book 4 pages 157 and 163

KLB secondary mathematics book 4 page 57

Golden tips

mathematics pages 198

3-4

Statistics

The variance and the standard deviation

By the end of the lesson, the learner should be able to

Define variance and standard deviation

Calculate variance and standard deviation of a distribution

Collecting data

Presenting data

Analyzing data

Assuming numbers

Discussions



Solving problems

Calculators

Square boards

Graph papers

Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 44-47

Teacher's book 4 pages 3-6,40

Longman explore mathematics students book 4 pages 157 and 163

KLB secondary mathematics book 4 page 57

Golden tips

mathematics pages 198-199

5-6

Statistics

Exercise

By the end of the lesson, the learner should be able to

Answer questions in previous exercises

Collecting data

Presenting data

Analyzing numbers

Discussions

Solving problems

Calculators

Square boards

Graph papers

Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 47-49

Teacher's book 4 pages 3-6,40

Longman explore mathematics students book 4 pages 168-169

KLB secondary mathematics book 4 page 59

Golden tips

mathematics pages 199-202

7

Statistics

Exercise



By the end of the lesson, the learner should be able to
Answer questions in previous exercises

Collecting data
Presenting data
Analyzing data
Assuming numbers
Discussions
Solving problems

Calculators
Square boards
Graph papers
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 47-49
Teacher's book 4 pages 3-6,40
Longman explore mathematics students book 4 pages 168-169
KLB secondary mathematics book 4 page 59
Golden tips
mathematics pages 199-202

11

1-2

Loci

Common Loci

By the end of the lesson, the learner should be able to
Define locus
Describe common types of loci

Drawing the locus of an item
Constructing loci
Measuring lengths/angles
Discussions
Solving problems

Graph papers
Charts
Commercial patterns
Pair of compasses
Ruler
Protractor
Square boards

Discovering secondary mathematics book 4 students book 4 pages 50-53
Teacher's book 4 pages 7-8,44-48
Longman explore mathematics students book 4 pages 269-271 and 41
KLB secondary mathematics book 4 page 68



Golden tips
mathematics pages 116

3-4

Loci

Common Loci

By the end of the lesson, the learner should be able to
Define locus and describe common types of loci

Drawing the locus of items
Construction loci
Measuring lengths/angles
Discussions
Solving problems

Graph papers
Square boards
Chard
Geometric patterns
Pair of compasses
Ruler
protractor
Discovering secondary mathematics book 4 students book 4 pages 50-53
Teacher's book 4 pages 7-8,44-48
Longman explore mathematics students book 4 pages 269-271 and 41
KLB secondary mathematics book 4 page 68
Golden tips
mathematics pages 116

5

Loci

Common Loci

By the end of the lesson, the learner should be able to
Define locus and describe common types of loci

Drawing the locus of an item
Constructing loci
Measuring lengths/angles
Discussions
Solving problems

Graph papers



Charts
Geometrical patterns
Pair of compasses
Ruler
Protractor

Discovering secondary mathematics book 4 students book 4 pages 50
Teacher's book 4 pages 7-8,44-48
Longman explore mathematics students book 4 pages 269-271 and 41
KLB secondary mathematics book 4 page 68
Golden tips
mathematics pages 116

6-7

Loci

Loci involving chords

By the end of the lesson, the learner should be able to
Construct and describe a locus involving chords

Drawing the locus from item
Constructing loci
Measuring lengths/angles
Discussions
Solving problems

Graph papers
Square boards
Charts
Geometrical patterns
Pair of compasses
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 54
Teacher's book 4 pages 7-8,49
Longman explore mathematics students book 4 pages 269-271 and 42
KLB secondary mathematics book 4 page 84
Golden tips
mathematics pages 116-120

12

1-2

Loci



Loci of inequality

By the end of the lesson, the learner should be able to

- Construct a locus of inequalities
- Describe locus of inequalities

- Drawing the locus of an item
- Constructing loci
- Measuring length/angles
- Discussions
- Solving problems

- Graph papers
- Square boards
- Charts
- Geometric patterns
- Pair of compasses
- Ruler
- protractors

Discovering secondary mathematics book 4 students book 4 pages 55
Teacher's book 4 pages 50-51
Longman explore mathematics students book 4 pages 269-271 and 49
KLB secondary mathematics book 4 page 81
Golden tips
mathematics pages 116-120

3

Loci

Loci of inequality

By the end of the lesson, the learner should be able to

- Construct and describe a locus for inequality

- Drawing the locus of a n item
- Constructing loci
- Measuring length/angles
- Discussions
- Solving problems

- Graph papers
- Square board
- Charts
- Geometric patterns
- Pair of compasses
- Ruler
- protractor



Discovering secondary mathematics book 4 students book 4 pages 57-58
Teacher's book 4 pages 53-54
Longman explore mathematics students book 4 pages 269-271 and 49
KLB secondary mathematics book 4 page 81
Golden tips
mathematics pages 116-120

4-5

Loci

Intersecting loci

By the end of the lesson, the learner should be able to
Describe and construct the intersecting loci

Drawing of the locus of an item
Constructing loci
Measuring lengths/angles
Discussions
Solving problems

Graph papers
Square boards
Charts
Geometric patterns
Pair of compasses
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 55-55
Teacher's book 4 pages 7-8,52
Longman explore mathematics students book 4 pages 269-271 and 50
KLB secondary mathematics book 4 page 75
Golden tips
mathematics pages 116-120

6-7

Loci

revision

By the end of the lesson, the learner should be able to
Answer the question in previous exercises

Drawing the locus of an item
Constructing loci



Measuring length/angle
Discussions
Solving problems

Graph papers
Square boards
Charts
Geometric patterns
Pair of compasses
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 57-58
Teacher's book 4 pages 7-8, 53-54
Longman explore mathematics students book 4 pages 271-272 and 52-54
KLB secondary mathematics book 4 page 85
Golden tips mathematics pages 122-124

EXAMINATIONS

MATHEMATICS FORM IV
SCHEMES OF WORK
TERM 2
1
1-2

Trigonometric ratios

Deviation of $\sin 2x$ 棋=1

By the end of the lesson, the learner should be able
to derive the trigonometric identity
 $\sin 2x = 1$ and use it to solve problems involving the trigonometric ratios

Drawing right-angle triangle
Measuring/angles/lengths
Squaring numbers
Discussions
Solving problems

Square boards
Graph papers
Charts illustrating aptitude period and phase angle
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 59-61
Teacher's book 4 pages 54-55
Longman explore mathematics students book 4 pages 64-65
KLB secondary mathematics book 4 page 90
Golden tips mathematics pages 134-140



3

Trigonometric ratios

Deviation of $\sin 2x$ 棋=1

By the end of the lesson, the learner should be able to
Identify $\sin 2x$ 棋=1 and use it to solve problems involving the
trigonometric ratios

Drawing right-angle triangles

Measuring angles lengths

Squaring numbers

Discussions

Solving problems

Square boards

Graphs papers

Charts illustrating amplitude period and phase angle

Ruler

protractor

Discovering secondary mathematics book 4 students book 4 pages 59-61
Teacher's book 4 pages 54-55

Longman explore mathematics students book 4 pages 64-65

KLB secondary mathematics book 4 page 90

Golden tips mathematical pages 134-146

4-5

Trigonometric ratios

The graph $y = \sin x$ and $y = a \sin x$

By the end of the lesson, the learner should be able to
Draw graphs of $y = \sin x$ and $y = a \sin x$ and determine their amplitudes, periods
and wavelengths

Drawing right-angle triangles

Measuring angles/lengths

Squaring numbers

Getting the square root of numbers

Discussion

Solving problems

Square boards

Graph papers

Charts illustrating amplitudes period and phase angle

Ruler



protractor

Discovering secondary mathematics book 4 students book 4 pages 61-62
Teacher's book 4 pages 54-55
Longman explore mathematics students book 4 pages 66-69
KLB secondary mathematics book 4 page 93
Golden tips mathematical pages 134-146

6-7

Trigonometric ratios

The graph $y = \cos x$ and $y = \cos x$

By the end of the lesson, the learner should be able to
Draw the graph $y = \cos x$ and $y = \cos x$ and determining their amplitudes the
periods and wavelengths

Drawing right-angle triangles
Measuring angles/lengths
Squaring numbers
Getting the square root of numbers
Discussion
Solving problems

Square board
Graph papers
Charts illustrating amplitudes period and phase-angle
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 61-62
Teacher's book 4 pages 59-11,54-55
Longman explore mathematics students book 4 pages 66-69
KLB secondary mathematics book 4 page 93
Golden tips mathematical pages 134-146

2

1-2

Trigonometric ratios

The graphs of $y = \sin bx$ and $y = \cos bx$

By the end of the lesson, the learner should be able to
Draw graphs of $y = \sin bx$ and $y = \cos bx$ and determine their amplitudes
periods and wavelengths



Drawing right-angle triangles
Measuring angles/lengths
Squaring numbers
Getting the square roots of numbers
Discussion
Solving problems

Square boards
Charts illustrating amplitude periods and phase angles
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 61-62
Teacher's book 4 pages 9-11,54-55
Longman explore mathematics students book 4 pages 66-69
KLB secondary mathematics book 4 page 93
Golden tips mathematical pages 134-146

3-4

Trigonometric ratios

The graphs $y = a \sin bx$ and $y = a \cos bx$

By the end of the lesson, the learner should be able to
Draw graphs of $y = a \cos bx$ and $y = a \sin bx$ and determine their amplitudes,
periods and wavelengths

Drawing right-angle triangles
Measuring angles/lengths
Squaring numbers
Getting the square roots of numbers
Discussion
Solving problems

Square boards
Charts illustrating amplitudes periods and phase angles
Ruler
Protractor

Discovering secondary mathematics book 4 students book 4 pages 63-66
Teacher's book 4 pages 9-11,54-55
Longman explore mathematics students book 4 pages 66-69
KLB secondary mathematics book 4 page 93
Golden tips mathematical pages 134-146

5-6



Trigonometric ratios

The graphs of $y = \sin(bx)$ and $y = \cos(bx)$

By the end of the lesson, the learner should be able to

Draw the graphs of $y = \sin(bx)$ and $y = \cos(bx)$ and determine their amplitudes, periods and wave lengths

Drawing right-angle triangles

Measuring angles/lengths

Squaring numbers

Getting the square roots of numbers

Discussion

Solving problems

Square boards

Graph papers

Charts illustrating amplitudes periods and phase angles

Ruler

protractor

Discovering secondary mathematics book 4 students book 4 pages 66-68
Teacher's book 4 pages 9-11, 54-55

Longman explore mathematics students book 4 pages 66-69

KLB secondary mathematics book 4 page 93

Golden tips mathematical pages 134-146

7

Trigonometric ratios

The graphs of $y = \sin(bx)$ and $y = \cos(bx)$

By the end of the lesson, the learner should be able to

Draw the graphs of $y = \sin(bx)$ and $y = \cos(bx)$ and determine their amplitudes, periods and wavelengths

Drawing right angle-triangles

Measuring angles/heights

Squaring numbers

Getting the square roots of numbers

Discussion

Solving problems

Square boards

Graphs papers

Charts illustrating amplitude periods and phase angles

Ruler

protractor

Discovering secondary mathematics book 4 students book 4 pages 66-68



Teacher's book 4 pages 9-11, 56-57

Longman explore mathematics students book 4 pages 66-69

KLB secondary mathematics book 4 page 93

Golden tips mathematical pages 134-146

3

1-2

Trigonometric ratios

The graphs of $y=\tan x$ and $y=\arctan x$

By the end of the lesson, the learner should be able to

Draw the graphs $y=\tan x$ and $y=\arctan x$ and determine their amplitudes, periods and wavelengths

Drawing right angle-triangles

Measuring angles/heights

Squaring numbers

Getting the square roots of numbers

Discussion

Solving problems

Square boards

Graph papers

charts illustrating amplitudes period and phase angle

Ruler

protractor

Discovering secondary mathematics book 4 students book 4 pages 68-69

Teacher's book 4 pages 9-11,

Longman explore mathematics students book 4 pages 66-69

KLB secondary mathematics book 4 page 93

Golden tips mathematical pages 134-146

3-4

Trigonometric ratios

The graphs of $y=\arctan(bx)$

By the end of the lesson, the learner should be able to

Draw the graph of $y=\arctan(bx)$ and determine its amplitude, periods and wavelengths

Drawing right angle-triangles

Measuring angles/lengths

Squaring numbers

Getting the square roots of numbers

Discussions



Solving problems

Square boards

Graphs papers

Discovering secondary mathematics book 4 students book 4 pages 69-70
Teacher's book 4 pages 9-11, 57-58
Longman explore mathematics students book 4 pages 66-69
KLB secondary mathematics book 4 page 93
Golden tips mathematical pages 134-146

5-6

Trigonometric ratios

exercise

By the end of the lesson, the learner should be able to
Answer questions in exercise 4.6

Drawing right-angle triangles
Measuring angles/lengths
Squaring numbers
Getting the square roots of numbers
Discussions
Solving problems

Square boards
Charts illustrating amplitude period and phase angle
Graph papers
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 71
Teacher's book 4 pages 9-11, 57-58
Longman explore mathematics students book 4 pages 77-78
KLB secondary mathematics book 4 page 102
Golden tips mathematical pages 134-146

7

Trigonometric ratios

Solving trigonometric problems

By the end of the lesson, the learner should be able to
Solve trigonometric equations by calculating and graphically

Drawing right-angle triangles



Measuring angles/lengths
Squaring numbers
Getting the square roots of numbers
Discussions
Solving problems

Square boards
Graph papers
Charts illustrating amplitude period and phase angle
Ruler
Protractor

Discovering secondary mathematics book 4 students book 4 pages 71-73
Teacher's book 4 pages 9-11,57-58
Longman explore mathematics students book 4 pages 77-78
KLB secondary mathematics book 4 page 102
Golden tips mathematical pages 134-146

4

1-2

Trigonometric ratios

Solving trigonometric equations

By the end of the lesson, the learner should be able to
Solve trigonometric equations by calculations and graphically

Drawing right-angle triangles
Measuring angles/lengths
Squaring numbers
Getting the square root of numbers
Discussions
Solving problems

Square boards
Graph papers
Charts illustrating amplitude period and phase angles
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 71-73
Teacher's book 4 pages 9-11,57-58
Longman explore mathematics students book 4 pages 77-78
KLB secondary mathematics book 4 page 100
Golden tips mathematical pages 134-147

3-4



Trigonometric ratios

Revision

By the end of the lesson, the learner should be able to
Answer the questions in further exercise 4

Drawing right angle triangles
Measuring angles/lengths
Squaring numbers
Getting the square root of numbers
Discussions
Solving problems

Square boards
Graph papers
Charts illustrating amplitude period and phase angles
Ruler
protractor

Discovering secondary mathematics book 4 students book 4 pages 71-73
Teacher's book 4 pages 9-11,58
Longman explore mathematics students book 4 pages 77-78
KLB secondary mathematics book 4 page 102
Golden tips mathematical pages 147-149

5-6

Three dimensional geometry

Geometrical properties of solids

By the end of the lesson, the learner should be able to
State the geometrical properties of column solids

Making models of column solids
Sketching nets of solids
Measuring angles/lengths
Discussions
Solving problems

3D models
Column solids
Net cents of solids
Nets of column solids

Discovering secondary mathematics book 4 students book 4 pages 74-75
Teacher's book 4 pages 11-13,59
Longman explore mathematics students book 4 pages 79
KLB secondary mathematics book 4 page 104
Golden tips mathematical pages 280



7

Three dimensional geometry

Geometric properties of solids

By the end of the lesson, the learner should be able to
State the geometric properties of column solids

Making models of column solids
Measuring angles/strengths
Discussions
Solving problems

3D models
Column solids
Net our of column solids
Nets of column solids

Discovering secondary mathematics book 4 students book 4 pages 74-75
Teacher's book 4 pages 11-13,59
Longman explore mathematics students book 4 pages 79
KLB secondary mathematics book 4 page 104
Golden tips mathematical pages 280

5

1-2

Three dimensional geometry

Projection of line on in pane

By the end of the lesson, the learner should be able to
Identify the projection of a line on a plane

Making models of column solids
Sketching nets of solids
Drawing the shape of solids
Measuring angles/lengths
Discussions
Solving problems

3D models
Column solids
Net cuts of column solids
Nets of column solids



Discovering secondary mathematics book 4 students book 4 pages 75-76
Teacher's book 4 pages 11-13,60
Longman explore mathematics students book 4 pages 85
KLB secondary mathematics book 4 page 106
Golden tips mathematical pages 280

3-4

Three dimensional geometry

An angle between two lines

By the end of the lesson, the learner should be able to
Identify and calculate the angle between two lines

Making models of column solids
Sketching nets of solids
Drawing the shape of solids
Measuring lengths
Discussions
Solving problems

3D models column solids
Nets of column solids

Discovering secondary mathematics book 4 students book 4 pages 76
Teacher's book 4 pages 11-13,60
Longman explore mathematics students book 4 pages 82
KLB secondary mathematics book 4 page 106
Golden tips mathematical pages 280

4-5

Three dimensional geometry

Skew lines

By the end of the learner should be able to
Identify skew lines and determine the angle between them

Making models of column solids
Sketching nets of solids
Drawing the shape of solids
Measuring angles/lengths
Discussions
Solving problems

3D models common solids



Cut out of common solids
Net of common solids

Discovering secondary mathematics book 4 students book 4 pages 76-78
Teacher's book 4 pages 11-13, 60
Longman explore mathematics students book 4 pages 90
KLB secondary mathematics book 4 page 118
Golden tips mathematical pages 281

6-7

Three dimensional geometry

Skew lines

By the end of the lesson, the learner should be able to
Identify skew lines and determine the angle between them

Making models of common solids
Drawing the shape of solids
Measuring angles/lengths
Discussions
Solving problems

3D models common solids
Cut out common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 76-78
Teacher's book 4 pages 11-13, 60
Longman explore mathematics students book 4 pages 90
KLB secondary mathematics book 4 page 118
Golden tips mathematical pages 281

6

1-2

Three dimensional geometry

The length of a line on a solid

By the end of the lesson, the learner should be able to
Identify and contribute the length of a line on a solid

Making models of common solids
Sketching nets of solids



Drawing the shapes solids
Measuring lengths/angles
Discussions
Solving problems

3D models
Common solids
Cut outs of common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 78-79
Teacher's book 4 pages 11-13,60
Longman explore mathematics students book 4 pages 87
KLB secondary mathematics book 4 page 106
Golden tips mathematical pages 281-286

3

Three dimensional geometry

The length of a line on a solid

By the end of the lesson, the learner should be able to
Identify and calculate the length of a line on solid

Making models of common solids
Sketching nets of solids
Drawing the shapes of solids
Measuring angles/lengths
Discussions
Solving problems

3D models
Common solids
Cut outs of common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 78-79
Teacher's book 4 pages 11-13,60
Longman explore mathematics students book 4 pages 87
KLB secondary mathematics book 4 page 106
Golden tips mathematical pages 281-286

4-5

Three dimensional geometry

An angle between a line and a plane



By the end of the lesson, the learner should be able to
Identify and calculate an angle between a line and a plane

Making models of common solids
Sketching nets of solids
Drawing nets of solids
Measuring lengths /angles
Discussions
Solving problems

3D models
Common solids
Cut out of common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 79-81
Teacher's book 4 pages 11-13,60-61
Longman explore mathematics students book 4 pages 82
KLB secondary mathematics book 4 page 106
Golden tips mathematical pages 281-282

6-7

Three dimensional geometry

An angle between a line plane

By the end of the lesson, the learner should be able to
Identify and calculate the length of an enable between a line and a plane

Making models of common solids
Sketching nets of solids
Drawing nets of solids
Measuring angles/lengths
Discussions
Solving problems

3D models of common solids
Cut outs of common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 79-81
Teacher's book 4 pages 11-13,60-61
Longman explore mathematics students book 4 pages 82
KLB secondary mathematics book 4 page 106
Golden tips mathematical pages 281-282



1-2

The dimensional geometry

An angle between two planes

By the end of the lesson, the learner should be able to
Identify and calculate an angle between two planes

Making models of common solids
Sketching nets of solids
Drawing the shapes of solids
Measuring angles/lengths
Discussions
Solving problems

3D models
Common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 81-83
Teacher's book 4 pages 11-13,61
Longman explore mathematics students book 4 pages 85
KLB secondary mathematics book 4 page 113
Golden tips mathematical pages 281

3-4

The dimensional geometry

An angle between two planes

By the end of the lesson, the learner should be able to
Identify and calculate on angle between two planes

Making models of common solids
Sketching nets of solids
Drawing the shapes of solids
Measuring lengths
Discussions
Solving problems

3D models
Common solids
Cut outs of common solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 81-83
Teacher's book 4 pages 11-13,61
Longman explore mathematics students book 4 pages 85
KLB secondary mathematics book 4 page 113



Golden tips mathematical pages 281

5-7

The dimensional geometry

Revision

By the end of the lesson, the learner should be able to
Answer the questions on the further Exercise 5

Making models of common solids
Sketching nets of solids
Drawing the shapes of solids
Measuring lengths
Discussions
Solving problems

3D models
Common solids
Cut outs column solids
Nets of common solids

Discovering secondary mathematics book 4 students book 4 pages 81-83
Teacher's book 4 pages 11-13,61
Longman explore mathematics students book 4 pages 85
KLB secondary mathematics book 4 page 113
Golden tips mathematical pages 281

8

1

Longitudes and latitude

Great and small circles

By the end of the lesson, the learner should be able to
Define the great and small circles in relation to a sphere

Drawing circles
Rolling balls
Spinning and writing time
Discussions
Solving problems

Globe calculations
Ball



Graph papers
Square boards
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 85-86
Teacher's book 4 pages 11-13, 63
Longman explore mathematics students book 4 pages 101
KLB secondary mathematics book 4 page 125
Golden tips mathematical pages 292

2

Longitudes and latitudes

Longitudes and latitudes

By the end of the lesson, the learner should be able to
Identify and define latitudes and longitudes

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussion
Solving problems

Globe
Calculators
Ball
Graph papers
Square boards
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 86-87
Teacher's book 4 pages 13-13, 63
Longman explore mathematics students book 4 pages 102
KLB secondary mathematics book 4 page 125
Golden tips mathematical pages 292

3-4

Longitudes and latitudes

Position on the surface of the earth

By the end of the lesson, the learner should be able to
Locate a place on earth's surface in terms of latitudes and longitudes



Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Globe
Calculators
Ball
Graph papers
Square boards
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 87-88
Teacher's book 4 pages 13-13,63
Longman explore mathematics students book 4 pages 103
KLB secondary mathematics book 4 page 128
Golden tips mathematical pages 293

5

Longitudes and latitude

Positions of points on the surface of the earth

By the end of the lesson, the learner should be able to
Locate a place on earth's surface in terms of latitudes and longitudes

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Globe
Calculators
Ball
Graph papers
Square boards
Mathematical tables
Multiplications tables

Discovering secondary mathematics book 4 students book 4 pages 87-88
Teacher's book 4 pages 13-13,63
Longman explore mathematics students book 4 pages 103
KLB secondary mathematics book 4 page 128
Golden tips mathematical pages 293



6-7

Longitudes and latitudes

The distance between two points along a great circle

By the end of the lesson, the learner should be able to

Calculate the distance between two points along a great circle in practical rules and kilometers and convert nautical miles to kilometers and vice versa.

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Drawing circles
Calculators
Ball
Graph papers
Square boards
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 89-91
Teacher's book 4 pages 13-13,63
Longman explore mathematics students book 4 pages 105
KLB secondary mathematics book 4 page 130
Golden tips mathematical pages 293-297

9

1-2

Longitudes and latitudes

The distance between two points along a great circle

By the end of the lesson, the learner should be able to

Calculate the distance between two points along a great circle in nautical miles and kilometers and convert nautical miles to kilometers

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions



Solving problems

Globe calculators

Ball

Graph papers

Square boards

Mathematical tables

Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 89-91
Teacher's book 4 pages 13-13,63

Longman explore mathematics students book 4 pages 105

KLB secondary mathematics book 4 page 130

Golden tips mathematical pages 293-297

3-4

Longitudes and latitudes

The distance between two points along in small circle

By the end of the lesson, the learner should be able to

Calculate the distance between two points along a small circle in
nautical miles and kilometers

Drawing circles

Rolling balls

Spinning the globe

Reading and writing time

Discussions

Solving problems

Globe

Calculators

Balls

Graph papers

Square boards

Mathematical tables

Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 91-93
Teacher's book 4 pages 13-45,63

Longman explore mathematics students book 4 pages 108

KLB secondary mathematics book 4 page 133

Golden tips mathematical pages 295-297



Longitudes and latitudes

The distance between two points along a small circle

By the end of the lesson, the learner should be able to

Calculate the distance between two points along a small circle in nautical miles and kilometers

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Globe
Calculator
Balls
Graph papers
Square boards
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 91-93
Teacher's book 4 pages 13-45,63
Longman explore mathematics students book 4 pages 108
KLB secondary mathematics book 4 page 133
Golden tips mathematical pages 293-297

6-7

Longitudes and latitudes

Time and longitude

By the end of the lesson, the learner should be able to

Calculate time in relation to longitudes

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Globe
Calculator
Balls
Graph papers
Square boards
Mathematical tables



Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 93-94
Teacher's book 4 pages 13-15, 63
Longman explore mathematics students book 4 pages 112
KLB secondary mathematics book 4 page 141
Golden tips mathematical pages 298

10

1-2

Longitudes and latitudes

speed

By the end of the lesson, the learner should be able to
Calculate speed in knots and kilometer per hour

Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Globe
Calculators
Balls
Graph paper
Square boards
Mathematical tables
Multiplications tables

Discovering secondary mathematics book 4 students book 4 pages 94-96
Teacher's book 4 pages, 63-64
Longman explore mathematics students book 4 pages 111
KLB secondary mathematics book 4 page 142
Golden tips mathematical pages 298

3-4

Longitudes and latitudes

Speed

By the end of the lesson, the learner should be able to
Calculate speed in knots and kilometers per hour



Drawing circles
Rolling balls
Spinning the globe
Reading and writing time
Discussions
Solving problems

Globe
Calculator
Graph papers
Square boards
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 94-96
Teacher's book 4 pages, 63-64
Longman explore mathematics students book 4 pages 111
KLB secondary mathematics book 4 page 142
Golden tips mathematical pages 298

5-6

Linear programming

Forming and solving inequalities

By the end of the lesson, the learner should be able to
Form linear inequalities based on real life situations

Forming inequalities
Forming algebraic equations
Shading unwanted regions
Discussions
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Multiplication tables
rulers

Discovering secondary mathematics book 4 students book 4 pages 94-96
Teacher's book 4 pages, 15-17, 64-67
Longman explore mathematics students book 4 pages 118
KLB secondary mathematics book 4 page 150
Golden tips mathematical pages 176



7

Linear programming

Forming and solving inequalities

By the end of the lesson, the learner should be able to
Form and represent linear inequalities in graph

Forming inequalities
Forming algebraic expressions
Plotting graphs
Discussions
Solving problems

Square boards
Papers
Calculator
Mathematical tables
Multiplication tables
Ruler

Discovering secondary mathematics book 4 students book 4 pages 97-100
Teacher's book 4 pages, 15-1763-64
Longman explore mathematics students book 4 pages 118
KLB secondary mathematics book 4 page 150
Golden tips mathematical pages 176

11

1-2

Linear programming

Forming and solving linear equations

By the end of the lesson, the learner should be able to
Form and represent linear inequalities in a graph

Forming inequalities
Forming algebraic equation
Plotting graphs
Shading unrated region
Discussions
Solving problems

Square boards
Graph papers
mathematical tables
ruler



Discovering secondary mathematics book 4 students book 4 pages 98-100
Teacher's book 4 pages, 67-73
Longman explore mathematics students book 4 pages 121-125
KLB secondary mathematics book 4 page 157
Golden tips mathematical pages 181

3-4

Linear programming

optimization

By the end of the lesson, the learner should be able to
Solve and interpret the optimum solution of linear inequalities

Forming inequalities
Forming algebraic expressions
Plotting graphs
Shading unwanted regions
Discussions
Solving problems

Square board
Graph papers
Calculators
Mathematical tables
ruler

Discovering secondary mathematics book 4 students book 4 pages 98-100
Teacher's book 4 pages, 67-73
Longman explore mathematics students book 4 pages 121-125
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5-6

Linear programming

The objective function

Forming inequalities
Forming algebraic expressions
Plotting graphs
Shading unwanted regions
Discussions
Solving problems



Square boards
Graph papers
Calculators
Mathematical tables

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181

7

Linear programming

The objective function

Forming inequalities
Forming algebraic expressions
Plotting graphs
Shading unwanted regions
Solving problems
discussions

square boards
graph papers
calculators
mathematical tables
ruler

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MATHEMATICS FORM IV
SCHEMES OF WORK
TERM 3

1

1-2

Differentiation

Average and instantaneous rate of change



By the end of the lesson, the learner should be able to
Define differentiation
Find the average rate of change and instantaneous rate of change

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square board
Graph paper
Calculators
Ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 105-107
Teacher's book 4 pages, 17-19,91
Longman explore mathematics students book 4 pages 170
KLB secondary mathematics book 4 page 162
Golden tips mathematical pages 302-303

3-4

differentiation

Average instantaneous rate of change

By the end of the lesson, the learner should be able to
Define differentiation
Find average rate of change and instantaneous rate of change

Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

square board
Graph papers
Calculators
Ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 105-107
Teacher's book 4 pages, 17-19,91
Longman explore mathematics students book 4 pages 170
KLB secondary mathematics book 4 page 162
Golden tips mathematical pages 302-303



5-6

Differentiation

The gradient of a curve at a point

By the end of the lesson the learner should be able to

Define a tangent

Find the gradient of a curve at a point using tangent

Sketching curves

Determining the gradient of a curve

Deriving formulae

Discussions

Solving problems

Square boards

Graph papers

Calculators

Ruler

Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 107-108

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KLB secondary mathematics book 4 page 163-164

Golden tips mathematical pages 304-305

7

differentiation

the gradient of $y=x^n$

By the end of the lesson, the learner should be able to

Find the gradient of a function in the form $y=x^n$ where n is a positive integer

Sketching curves

Determining the gradient of a curve

Deriving formulae

Discussions

Solving problems

Square boards

Graph papers

Calculators

Ruler

Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 108-109

Teacher's book 4 pages, 17-19,91



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KLB secondary mathematics book 4 page 164-166
Golden tips mathematical pages 304-305

2

1

Differentiation

The gradient of $y=x^n$

By the end of the lesson, the learner should be able to

Find the gradient function of a function in the form $y=x^n$ where n is a positive integer

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
Calculators
Ruler
Mathematical tables

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Teacher's book 4 pages, 17-19,91
Longman explore mathematics students book 4 pages 174-177
KLB secondary mathematics book 4 page 164-166
Golden tips mathematical pages 304-305

2-3

Differentiation

The gradient of $y=ax^n$

By the end of the lesson, the learner should be able to

Find the gradient function of a function in the form $y=ax^n$ where n is a positive integer

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems



Square boards
Graph papers
Calculators
Ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 109-110
Teacher's book 4 pages, 17-19,91
Longman explore mathematics students book 4 pages 177-180
KLB secondary mathematics book 4 page 166-167
Golden tips mathematical pages 304-305

4-5

Differentiation

The delta notation

By the end of the lesson, the learner should be able to
Find the derivative of a function using delta notation

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Discovering secondary mathematics book 4 students book 4 pages 110-111
Teacher's book 4 pages, 17-19,91
Longman explore mathematics students book 4 pages 172-177
KLB secondary mathematics book 4 page 167-170
Golden tips mathematical pages 304-305

6-7

Differentiation

The derivative a polynomial

By the end of the lesson, the learner should be able to
Determine the derivative of a polynomial



Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square boards
Graph calculators
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 111-112
Teacher's book 4 pages, 17-19, 91-92
Longman explore mathematics students book 4 pages 178-180
KLB secondary mathematics book 4 page 170-172
Golden tips mathematical pages 305

3

1-2

Differentiation

The derivative of polynomial

By the end of the lesson, the learner should be able to
Determine the derivative of a polynomial

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
Calculators
Ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 111-112
Teacher's book 4 pages, 17-19, 91-92
Longman explore mathematics students book 4 pages 178-180
KLB secondary mathematics book 4 page 170-172
Golden tips mathematical pages 305

3-4

Differentiation

Equations of tangents and normal to curves



By the end of the lesson, the learner should be able to
Find the equation of a tangent
Find the equation of a normal to a curve

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square boards
Papers
Calculators
Ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 112-113
Teacher's book 4 pages, 17-19,93
Longman explore mathematics students book 4 pages 180-182
KLB secondary mathematics book 4 page 173-174
Golden tips mathematical pages 305-306

5-6

Differentiation

Stationary points

By the end of the lesson, the learner should be able to
Determine the stationary points of a curve

Sketching curves
Graph papers
Calculators
Ruler
Mathematical tables

Ruler
Mathematical tables
Graph papers
Square boards

Discovering secondary mathematics book 4 students book 4 pages 114-116
Teacher's book 4 pages, 17-19,92
Longman explore mathematics students book 4 pages 186-188
KLB secondary mathematics book 4 page 174-180
Golden tips mathematical pages 306-307



differentiation

Stationary points

By the end of the lesson, the learner should be able to
Determine the stationary points of a curve

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 114-116
Teacher's book 4 pages, 17-19,92
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Golden tips mathematical pages 306-307

4

1-2

differentiation

Curve sketching

By the end of the lesson, the learner should be able to
Sketch a curve

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions
Problem solving

Square boards
Graph paper
Calculators
Ruler
Mathematical table

Discovering secondary mathematics book 4 students book 4 pages 116-117
Teacher's book 4 pages, 17-19,93
Longman explore mathematics students book 4 pages 188-190
KLB secondary mathematics book 4 page 180-182
Golden tips mathematical pages 306-309



3-4

Differentiation

Differentiation in kinematics

By the end of the lesson, the learner should be able to

- Apply differentiation in calculating distance
- Apply differentiation in calculating distance
- Apply differentiation in calculating velocity
- Apply differentiation in calculating acceleration

Sketching curves

Determining the gradient of a curve

Deriving formulae

Discussions

Solving problems

Square boards

Graph papers

Ruler

Calculators

Mathematical tables

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Teacher's book 4 pages, 17-19, 94-95
Longman explore mathematics students book 4 pages 182-185
KLB secondary mathematics book 4 page 182-186
Golden tips mathematical pages 309-310

5

Differentiation

Differentiation in kinematic

By the end of the lesson, the learner should be able to

- Apply differentiation in calculating distance
- Apply differentiation in calculating velocity
- Apply differentiation in calculating acceleration

Sketching curves

Determine the gradient of a curve

Deriving formulae

Discussions

Problems solving

Graph papers

Square boards

Ruler



Calculators
Mathematical tables

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KLB secondary mathematics book 4 page 182-186
Golden tips mathematical pages 309-310

6-7

Differentiation

Maxima and minima

By the end of the lesson, the learner should be able to
Apply differentiation in finding the maximum of a function
Apply differentiation in finding of a function

Sketching curves
Determining the gradient of a curve
Discussions
Solving problems

Boards
Graph papers
Calculators
Ruler
Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 119-121
Teacher's book 4 pages, 17-19, 94-95
Longman explore mathematics students book 4 pages 182-185
KLB secondary mathematics book 4 page 186-189
Golden tips mathematical pages 310-313

5

1-2

Differentiation

Revision

By the end of the lesson, the learner should be able to
Answer questions in differentiations

Sketching curves
Determining the gradient of a curve
Deriving formulae
Discussions



Solving problems

Square boards

Graph papers

Calculators

Rulers

Mathematical tables

Discovering secondary mathematics book 4 students book 4 pages 119-121
Teacher's book 4 pages, 17-19, 94-95
Longman explore mathematics students book 4 pages 341-354
KLB secondary mathematics book 4 page 232-283
Golden tips mathematical pages 322-457

3

Approximation of area

Approximating area by counting

By the end of the lesson, the learner should be able to

Approximate the area of an irregular shape by counting the number of squares it curves

Approximating area

Counting

Sketching/drawing shapes

Tracing objects

Plotting curves

Discussions

Solving problems

Square boards

Graph papers

Tracing papers

Calculators

Mathematical tables

Irregular and regular shapes

Discovering secondary mathematics book 4 students book 4 pages 122-124
Teacher's book 4 pages, 19-20, 95
Longman explore mathematics students book 4 pages 197-200
KLB secondary mathematics book 4 page 195-201
Golden tips mathematics pages 316-317

4-5

Approximation of area



The trapezium rule

By the end of the lesson, the learner should be able to

Derive the formula for the trapezium rule

Use the trapezium rule to solve problems

Approximating area

Counting

Sketching/drawing shapes

Tracing objects

Plotting curves

Discussions

Solving problems

Square boards

Graph papers

Tracing papers

Calculators

Mathematical tables

Irregular and regular shapes

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Longman explore mathematics students book 4 pages 197-200

KLB secondary mathematics book 4 page 195-201

Golden tips mathematics pages 316-317

6-7

Approximation of area

The trapezium rule

By the end of the lesson, the learner should be able to

Derive the formulae for the trapezium rule

Use the trapezium rule to solve problems

Approximating area

Counting

Sketching/drawing shapes

Tracing objects

Plotting curves

Discussions

Solving problems

Square boards

Graph papers

Tracing papers

Calculators

Mathematical tables

Irregular and regular shapes



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KLB secondary mathematics book 4 page 195-201
Golden tips mathematics pages 316-317

6

1-2

Approximation of area

The mid ordinate rule

By the end of the lesson, the learner should be able to
Derive the mid-ordinate rule
Use mid-ordinate rule to solve problems

Approximating area
Counting sketching/drawing shapes
Tracing objects
Plotting curves
Discussions
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Irregular and regular shapes

Discovering secondary mathematics book 4 students book 4 pages 127-128
Teacher's book 4 pages, 19-20,97-100
Longman explore mathematics students book 4 pages 201-202
KLB secondary mathematics book 4 page 202-206
Golden tips mathematics pages 317

3

Approximation of area

Revision

By the end of the lesson, the learner should be able to
Answer questions on area approximation

Approximating area
Counting
Tracing objects
Plotting curves



Discussions
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Irregular and regular shapes

Discovering secondary mathematics book 4 students book 4 pages 129-130
Teacher's book 4 pages, 19-20, 97-100
Longman explore mathematics students book 4 pages 341-354
KLB secondary mathematics book 4 page 232-283
Golden tips mathematics pages 322-457

4

Integration

differentiation

By the end of the lesson, the learner should be able to
Carry out the process of differentiation

Sketching curves
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 131
Teacher's book 4 pages, 21-22, 100
Longman explore mathematics students book 4 pages 341-354
KLB secondary mathematics book 4 page 207-208
Golden tips mathematics pages 314

5

Integration

Reverse differentiation

By the end of the lesson, the learner should be able to
Interpret integration as the reverse of differentiation



Sketching curves
Discussions
Deriving formulae
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 131-132
Teacher's book 4 pages, 21-22,100
Longman explore mathematics students book 4 pages 208-209
KLB secondary mathematics book 4 page 208-212
Golden tips mathematics pages 314

6

integration

The integration notation

By the end of the lesson, the learner should be able to
Use the integration notation to carry out integration

Sketching curves
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 132
Teacher's book 4 pages, 21-22,100-101
Longman explore mathematics students book 4 pages 213
KLB secondary mathematics book 4 page 212-216

7

integration

Definite integrals

By the end of the lesson, the learner should be able to



Integrate a polynomial

Sketching curves

Deriving formulae

Discussions

Solving problems

Square boards

Graph paper

Calculators

Mathematical tables

Multiplication tables

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Longman explore mathematics students book 4 pages 214-216

KLB secondary mathematics book 4 page 212-216

Golden tips mathematics 314-315

7

1-2

Integration

Area under a curve

By the end of the lesson, the learner should be able to

Apply integration to find the area under a curve

Sketching curves

Deriving formulae

Discussions

Solving problems

Square boards

Deriving formulae

Discussions

Solving problems

Discovering secondary mathematics book 4 students book 4 pages 134-137

Teacher's book 4 pages, 21-22,101-102

Longman explore mathematics students book 4 pages 212-217

KLB secondary mathematics book 4 page 217-222

Golden tips mathematics 318-321

3-4



Integration

Area under a curve

By the end of the lesson, the learner should be able to
Apply integration to find the area under a curve

Sketching curves
Deriving formulae
Discussions
Solving problems

Square boards
Graphs papers
Mathematical tables
Calculators
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 134-137
Teacher's book 4 pages, 21-22,101-102
Longman explore mathematics students book 4 pages 212-217
KLB secondary mathematics book 4 page 217-2222
Golden tips mathematics 318-321

5-6

integration

Integration in kinematics

By the end of the lesson, the learner should be able to
Apply integration in kinematics

Sketching curves
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
Calculators
Mathematical tables
Multiplication tables

Discovering secondary mathematics book 4 students book 4 pages 138-139
Teacher's book 4 pages, 21-22,101-102
Longman explore mathematics students book 4 pages 210-212
KLB secondary mathematics book 4 page 223-231
Golden tips mathematics 315-316



7

integration

Integration in kinematics

By the end of the lesson, the learner should be able to
apply integration in kinematics

Sketching curves
Deriving formulae
Discussions
Solving problems

Square boards
Graph papers
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Mathematical tables
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Teacher's book 4 pages, 21-22,101-102
Longman explore mathematics students book 4 pages 210-212
KLB secondary mathematics book 4 page 223-231
Golden tips mathematics 315-316

K.C.S.E EXAMINATIONS